MATHEMATICS

TRIGONOMETRIC RATIO

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TRIGONOMETRIC RATIO

JEE EXERCISES

Exercise J-1

INTRODUCTION TO TRIGONOMETRY RATIOS

1. The expression $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$ can be written as : **[JEE(Main)-2013]**

व्यजंक $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$ को लिखा जा सकता है : **[JEE(Main)-2013]**

(A) sinA cosA + 1 (B) secA cosecA + 1 (C) tanA + cotA (D) secA + cosecA

Ans. B

Sol. Given expression

$$= \frac{\sin A}{\cos A}\times\frac{\sin A}{\sin A-\cos A}+\frac{\cos A}{\sin A}\times\frac{\cos A}{\cos A-\sin A}$$

$$= \frac{1}{\sin A-\cos A}\left\{\frac{\sin^3 A-\cos^3 A}{\cos A\sin A}\right\}$$

$$\frac{\sin^2 A+\sin A\cos A+\cos^2 A}{\sin A\cos A}=1+\sec A\,\text{cosec}\,A$$

2. Let $f_k(x)=\frac{1}{k}(\sin^k x+\cos^k x)$ where $x\in R$ and $k\geq 1$. Then $f_4(x)-f_6(x)$ equals : **[JEE(Main)-2014]**

माना $f_k(x)=\frac{1}{k}(\sin^k x+\cos^k x)$ जहाँ $x\in R$ तथा $k\geq 1$ तो $f_4(x)-f_6(x)$ का मान होगा : **[JEE(Main)-2014]**

(A) 1/4 (B) 1/12 (C) 1/6 (D) 1/3

Ans. B

Sol. $f_k(x)=\frac{1}{k}\left\{\sin^k x+\cos^k x\right\}$

$\because \quad f_4(x)-f_6(x)=\frac{1}{4}\left\{\sin^4 x+\cos^4 x\right\}-\frac{1}{6}\left\{\sin^6 x+\cos^6 x\right\}$

$=\frac{1}{4}\left\{1-2\sin^2 x\cos^2 x\right\}-\frac{1}{6}\left\{1-3\sin^2 x\cos^2 x\right\}=\frac{1}{12}$

BASED ON COMPOUND ANGLES

3. If $a\cos\theta+b\sin\theta=3$ and $a\sin\theta-b\cos\theta=4$, then the value of a^2+b^2 is :

(A) 9 (B) 16 (C) 25 (D) None of these

यदि $a\cos\theta + b\sin\theta = 3$ एवं $a\sin\theta - b\cos\theta = 4$ तब $a^2 + b^2$ का मान होगा :

(A) 9 (B) 16 (C) 25 (D) इनमें से कोई नहीं

Ans. C

Sol. $a\cos\theta + b\sin\theta = 3$

$\Rightarrow (a\cos\theta + b\sin\theta)^2 = 9$

$\Rightarrow a^2\cos^2\theta + b^2\sin^2\theta + 2ab\sin\theta\cos\theta = 9$ ...(i)

$a\sin\theta - b\cos\theta = 4$

$\Rightarrow (a\sin\theta - b\cos\theta)^2 = 16$

$\Rightarrow a^2\sin^2\theta + b^2\cos^2\theta - 2ab\sin\theta\cos\theta = 16$ ...(ii)

By (i) + (ii); we get

$a^2 + b^2 = 25$

4. The expression $\sec^2\theta \cdot \text{cosec}^2\theta$ is equal to :

(A) $\sin^2\theta + \cos^2\theta$ (B) $\sin^2\theta - \cos^2\theta$ (C) $\sec^2\theta + \text{cosec}^2\theta$ (D) None of these

व्यंजक $\sec^2\theta \cdot \text{cosec}^2\theta$ किस के तुल्य है :

(A) $\sin^2\theta + \cos^2\theta$ (B) $\sin^2\theta - \cos^2\theta$ (C) $\sec^2\theta + \text{cosec}^2\theta$ (D) इनमें से कोई नहीं

Ans. C

Sol. $\sec^2\theta\,\text{cosec}^2\theta = \dfrac{1}{\sin^2\theta\cos^2\theta} = \dfrac{\sin^2\theta + \cos^2\theta}{\sin^2\theta\cos^2\theta}$

$= \sec^2\theta + \text{cosec}^2\theta$

5. If $\tan 50^\circ - \tan 40^\circ = k\tan 10^\circ$, then the value of k is :

(A) 2 (B) 1 (C) 3 (D) None of these

यदि $\tan 50^\circ - \tan 40^\circ = k\tan 10^\circ$ तब k का मान क्या होगा :

(A) 2 (B) 1 (C) 3 (D) इनमें से कोई नहीं

Ans. A

Sol. $\because \tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A\tan B}$

$\Rightarrow \tan 50^\circ - \tan 40^\circ = \tan(50^\circ - 40^\circ)(1 + \tan 50^\circ\tan 40^\circ)$

$\therefore k = 1 + \tan 50^\circ\tan 40^\circ = \dfrac{\cos 40^\circ\cos 50^\circ + \sin 40^\circ\sin 50^\circ}{\cos 40^\circ\cos 50^\circ}$

$$= \frac{2\cos(50°-40°)}{2\cos 40°\cos 50°} = \frac{2\cos(10°)}{\cos(40°+50°)+\cos(40°-50°)}$$

$$\Rightarrow K = \frac{2\cos 10°}{0+\cos 10°} = 2$$

6. The exact value of $\frac{\sin\ 24^o\ \cos\ 6^o\ -\ \sin\ 6^o\ \sin\ 66^o}{\sin\ 21^o\ \cos\ 39^o\ -\ \cos\ 51^o\ \sin\ 69^o}$ is :

(A) 0 (B) 1 (C) – 1 (D) None of these

$\frac{\sin\ 24^o\ \cos\ 6^o\ -\ \sin\ 6^o\ \sin\ 66^o}{\sin\ 21^o\ \cos\ 39^o\ -\ \cos\ 51^o\ \sin\ 69^o}$ का निश्चित मान क्या होगा :

(A) 0 (B) 1 (C) – 1 (D) इनमें से कोई नहीं

Ans. C

Sol. $\frac{\sin 24°\cos 6° - -\sin 6°\sin 66°}{\sin 21°\cos 39° - \cos 51°\sin 69°}$

$$= \frac{\sin 24°\cos 6° - \sin 6°\sin(90°-24°)}{\sin 21°\cos 39° - \cos(90°-39°)\sin(90°-21°)}$$

$$= \frac{\sin 24°\cos 6° - \sin 6°\cos 24°}{\sin 21°\cos 39° - \cos(21°)\sin(39°)}$$

$$= \frac{\sin(24°-6°)}{\sin(21°-39°)} = \frac{\sin(18°)}{\sin(-18°)} = -1$$

7. If $\tan\alpha + \cot\alpha = a$, then the value of $\tan^4\alpha + \cot^4\alpha$ is :

(A) $a^2 - 4a + 2$ (B) $a^4 - 4a^2 + 2$ (C) $a^4 + 4a^2 - 2$ (D) None of these

यदि $\tan\alpha + \cot\alpha = a$ तब $\tan^4\alpha + \cot^4\alpha$ किस के तुल्य होगा :

(A) $a^2 - 4a + 2$ (B) $a^4 - 4a^2 + 2$ (C) $a^4 + 4a^2 - 2$ (D) इनमें से कोई नहीं

Ans. B

Sol. Given $\tan\alpha + \cot\alpha = a$

$\tan^4\alpha + \cot^4\alpha = (\tan^2\alpha + \cot^2\alpha)^2 - 2\tan^2\alpha\cot^2\alpha = [(\tan\alpha + \cot\alpha)^2 - 2]^2 - 2$

$= (a^2 - 2)^2 - 2$

$= a^4 - 4a^2 + 4 - 2$

$= a^4 - 4a^2 + 2$

8. The simplified value of, $\frac{2\left(\sin 2\theta + 2\cos^2\theta - 1\right)}{\cos\theta - \sin\theta - \cos 3\theta + \sin 3\theta}$ is :

$\frac{2\left(\sin 2\theta + 2\cos^2\theta - 1\right)}{\cos\theta - \sin\theta - \cos 3\theta + \sin 3\theta}$ का सरलतम रूप क्या होगा :

(A) $\sin\theta$ (B) $\cos\theta$ (C) $\text{cosec}\,\theta$ (D) $\sec\theta$

Ans. C

Sol. $\frac{2\left(\sin 2\theta + 2\cos^2\theta - 1\right)}{\cos\theta - \sin\theta - \cos 3\theta + \sin 3\theta}$

$= \frac{2\left(\sin 2\theta + \cos 2\theta\right)}{\cos\theta - \cos 3\theta + \sin 3\theta - \sin\theta}$

$= \frac{2\left(\sin 2\theta + \cos 2\theta\right)}{2\sin 2\theta\sin\theta + 2\cos 2\theta\sin\theta}$

$= \frac{2\left(\sin 2\theta + \cos 2\theta\right)}{2\sin\theta\left(\sin 2\theta + \cos 2\theta\right)} = \text{cosec}\,\theta$

9. If $\cos\theta = \frac{1}{2}\left(a + \frac{1}{a}\right)$, then the value of $\cos 3\theta$ in terms of 'a' is :

(A) $\frac{1}{4}\left(a^3 + \frac{1}{a^3}\right)$ (B) $\frac{1}{2}\left(a^3 + \frac{1}{a^3}\right)$ (C) $4\left(a^3 + \frac{1}{a^3}\right)$ (D) None of these

यदि $\cos\theta = \frac{1}{2}\left(a + \frac{1}{a}\right)$ तब $\cos 3\theta$ का मान 'a' के रूप में क्या होगा :

(A) $\frac{1}{4}\left(a^3 + \frac{1}{a^3}\right)$ (B) $\frac{1}{2}\left(a^3 + \frac{1}{a^3}\right)$ (C) $4\left(a^3 + \frac{1}{a^3}\right)$ (D) इनमें से कोई नहीं

Ans. B

Sol. $\cos\theta = \frac{1}{2}\left(a + \frac{1}{a}\right)$

$\Rightarrow \cos 3\theta = 4\cos^3\theta - 3\cos\theta$

$= \cos\theta\left(4\cos^2\theta - 3\right)$

$$=\left(a+\frac{1}{a}\right)\frac{1}{2}\left[4\left(\frac{1}{2}\left(a+\frac{1}{a}\right)\right)^2-3\right]$$

$$=\frac{1}{2}\left(a+\frac{1}{a}\right)\left[a^2+\frac{1}{a^2}+2-3\right]$$

$$=\frac{1}{2}\left(a+\frac{1}{a}\right)\left(a^2+\frac{1}{a^2}-1\right)$$

$$=\frac{1}{2}\left(a^3+\frac{1}{a}-a+a+\frac{1}{a^3}-\frac{1}{a}\right)=\frac{1}{2}\left(a^3+\frac{1}{a^3}\right)$$

10. $(1+\tan\alpha\tan\beta)^2+(\tan\alpha-\tan\beta)^2$ is equal to :

$(1+\tan\alpha\tan\beta)^2+(\tan\alpha-\tan\beta)^2$ किस के बराबर है :

(A) $\tan^2\alpha+\tan^2\beta$ (B) $\cos^2\alpha\ \cos^2\beta$ (C) $\sec^2\alpha\ \sec^2\beta$ (D) $\tan^2\alpha\ \tan^2\beta$

Ans. C

Sol. $(1+\tan\alpha\tan\beta)^2+(\tan\alpha-\tan\beta)^2$

$=1+\tan^2\alpha\tan^2\beta+2\tan\alpha\tan\beta+\tan^2\alpha+\tan^2\beta-2\tan\alpha\tan\beta$

$=1+\tan^2\alpha+\tan^2\beta(1+\tan^2\alpha)$

$=(1+\tan^2\alpha)(1+\tan^2\beta)$

$=\sec^2\alpha\cdot\sec^2\beta$

11. If $\frac{1+\sin 2x}{1-\sin 2x}=\cot^2(a+x)\ \forall\ x\in R-\left(n\pi+\frac{\pi}{4}\right), n\in N$. Then 'a' equals :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) None of these

यदि $\frac{1+\sin 2x}{1-\sin 2x}=\cot^2(a+x)\ \forall\ x\in R-\left(n\pi+\frac{\pi}{4}\right), n\in N$ तब 'a' किस के बराबर होगा :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) इनमें से कोई नहीं

Ans. C

Sol. Given

$$\frac{1+\sin 2x}{1-\sin 2x}=\cot^2(a+x)\ \forall x\in R-\left\{n\pi+\frac{\pi}{4}\right\}$$

$$\Rightarrow \frac{\sin^2 x+\cos^2 x+2\sin x\cos x}{\sin^2 x+\cos^2 x-2\sin x\cos x}=\cot^2(a+x)$$

$$\Rightarrow \left[\frac{\sin x+\cos x}{\sin x-\cos x}\right]^2=\cot^2(a+x)$$

$$\Rightarrow \left(\frac{1+\tan x}{\tan x-1}\right)^2=\frac{1}{\tan^2(a+x)}$$

$$\Rightarrow \tan^2\left(\frac{\pi}{4}+x\right)\cdot\tan^2(a+x)=1$$

$$\Rightarrow \left[\tan\left(\frac{\pi}{4}+x\right)\cdot\tan(a+x)\right]^2=1$$

$$\Rightarrow a=\frac{\pi}{2}+\frac{\pi}{4}=\frac{3\pi}{4}$$

12. If $\tan\alpha=\frac{5}{6}$ and $\tan\beta=\frac{1}{11}$, then $\left(0<\alpha,\beta<\frac{\pi}{2}\right)$:

(A) $\alpha+\beta=\frac{\pi}{6}$ (B) $\alpha+\beta=\frac{\pi}{4}$ (C) $\alpha+\beta=\frac{\pi}{3}$ (D) None of these

यदि $\tan\alpha=\frac{5}{6}$ एवं $\tan\beta=\frac{1}{11}$, तब $\left(0<\alpha,\beta<\frac{\pi}{2}\right)$:

(A) $\alpha+\beta=\frac{\pi}{6}$ (B) $\alpha+\beta=\frac{\pi}{4}$ (C) $\alpha+\beta=\frac{\pi}{3}$ (D) इनमें से कोई नहीं

Ans. B

Sol. Given $\tan\alpha=\frac{5}{6}$ $\tan\beta=\frac{1}{11}$ $\left(0<\alpha,\beta<\frac{\pi}{2}\right)$

$\because \tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}$

$$\Rightarrow \tan(\alpha+\beta)=\frac{\frac{5}{6}+\frac{1}{11}}{1-\frac{5}{66}}=\frac{61}{61}=1$$

$$\Rightarrow \alpha+\beta=\frac{\pi}{4} \qquad \{\because 0<\alpha+\beta<\pi\}$$

13. If $\tan x \tan y = a$ and $x + y = \frac{\pi}{6}$, then $\tan x$ and $\tan y$ satisfy the equation :

यदि $\tan x \tan y = a$ एवं $x + y = \frac{\pi}{6}$ तब $\tan x$ एवं $\tan y$ निम्न में से किस समीकरण को संतुष्ट कटेंगे :

(A) $x^2 - \sqrt{3}(1 - a)x + a = 0$ (B) $\sqrt{3}x^2 - (1 - a)x + a\sqrt{3} = 0$

(C) $x^2 + \sqrt{3}(1 - a)x - a = 0$ (D) $\sqrt{3}x^2 + (1 + a)x - a\sqrt{3} = 0$

Ans. B

Sol. Given

$x + y = \frac{\pi}{6}$ & $\tan x \tan y = a$

Let equation whose roots will be $\tan x$ & $\tan y$ is

$t^2 - (\tan x + \tan y)t + \tan x \tan y = 0$

$t^2 - (\tan(x + y)(1 - \tan x \tan y))t + a = 0$

$$\Rightarrow t^2 - \left(\tan\left(\frac{\pi}{6}\right)\cdot(1-a)\right)t + a = 0$$

$$\Rightarrow t^2 - \left(\frac{1-a}{\sqrt{3}}\right)t + a = 0$$

$$\Rightarrow \sqrt{3}t^2 - (1-a)t + a\sqrt{3} = 0$$

$\Rightarrow$ equation is $\sqrt{3}x^2 - (1-a)x + \sqrt{3}a = 0$

14. Which of the following number(s) is/are rational?

निम्न में से कौनसी संख्या परिमेय हैं?

(A) $\sin 15^\circ$ (B) $\cos 15^\circ$ (C) $\sin 15^\circ \cos 15^\circ$ (D) $\sin 15^\circ \cos 75^\circ$

Ans. C

Sol. (A) $\sin 15° = \frac{\sqrt{3}-1}{2\sqrt{2}} \in Q^c$ (Irrational)

(B) $\cos 15° = \frac{\sqrt{3}+1}{2\sqrt{2}} \in Q^c$ (Irrational)

(C) $\sin 15° \cos 15° = \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right) = \frac{1}{4} \in Q$ (Rational)

(D) $\sin 15° \cos 75° = \sin^2 15° = \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^2 \in Q^c$ (Irrational)

15. If $\cos x + \cos y + \cos \alpha = 0$ and $\sin x + \sin y + \sin \alpha = 0$, then $\cot\left(\frac{x+y}{2}\right) =$

यदि $\cos x + \cos y + \cos \alpha = 0$ तथा $\sin x + \sin y + \sin \alpha = 0$, तब $\cot\left(\frac{x+y}{2}\right) =$

(A) $\sin \alpha$ (B) $\cos \alpha$ (C) $\cot \alpha$ (D) $2 \sin \alpha$

Ans. C

Sol. Given $\cos x + \cos y + \cos \alpha = 0$
& $\sin x + \sin y + \sin \alpha = 0$

$\Rightarrow 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) = -\cos \alpha$

& $2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) = -\sin \alpha$

$\cot\left(\frac{x+y}{2}\right) = \cot \alpha$

16. If the roots of the quadratic equation, $x^2 + px + q = 0$ are $\tan 30^\circ$ and $\tan 15^\circ$ respectively, then the value of $2 + q - p$ is : **[AIEEE-2006]**

द्विघात समीकरण $x^2 + px + q = 0$ के मूल $\tan 30^\circ$ एवं $\tan 15^\circ$ हैं। तब $2 + q - p$ का मान क्या होगा : **[AIEEE-2006]**

(A) 2 (B) 3 (C) 0 (D) 1

Ans. B

Sol. $x^2 + px + q = 0$
$\tan 30° + \tan 15° = -p$
$\tan 30° \cdot \tan 15° = q$

$$\tan 45° = \frac{\tan 30° + \tan 15°}{1 - \tan 30° \tan 15°} = \frac{-p}{1-q} = 1$$

$\Rightarrow \quad -p = 1 - q$

$\Rightarrow \quad q - p = 1 \qquad \therefore 2 + q - p = 3$

17. Let $\cos(\alpha + \beta) = \frac{4}{5}$ and let $\sin(\alpha - \beta) = \frac{5}{13}$, where $0 \le \alpha, \beta \le \frac{\pi}{4}$. Then $\tan 2\alpha =$ **[AIEEE-2010]**

माना $\cos(\alpha + \beta) = \frac{4}{5}$ तथा माना $\sin(\alpha - \beta) = \frac{5}{13}$, जहाँ $0 \le \alpha, \beta \le \frac{\pi}{4}$ हैं तो $\tan 2\alpha =$ **[AIEEE-2010]**

(A) $\frac{25}{16}$ (B) $\frac{56}{33}$ (C) $\frac{19}{12}$ (D) $\frac{20}{7}$

Ans. B

Sol. Given $\cos(\alpha+\beta) = \frac{4}{5}$ $\sin(\alpha-\beta) = \frac{5}{13}$ $0 \le \alpha, \beta \le \frac{\pi}{4}$

$\Rightarrow \tan(\alpha+\beta) = \frac{3}{4}$ & $\tan(\alpha-\beta) = \frac{5}{12}$

$\Rightarrow \tan(2\alpha) = \tan(\alpha+\beta+\alpha-\beta)$

$$= \frac{\tan(\alpha+\beta) + \tan(\alpha-\beta)}{1 - \tan(\alpha+\beta)\tan(\alpha-\beta)}$$

$$= \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{5}{16}} = \frac{14}{12} \times \frac{16}{11}$$

$$= \frac{56}{33}$$

18. In a ΔPQR, if $3\sin P + 4\cos Q = 6$ and $4\sin Q + 3\cos P = 1$, then the angle R is equal to : **[AIEEE-2012]**

ΔPQR में यदि $3\sin P + 4\cos Q = 6$ तथा $4\sin Q + 3\cos P = 1$ है, तो कोण R बराबर हैं : **[AIEEE-2012]**

(A) $\frac{3\pi}{4}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

Ans. C

Sol. $3\sin P + 4\cos Q = 6$ ..... (1)

$4\sin Q + 3\cos P = 1$ ..... (2)

Squaring (1) and (2). Then adding

$9 + 16 + 24\sin(P + Q) = 37$

$\Rightarrow 24\sin(P + Q) = 12$

$\Rightarrow \sin(P + Q) = \frac{1}{2} \Rightarrow \sin R = \frac{1}{2}$

$\therefore R = \frac{\pi}{6}$ or $\frac{5\pi}{6}$

But $R = \frac{5\pi}{6}$ does't satisfy the given equation so

$R = \frac{\pi}{6}$

COMPONENDO AND DIVIDENDO

19. If $3\sin\beta = \sin(2\alpha + \beta)$ then :

(A) $\tan(\alpha + \beta) = 2\tan\alpha$ (B) $2\sin\beta = \sin(\alpha - \beta)\cos\alpha$

(C) $\sin\beta = \cos(\alpha + \beta)\cos\alpha$ (D) None of these

यदि $3\sin\beta = \sin(2\alpha + \beta)$ तब :

(A) $\tan(\alpha + \beta) = 2\tan\alpha$ (B) $2\sin\beta = \sin(\alpha - \beta)\cos\alpha$

(C) $\sin\beta = \cos(\alpha + \beta)\cos\alpha$ (D) इनमें से कोई नहीं

Ans. A

Sol. $3\sin\beta = \sin(2\alpha + \beta)$

$\Rightarrow \frac{\sin(2\alpha + \beta)}{\sin\beta} = \frac{3}{1}$

Apply componendo and dividendo

$\frac{\sin(2\alpha + \beta) + \sin\beta}{\sin(\beta + 2\alpha) - \sin\beta} = \frac{4}{2}$

$\frac{2\sin(\alpha + \beta)\cos\alpha}{2\cos(\alpha + \beta)\sin(\alpha)} = 2$

$\Rightarrow \tan(\alpha + \beta) = 2\tan\alpha$

20. If $\sin(x + 3\alpha) = 3\sin(\alpha - x)$, then :

यदि $\sin(x + 3\alpha) = 3\sin(\alpha - x)$ तब :

(A) $\tan x = \tan\alpha$ (B) $\tan x = \tan^2\alpha$ (C) $\tan x = \tan^3\alpha$ (D) $\tan x = 3\tan\alpha$

Ans. C

Sol. $\sin(x+3\alpha) = 3\sin(\alpha - x)$

$$\Rightarrow \frac{\sin(x+3\alpha)}{\sin(\alpha - x)} = 3$$

Apply componendo and dividendo

$$\Rightarrow \frac{\sin(x+3\alpha)+\sin(\alpha - x)}{\sin(x+3\alpha)-\sin(\alpha - x)} = \frac{4}{2} = 2$$

$$\Rightarrow \frac{2\sin(2\alpha)\cos(x+\alpha)}{2\cos(2\alpha)\sin(x+\alpha)} = 2$$

$$\Rightarrow 2\tan(x+\alpha) = \tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}$$

$$\Rightarrow \left(\frac{\tan x + \tan\alpha}{1 - \tan x\tan\alpha}\right)\cdot 2 = 2\left(\frac{\tan\alpha}{1-\tan^2\alpha}\right)$$

$\tan\alpha - \tan x\tan^2\alpha = \tan x + \tan\alpha - \tan^2\alpha\tan x - \tan^3\alpha$
$\Rightarrow \tan x = \tan^3\alpha$

21. If $\cos x = \frac{2\cos y - 1}{2 - \cos y}$, where $x, y \in (0, \pi)$. Then $\tan\frac{x}{2} \cdot \cot\frac{y}{2}$ is equal to :

यदि $\cos x = \frac{2\cos y - 1}{2 - \cos y}$, जहाँ $x, y \in (0, \pi)$. तब $\tan\frac{x}{2} \cdot \cot\frac{y}{2}$ बराबर होगा :

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{\sqrt{3}}$

Ans. B

Sol. $\frac{1}{\cos x} = \frac{2-\cos y}{2\cos y - 1}$ Applying C/D $\Rightarrow \frac{1-\cos x}{1+\cos x} = \frac{3(1-\cos y)}{1+\cos y}$

$\Rightarrow \tan^2\frac{x}{2} = 3\tan^2\frac{y}{2} \Rightarrow \tan^2\frac{x}{2}\cot^2\frac{y}{2} = 3$

CONTINUED PRODUCT

22. $\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$ is equal to :

$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$ बराबर होगा :

(A) $\frac{1}{2}$ (B) $\cos\frac{\pi}{8}$ (C) $\frac{1}{8}$ (D) $\frac{1+\sqrt{2}}{2\sqrt{2}}$

Ans. C

Sol. $\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$

$=\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\left(\pi-\frac{3\pi}{8}\right)\right)\left(1+\cos\left(\pi-\frac{\pi}{8}\right)\right)$

$=\left(1+\cos\frac{\pi}{8}\right)\left(1-\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1-\cos\frac{3\pi}{8}\right)$

$=\left(1-\cos^2\frac{\pi}{8}\right)\left(1-\cos^2\frac{3\pi}{8}\right)=\left[\sin\frac{\pi}{8}\sin\left(\frac{3\pi}{8}\right)\right]^2$

$=\left[\sin\left(\frac{\pi}{8}\right)\sin\left(\frac{\pi}{2}-\frac{\pi}{8}\right)\right]^2=\left[\sin\frac{\pi}{8}\cos\frac{\pi}{8}\right]^2$

$=\frac{1}{4}\left[\sin\left(2\cdot\frac{\pi}{8}\right)\right]^2=\frac{1}{4}\left[\sin\frac{\pi}{4}\right]^2=\frac{1}{8}$

23. $\cos 1°.\cos 2°.\cos 3°....\cos 179°$ =

(A) 0 (B) 1 (C) 2 (D) 3

Ans. A

Sol. $\cos 1°\cos 2°\ .....\ \cos(179°)$

$=\cos 1°\cos 2°\ .....\ \cos 89°\cos 90°\ .....\ \cos(179°)$

$=0$

RANGE OF TRIGONOMETRIC FUNCTIONS

24. If $A = \sin^2 x + \cos^4 x$, then for all real x : **[AIEEE-2011]**

यदि $A = \sin^2 x + \cos^4 x$, तो सभी वास्तविक x के लिये मान होंगे : **[AIEEE-2011]**

(A) $1 \le A \le 2$ (B) $\frac{3}{4} \le A \le \frac{13}{16}$ (C) $\frac{3}{4} \le A \le 1$ (D) $\frac{13}{16} \le A \le 1$

Ans. C

Sol. $A = \sin^2 x + \cos^4 x$

$= \cos^4 x - \cos^2 x + 1$

$= (\cos^2 x)^2 - 2.\cos^2 x.\frac{1}{2} + \frac{1}{4} - \frac{1}{4} + 1$

$= (\cos^2 x - \frac{1}{2})^2 + \frac{3}{4}$

$\because 0 \le \cos^2 x \le 1$

$\Rightarrow -\frac{1}{2} \le \cos^2 x - \frac{1}{2} \le \frac{1}{2}$

$\Rightarrow 0 \le \left(\cos^2 x - \frac{1}{2}\right)^2 \le \frac{1}{4}$

$\Rightarrow \frac{3}{4} \le A \le 1$

SUMMATION OF SERIES

25. The sum of $\sin^3\theta + \frac{1}{3}\sin^3 3\theta + \frac{1}{3^2}\sin^3 9\theta + ... +$ to n terms is :

$\sin^3\theta + \frac{1}{3}\sin^3 3\theta + \frac{1}{3^2}\sin^3 9\theta + ... +$ n के पदों का मान होगा :

(A) $\frac{1}{2}\left[3\sin\theta - \frac{1}{3^{n-1}}\sin(3^n\theta)\right]$ (B) $\frac{1}{2}\left[3\cos\theta - \frac{1}{3^{n-1}}\sin(3^n\theta)\right]$

(C) $\frac{1}{4}\left[3\cos\theta - \frac{1}{3^{n-1}}\sin(3^n\theta)\right]$ (D) $\frac{1}{4}\left[3\sin\theta - \frac{1}{3^{n-1}}\sin(3^n\theta)\right]$

Ans. D

Sol. $\because \sin 3\theta = 3\sin\theta - 4\sin^3\theta$

$$\Rightarrow \sin^3\theta = \frac{3\sin\theta - \sin 3\theta}{4}$$

$$\Rightarrow \sin^3\theta + \frac{1}{3}\sin^3(3\theta) + \frac{1}{3^2}\sin^3(9\theta) + ... + \frac{1}{3^{n-1}}\sin^3\left(3^{n-1}\theta\right)$$

$$= \left(\frac{3\sin\theta - \sin 3\theta}{4}\right) + \frac{1}{3}\left(\frac{3\sin 3\theta - \sin 9\theta}{4}\right) + \frac{1}{3^2}\left(\frac{3\sin 9\theta - \sin 3^3\theta}{4}\right) + ... + \frac{1}{3^{n-1}}\left(\frac{3\sin\left(3^{n-1}\theta\right) - \sin\left(3^n\theta\right)}{4}\right)$$

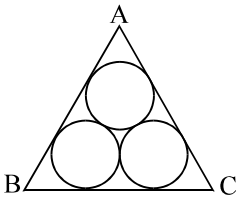
$$= \frac{3\sin\theta}{4} - \frac{1}{4}\sin 3\theta + \frac{1}{4}\sin 3\theta - \frac{1}{12}\sin(9\theta) + \frac{1}{12}\sin 9\theta - \frac{1}{36}\sin(27\theta) + ... + \frac{1}{3^{n-2}}\cdot\frac{1}{4}\sin\left(3^{n-1}\theta\right) - \frac{1}{3^{n-1}\cdot 4}\sin\left(3^n\theta\right)$$

$$= \left[\frac{3\sin\theta}{4} - \frac{1}{3^{n-1}}\frac{\sin\left(3^n\theta\right)}{4}\right]$$

OTHERS

26. In an equilateral triangle, 3 coins of radii 1 unit each are kept so that they touch each other and also the sides of the triangle. Area of the triangle is :

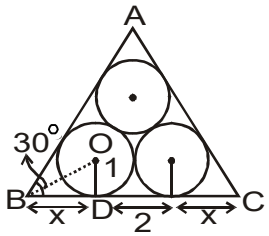
इकाई त्रिज्या के तीन सिक्के (coins) एक समबाहु त्रिभुज में इस प्रकार से रखे जाते है कि वे परस्पर एक दूसरे को तथा त्रिभुज की भुजाओं को स्पर्श करते है तो त्रिभुज का क्षेत्रफल होगा :



(A) $4 + 2\sqrt{3}$ (B) $6 + 4\sqrt{3}$ (C) $12 + \frac{7\sqrt{3}}{4}$ (D) $3 + \frac{7\sqrt{3}}{4}$

Ans. B

Sol. In ΔBOD



$$\tan 30° = \frac{1}{x}$$

$$\Rightarrow x = \sqrt{3}$$

$$\Rightarrow BC = 2 + 2\sqrt{3}$$

$$\text{area of } \Delta ABC = \frac{\sqrt{3}}{4}(2 + 2\sqrt{3})^2 = 4\sqrt{3} + 6$$

27. Let $\theta \in (0, \pi/4)$ and $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$, then :

माना कि $\theta \in (0, \pi/4)$ तथा $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$ तो :

(A) $t_1 > t_2 > t_3 > t_4$ (B) $t_4 > t_3 > t_1 > t_2$ (C) $t_3 > t_1 > t_2 > t_4$ (D) $t_2 > t_3 > t_1 > t_4$

Ans. B

Sol. $\because \quad \theta \in \left(0, \frac{\pi}{4}\right) \Rightarrow \tan\theta < 1$ and $\cot\theta > 1$

let $\tan\theta = 1 - x$ and $\cot\theta = 1 + y$

when $x, y > 0$ and are very small

$\therefore t_1 = (1-x)^{1-x}$, $t_2 = (1-x)^{1+y}$

$t_3 = (1+y)^{1-x}$ $t_4 = (1+y)^{1+y}$

clearly $t_4 > t_3$ & $t_1 > t_2$

also $t_3 > t_1$

Then $t_4 > t_3 > t_1 > t_2$

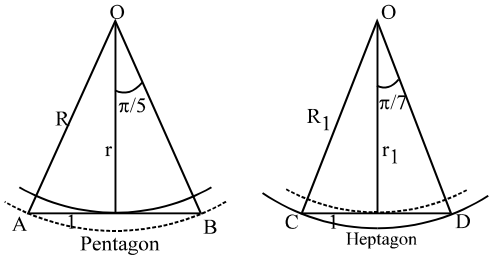
28. A circle is inscribed inside a regular pentagon and another circle is circumscribed about this pentagon. Similarly a circle is inscribed in a regular heptagon and another circumscribed about the heptagon. The area of the regions between the two circles in two cases are A_1 and A_2 respectively. If each polygon has a side length of 2 units then which one of the following is true?

एक सम पंचभुज तथा एक सम सप्तभुज के भीतर तथा बाहर वृत खींचा जाता है यदि दोनों वृत के मध्य का क्षेत्रफल दोनों बहुभुज के लिये यदि A_1 तथा A_2 है तथा प्रत्येक बहुभुज की भुजा 2 सेमी है, तो निम्न में से कौनसा सत्य होगा?

(A) $A_1 = \frac{5}{7}A_2$ (B) $A_1 = \frac{25}{49}A_2$ (C) $A_1 = \frac{49}{25}A_2$ (D) $A_1 = A_2$

Ans. D

Sol. in 1st case $r = \cot\frac{\pi}{5}$; $R = \operatorname{cosec}\frac{\pi}{5}$



2nd case $r_1 = \cot\frac{\pi}{7}$; $R_1 = \operatorname{cosec}\frac{\pi}{7}$

$\therefore \quad A_1 = \pi(R^2 - r^2) = \pi\left(\operatorname{cosec}^2\frac{\pi}{5} - \cot^2\frac{\pi}{5}\right) = \pi$

|||ly $\quad A_2 = \pi(R_1^2 - r_1^2) = \pi\left(\operatorname{cosec}^2\frac{\pi}{7} - \cot^2\frac{\pi}{7}\right) = \pi$ $\Rightarrow A_1 = A_2$ Ans.

29. If θ and ϕ are acute angles satisfying $\sin\theta = \frac{1}{2}$, $\cos\phi = \frac{1}{3}$, then $\theta + \phi \in$:

यदि $\sin\theta = \frac{1}{2}$ तथा $\cos\phi = \frac{1}{3}$ एवं θ व ϕ न्यूनकोण हो तो $\theta + \phi \in$:

(A) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$ (B) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ (C) $\left(\frac{2\pi}{3}, \frac{5\pi}{6}\right)$ (D) $\left(\frac{5\pi}{6}, \pi\right)$

Ans. B

Sol. Given

$$\sin\theta = \frac{1}{2}; 0 < \theta < \frac{\pi}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

Given $\cos\phi = \frac{1}{3}; 0 < \phi < \frac{\pi}{2}$

$$\Rightarrow \phi \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$$

$$\Rightarrow \theta + \phi \in \left(\frac{\pi}{6} + \frac{\pi}{3}, \frac{\pi}{6} + \frac{\pi}{2}\right)$$

$$\Rightarrow \theta + \phi \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$$

Exercise J-2

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON COMPOUND ANGLES

1. Prove that : $\cos^2\alpha + \cos^2(\alpha + \beta) - 2\cos\alpha \cdot \cos\beta \cos(\alpha + \beta) = \sin^2\beta$

   सिद्ध करो कि : $\cos^2\alpha + \cos^2(\alpha + \beta) - 2\cos\alpha \cdot \cos\beta \cos(\alpha + \beta) = \sin^2\beta$

Ans.

Sol. L.H.S. =

$$\cos^2\alpha + \cos^2(\alpha+\beta) - 2\cos\alpha\cos\beta\cos(\alpha+\beta)$$

$$= \cos^2\alpha + \cos(\alpha+\beta)\left[\cos(\alpha+\beta) - 2\cos\alpha\cos\beta\right]$$

$$= \cos^2\alpha + \cos(\alpha+\beta)\left(\cos\alpha\cos\beta - \sin\alpha\sin\beta - 2\cos\alpha\cos\beta\right)$$

$$= \cos^2\alpha - \cos(\alpha+\beta)\cos(\alpha-\beta)$$

$$= \cos^2\alpha - \left(\cos^2\alpha - \sin^2\beta\right)$$

$$= \sin^2\beta \quad \left\{\because \cos^2\alpha - \sin^2\beta = \cos(\alpha+\beta)\cos(\alpha-\beta)\right\}$$

$$= \text{R.H.S.}$$

2. Prove that : $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha = \cot\alpha$.

   सिद्ध करो कि : $\tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha = \cot\alpha$.

Ans.

Sol. L.H.S.

$$= \tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + 8\cot 8\alpha$$

$$= \tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + \frac{8}{\tan 8\alpha}$$

$$= \tan\alpha + 2\tan 2\alpha + 4\tan 4\alpha + \frac{8\left(1-\tan^2 4\alpha\right)}{2\tan 4\alpha}$$

$$= \tan\alpha + 2\tan 2\alpha + \frac{8\tan^2 4\alpha + 8 - 8\tan^2 4\alpha}{2\tan 4\alpha}$$

$$= \tan\alpha + 2\tan 2\alpha + \frac{4}{\tan 4\alpha}$$

$$= \tan\alpha + 2\tan 2\alpha + \frac{4\left(1-\tan^2 2\alpha\right)}{2\tan 2\alpha}$$

$$= \tan\alpha + \frac{2}{\tan 2\alpha}$$

$$= \frac{2\tan^2\alpha + 2\left(1-\tan^2\alpha\right)}{2\tan\alpha} = \frac{1}{\tan\alpha} = \cot\alpha = \text{R.H.S.}$$

3. If $X = \sin\left(\theta+\frac{7\pi}{12}\right) + \sin\left(\theta-\frac{\pi}{12}\right) + \sin\left(\theta+\frac{3\pi}{12}\right)$, $Y = \cos\left(\theta+\frac{7\pi}{12}\right) + \cos\left(\theta-\frac{\pi}{12}\right) + \cos\left(\theta+\frac{3\pi}{12}\right)$ then prove that $\frac{X}{Y} - \frac{Y}{X} = 2\tan 2\theta$.

यदि $X = \sin\left(\theta+\frac{7\pi}{12}\right) + \sin\left(\theta-\frac{\pi}{12}\right) + \sin\left(\theta+\frac{3\pi}{12}\right)$, $Y = \cos\left(\theta+\frac{7\pi}{12}\right) + \cos\left(\theta-\frac{\pi}{12}\right) + \cos\left(\theta+\frac{3\pi}{12}\right)$ तो सिद्ध कीजिए कि $\frac{X}{Y} - \frac{Y}{X} = 2\tan 2\theta$.

Ans.

Sol. $X = \sin\left(\theta+\frac{7\pi}{12}\right) + \sin\left(\theta-\frac{\pi}{12}\right) + \sin\left(\theta+\frac{3\pi}{12}\right)$

$$\Rightarrow X = 2\sin\left(\frac{2\theta+\frac{\pi}{2}}{2}\right)\cos\left(\frac{\pi}{3}\right) + \sin\left(\theta+\frac{\pi}{4}\right)$$

$$\Rightarrow X = \sin\left(\theta+\frac{\pi}{4}\right)\left(2\cdot\cos\frac{\pi}{3}+1\right)$$

$$\Rightarrow X = 2\sin\left(\theta+\frac{\pi}{4}\right)$$

Now

$$Y = \cos\left(\theta+\frac{7\pi}{12}\right) + \cos\left(\theta-\frac{\pi}{12}\right) + \cos\left(\theta+\frac{3\pi}{12}\right)$$

$$\Rightarrow Y = 2\cos\left(\theta+\frac{\pi}{4}\right)\cos\left(\frac{\pi}{3}\right) + \cos\left(\theta+\frac{\pi}{4}\right)$$

$$\Rightarrow Y = 2\cos\left(\theta+\frac{\pi}{4}\right)$$

Now $$\frac{X}{Y}-\frac{Y}{X}=\frac{2\sin\left(\theta+\frac{\pi}{4}\right)}{2\cos\left(\theta+\frac{\pi}{4}\right)}-\frac{2\cos\left(\theta+\frac{\pi}{4}\right)}{2\sin\left(\theta+\frac{\pi}{4}\right)}$$

$$=\tan\left(\theta+\frac{\pi}{4}\right)-\cot\left(\theta+\frac{\pi}{4}\right)$$

$$=-\left[\cot\left(\theta+\frac{\pi}{4}\right)-\tan\left(\theta+\frac{\pi}{4}\right)\right]$$

$$=-\left[2\cot\left(2\theta+\frac{2\pi}{4}\right)\right]$$

$$=-2\cot\left(\frac{\pi}{2}+2\theta\right)=2\tan 2\theta$$

4. Find the positive integers p, q, r, s satisfying $\tan\frac{\pi}{24}=\left(\sqrt{p}-\sqrt{q}\right)\left(\sqrt{r}-s\right)$.

धनात्मक पूर्णांक p, q, r, तथा s को ज्ञात करो यदि $\tan\frac{\pi}{24}=\left(\sqrt{p}-\sqrt{q}\right)\left(\sqrt{r}-s\right)$

Ans. p = 3, q = 2; r = 2; s = 1

Sol. $\because \tan^2 A=\frac{1-\cos 2A}{1+\cos 2A}$

$$\Rightarrow \tan^2\left(\frac{\pi}{24}\right)=\frac{1-\cos\left(\frac{\pi}{12}\right)}{1+\cos\left(\frac{\pi}{12}\right)}=\frac{1-\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)}{1+\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)}$$

$$=\left(\frac{2\sqrt{2}-\left(\sqrt{3}+1\right)}{2\sqrt{2}+\left(\sqrt{3}+1\right)}\right)\cdot\left(\frac{2\sqrt{2}-\left(\sqrt{3}+1\right)}{2\sqrt{2}-\left(\sqrt{3}+1\right)}\right)$$

$$=\frac{\left(2\sqrt{2}-\left[\sqrt{3}+1\right]\right)^2}{8-4-2\sqrt{3}}=\frac{\left(2\sqrt{2}-\sqrt{3}-1\right)^2}{4-2\sqrt{3}}$$

$$= \frac{\left(4+2\sqrt{3}\right)\left(2\sqrt{2}-\sqrt{3}-1\right)^2}{\left(4+2\sqrt{3}\right)\left(4-2\sqrt{3}\right)}$$

$$= \frac{\left(4+2\sqrt{3}\right)\left[8+4+2\sqrt{3}-4\sqrt{2}\left(\sqrt{3}+1\right)\right]}{4}$$

$$= \frac{\left(4+2\sqrt{3}\right)}{4}\left[12+2\sqrt{3}-4\sqrt{6}-4\sqrt{2}\right]$$

$$= \left[\frac{48+8\sqrt{3}-16\sqrt{6}-16\sqrt{2}+24\sqrt{3}+12-24\sqrt{2}-8\sqrt{6}}{4}\right]$$

$$= 15-6\sqrt{6}+8\sqrt{3}-10\sqrt{2}$$

$$= \left(5-2\sqrt{6}\right)\left(3-2\sqrt{2}\right)$$

$$= \left(\sqrt{3}-\sqrt{2}\right)^2\left(\sqrt{2}-1\right)^2$$

$$\Rightarrow \tan\left(\frac{\pi}{24}\right) = \left(\sqrt{3}-\sqrt{2}\right)\left(\sqrt{2}-1\right)$$

$\Rightarrow$ p = 3, q = 2, r = 2, s = 1

5. If the value of the expression $\sin 25° \cdot \sin 35° \cdot \sin 85°$ can be expressed as $\frac{\sqrt{a}+\sqrt{b}}{c}$ where a, b, c $\in$ N and are in their lowest form, find the value of (a + b + c).

यदि व्यंजक $\sin 25° \cdot \sin 35° \cdot \sin 85°$ को $\frac{\sqrt{a}+\sqrt{b}}{c}$ के रूप में व्यक्त किया जाता है जहां a, b, c $\in$ N (न्यूनतम रूप) हो तो a + b + c का मान बताओं।

Ans. 24

Sol. $\sin(25°) \cdot \sin(35°) \cdot \sin(85°)$
$= \sin(25°)\sin(60° - 25°)\sin(60° + 25°)$

$$= \frac{1}{4}\sin\left(75°\right)$$

$$= \frac{1}{4}\cos\left(15°\right) = \frac{\sqrt{3}+1}{8\sqrt{2}} = \frac{\sqrt{\frac{3}{2}}+\sqrt{\frac{1}{2}}}{8}$$

$= \frac{\sqrt{6}+\sqrt{2}}{16}$

$\Rightarrow a = 6, b = 2, c = 16$

$\Rightarrow a + b + c = 24$

6. Prove that :

सिद्ध करो कि :

(a) $\tan 20° \cdot \tan 40° \cdot \tan 60° \cdot \tan 80° = 3$

(b) $\tan 9° - \tan 27° - \tan 63° + \tan 81° = 4.$

(c) $\sin^4 \frac{\pi}{16} + \sin^4 \frac{3\pi}{16} + \sin^4 \frac{5\pi}{16} + \sin^4 \frac{7\pi}{16} = \frac{3}{2}$

Ans.

Sol. (a) $\tan 20° \cdot \tan 40° \cdot \tan 60° \cdot \tan 80°$

$= \tan 20° \cdot \tan(60° - 40°) \cdot \tan(60° + 20°) \cdot \sqrt{3}$

$= \tan(3 \cdot 20°) \cdot \sqrt{3} = 3$

(b) $\tan 9° - \tan 27° - \tan 63° + \tan 81°$

$= \tan 9° + \tan 81° - (\tan 27° + \tan 63°)$

$= \tan 9° + \cot 9° - (\tan 27° + \cot 27°)$

$= 2 \operatorname{cosec}(18°) - 2 \operatorname{cosec}(54°)$

$= \frac{2}{\sin(18°)} - \frac{2}{\sin 54°} = \frac{2}{\sin 18°} - \frac{2}{\cos 36°} = \frac{2}{\left(\frac{\sqrt{5}-1}{4}\right)} - \frac{2}{\left(\frac{\sqrt{5}+1}{4}\right)} = 4$

(C) $\sin^4\left(\frac{\pi}{16}\right) + \sin^4\left(\frac{3\pi}{16}\right) + \sin^4\left(\frac{5\pi}{16}\right) + \sin^4\left(\frac{7\pi}{16}\right)$

$= \sin^4\left(\frac{\pi}{16}\right) + \sin^4\left(\frac{3\pi}{16}\right) + \sin^4\left(\frac{\pi}{2} - \frac{3\pi}{16}\right) + \sin^4\left(\frac{\pi}{2} - \frac{\pi}{16}\right)$

$= \sin^4\left(\frac{\pi}{16}\right) + \cos^4\left(\frac{\pi}{16}\right) + \sin^4\left(\frac{3\pi}{16}\right) + \cos^4\left(\frac{3\pi}{16}\right)$

$= \left[\sin^2\left(\frac{\pi}{16}\right) + \cos^2\left(\frac{\pi}{16}\right)\right]^2 + \left[\sin^2\left(\frac{3\pi}{16}\right) + \cos^2\left(\frac{3\pi}{16}\right)\right] - 2\left(\sin^2\frac{\pi}{16}\cos^2\frac{\pi}{16} + \sin^2\frac{3\pi}{16} + \cos^2\frac{3\pi}{16}\right)$

$= 2 - \frac{1}{2}\left(\sin^2\frac{\pi}{8} + \sin^2\left(\frac{\pi}{2} - \frac{\pi}{8}\right)\right)$

$= 2 - \frac{1}{2} = \frac{3}{2}$

7. Calculate without using trigonometric tables : $4\cos 20° - \sqrt{3}\cot 20°$

बिना त्रिकोणमितीय सारणी के निम्नलिखित को सरल करो : $4\cos 20° - \sqrt{3}\cot 20°$

Ans. -1

Sol. $4\cos 20^o - \sqrt{3}\cot 20^o$

$$= \frac{4\cos 20^o \sin 20^o - \sqrt{3}\cos 20^o}{\sin 20^o}$$

$$= \frac{1}{\sin 20^o} \cdot 2 \cdot \left[\sin 40^o - \frac{\sqrt{3}}{2}\cos 20^o \right]$$

$$= \frac{1}{\sin 20^o}\left[2\sin 40^o - 2\sin 60^o \cos 20^o\right]$$

$$= \frac{2\sin 40^o - \sin(80^o) - \sin(40^o)}{\sin 20^o}$$

$$= \frac{2\cos(60^o)\sin(-20^o)}{\sin 20^o} = -1$$

8. Calculate without using trigonometric tables : $\dfrac{2\cos 40° - \cos 20°}{\sin 20°}$

बिना त्रिकोणमितीय सारणी के निम्नलिखित को सरल करो : $\dfrac{2\cos 40° - \cos 20°}{\sin 20°}$

Ans. $\sqrt{3}$

Sol. $\dfrac{2\cos 40^o - \cos 20^o}{\sin 20^o} = \dfrac{2}{\sin 20^o}\left[\cos 40^o - \dfrac{1}{2}\cos 20^o\right]$

$$= \frac{2\cos 40^o - 2\cos 60^o \cos 20^o}{\sin 20^o}$$

$$= \frac{2\cos 40^o - \cos 80^o - \cos 40^o}{\sin 20^o} = \frac{2\sin 60^o \sin 20^o}{\sin 20^o}$$

$$= \sqrt{3}$$

9. Calculate without using trigonometric tables : $\cos^6\frac{\pi}{16}+\cos^6\frac{3\pi}{16}+\cos^6\frac{5\pi}{16}+\cos^6\frac{7\pi}{16}$

बिना त्रिकोणमितीय सारणी के निम्नलिखित को सरल करो : $\cos^6\frac{\pi}{16}+\cos^6\frac{3\pi}{16}+\cos^6\frac{5\pi}{16}+\cos^6\frac{7\pi}{16}$

Ans. $\frac{5}{4}$

Sol. $\cos^6\frac{\pi}{16}+\cos^6\frac{3\pi}{16}+\cos^6\frac{5\pi}{16}+\cos^6\frac{7\pi}{16}$

$=\cos^6\frac{\pi}{16}+\cos^6\frac{3\pi}{16}+\cos^6\left(\frac{\pi}{2}-\frac{3\pi}{16}\right)+\cos^6\left(\frac{\pi}{2}-\frac{7\pi}{16}\right)$

$=\cos^6\frac{\pi}{16}+\sin^6\frac{\pi}{16}+\cos^6\frac{3\pi}{16}+\sin^6\frac{3\pi}{16}$

$=\left(\sin^2\frac{\pi}{16}+\cos^2\frac{\pi}{16}\right)^3-3\sin^2\frac{\pi}{16}\cos^2\frac{\pi}{16}\left(\sin^2\frac{\pi}{16}+\cos^2\frac{\pi}{16}\right)$

$+\left(\sin^2\frac{3\pi}{16}+\cos^2\frac{3\pi}{16}\right)^3-3\sin^2\frac{3\pi}{16}\cos^2\frac{3\pi}{16}\left(\sin^2\frac{3\pi}{16}+\cos^2\frac{3\pi}{16}\right)$

$=2-3\left(\sin^2\frac{\pi}{16}\cos^2\frac{\pi}{16}\right)-3\sin^2\frac{3\pi}{16}\cos^2\frac{3\pi}{16}$

$=2-\frac{3}{4}\left(\sin^2\frac{\pi}{8}\right)-\frac{3}{4}\left(\sin^2\frac{3\pi}{8}\right)$

$=2-\frac{3}{8}\left(1-\cos\frac{\pi}{4}\right)-\frac{3}{8}\left(1-\cos\frac{3\pi}{4}\right)$

$=2-\frac{3}{4}+\frac{3}{8}\left(\cos\frac{\pi}{4}+\cos\frac{3\pi}{4}\right)$

$=2-\frac{3}{4}+\frac{3}{8}\left(\cos\frac{\pi}{4}-\cos\frac{\pi}{4}\right)$

$=\frac{5}{4}$

10. If α, β and γ are in A.P., show that $\cot\beta=\frac{\sin\alpha-\sin\gamma}{\cos\gamma-\cos\alpha}$

यदि α, β तथा γ समान्तर श्रेणी में हो, तो सिद्ध कीजिए कि $\cot\beta = \dfrac{\sin\alpha - \sin\gamma}{\cos\gamma - \cos\alpha}$

Ans.

Sol. Given $\alpha, \beta, \gamma \rightarrow$ A.P.

$\Rightarrow 2\beta = \alpha + \gamma$

R.H.S.

$$= \frac{\sin\alpha - \sin\gamma}{\cos\gamma - \cos\alpha} = \frac{2\cos\left(\frac{\alpha+\gamma}{2}\right)\sin\left(\frac{\alpha-\gamma}{2}\right)}{2\sin\left(\frac{\alpha+\gamma}{2}\right)\sin\left(\frac{\alpha-\gamma}{2}\right)}$$

$$= \cot\left(\frac{\alpha+\gamma}{2}\right) = \cot(\beta)$$

11. If $\tan\beta = \dfrac{\tan\alpha + \tan\gamma}{1 + \tan\alpha\tan\gamma}$, prove that $\sin 2\beta = \dfrac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$

यदि $\tan\beta = \dfrac{\tan\alpha + \tan\gamma}{1 + \tan\alpha\tan\gamma}$ हो, तो सिद्ध कीजिए कि $\sin 2\beta = \dfrac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$

Ans.

Sol. Given

$$\tan\beta = \frac{\tan\alpha + \tan\gamma}{1 + \tan\alpha\tan\gamma}$$

$$\Rightarrow \tan\beta = \frac{\sin(\alpha+\gamma)}{\cos(\alpha-\gamma)}$$

$$\Rightarrow \sin 2\beta = \frac{2\tan\beta}{1+\tan^2\beta} = \frac{2\sin(\alpha+\gamma)\cdot\cos(\alpha-\gamma)}{\cos^2(\alpha-\gamma)+\sin^2(\alpha+\gamma)}$$

$$\Rightarrow \sin 2\beta = \frac{\sin 2\alpha + \sin 2\gamma}{(\cos\alpha\cos\gamma + \sin\alpha\sin\gamma)^2 + (\sin\alpha\cos\gamma + \cos\alpha\sin\gamma)^2}$$

$$\Rightarrow \sin 2\beta = \frac{\sin 2\alpha + \sin 2\gamma}{\cos^2\gamma + \sin^2\gamma + \sin 2\alpha \sin 2\gamma} = \frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$$

COMPONENDO AND DIVIDENDO

12. If $m\tan(\theta-30°)=n\tan(\theta+120°)$, show that $\cos 2\theta=\frac{m+n}{2(m-n)}$.

यदि $m\tan(\theta-30°)=n\tan(\theta+120°)$, तो सिद्ध करो कि $\cos 2\theta=\frac{m+n}{2(m-n)}$

Ans.

Sol. $m\tan(\theta-30^o)=n\tan(\theta+120^o)$

$$\Rightarrow \frac{m}{n}=\frac{\tan(\theta+120^o)}{\tan(\theta-30^o)}=\frac{\sin(\theta+120^o)\cos(\theta-30^o)}{\sin(\theta-30^o)\cos(\theta+120^o)}$$

By C & D Rule

$$\Rightarrow \frac{m+n}{m-n}=\frac{\sin(\theta+120^o+\theta-30^o)}{\sin(\theta+120^o-\theta+30^o)}=\frac{\sin(90^o+2\theta)}{\sin(150^o)}$$

$$\Rightarrow \frac{m+n}{m-n}=\frac{\cos(2\theta)}{\cos(60^o)}\Rightarrow \cos(2\theta)=\frac{m+n}{2(m-n)}$$

CONTINUED PRODUCT

13. Let $P(k)=\left(1+\cos\frac{\pi}{4k}\right)\left(1+\cos\frac{(2k-1)\pi}{4k}\right)\left(1+\cos\frac{(2k+1)\pi}{4k}\right)\left(1+\cos\frac{(4k-1)\pi}{4k}\right)$ then find the value of (a) P(5) and (b) P(6).

माना कि $P(k)=\left(1+\cos\frac{\pi}{4k}\right)\left(1+\cos\frac{(2k-1)\pi}{4k}\right)\left(1+\cos\frac{(2k+1)\pi}{4k}\right)\left(1+\cos\frac{(4k-1)\pi}{4k}\right)$ तो (a) P(5) तथा (b) P(6) का मान बताइये।

Ans. (a) $\frac{3-\sqrt{5}}{32}$; (b) $\frac{2-\sqrt{3}}{16}$

Sol. $$P(k)=\left(1+\cos\frac{\pi}{4k}\right)\left(1+\cos\frac{(2k-1)\pi}{4k}\right)\left(1+\cos\frac{(2k+1)\pi}{4k}\right)\left(1+\cos\left(\frac{(4k-1)\pi}{4k}\right)\right)$$

$$=\left(1+\cos\frac{\pi}{4k}\right)\left(1+\cos\left(\frac{\pi}{2}-\frac{\pi}{4k}\right)\right)\left(1+\cos\left(\frac{\pi}{2}+\frac{\pi}{4k}\right)\right)\left(1+\cos\left(\pi-\frac{\pi}{4k}\right)\right)$$

$$=\left(1+\cos\frac{\pi}{4k}\right)\left(1-\cos\frac{\pi}{4k}\right)\left(1+\sin\left(\frac{\pi}{4k}\right)\right)\cdot\left(1-\sin\left(\frac{\pi}{4k}\right)\right)$$

$$=\left(1-\cos^2\left(\frac{\pi}{4k}\right)\right)\left(1-\sin^2\left(\frac{\pi}{4k}\right)\right)$$

$$=\sin^2\frac{\pi}{4k}\cdot\cos^2\frac{\pi}{4k}$$

$$=\frac{1}{4}\left(\sin\frac{\pi}{2k}\right)^2$$

(a) $P(5)=\frac{1}{4}\left(\sin\frac{\pi}{10}\right)^2=\frac{1}{4}(\sin 18^\circ)^2$

$$=\frac{1}{4}\left(\frac{\sqrt{5}-1}{4}\right)^2=\frac{1}{64}\left(6-2\sqrt{5}\right)$$

$$=\frac{1}{32}\left(3-\sqrt{5}\right)$$

(b) $P(6)=\frac{1}{4}\left[\sin\left(\frac{\pi}{12}\right)\right]^2$

$$=\frac{1}{4}[\sin 15^\circ]^2=\frac{1}{4}\left[\frac{\sqrt{3}-1}{2\sqrt{2}}\right]^2$$

$$=\frac{2-\sqrt{3}}{16}$$

14. Given that $(1 + \tan 1°)(1 + \tan 2°).....(1 + \tan 45°) = 2^n$, find n.

दिया है कि $(1 + \tan 1°)(1 + \tan 2°).....(1 + \tan 45°) = 2^n$ तो n का मान बताओं।

Ans. $n = 23$

Sol. $(1+\tan 1^\circ)(1+\tan 2^\circ).....(1+\tan 44^\circ)(1+\tan 45^\circ)=2^n$

$\because (1+\tan A)(1+\tan B)=2 \quad \text{If } A+B=45^\circ$

$\Rightarrow 2^{22}\cdot(1+\tan 45^\circ)=2^n$

$\Rightarrow n=23$

15. The value of : $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\sin\frac{7\pi}{14}\sin\frac{9\pi}{14}\sin\frac{11\pi}{14}\sin\frac{13\pi}{14}$ is equal to _____.

$\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\sin\frac{7\pi}{14}\sin\frac{9\pi}{14}\sin\frac{11\pi}{14}\sin\frac{13\pi}{14}$ का मान बराबर होगा _____

Ans. $\frac{1}{64}$

Sol. $\sin\left(\frac{\pi}{14}\right)\sin\left(\frac{3\pi}{14}\right)\sin\left(\frac{5\pi}{14}\right)\sin\left(\frac{7\pi}{14}\right)\cdot\sin\left(\frac{9\pi}{14}\right)\cdot\sin\left(\frac{11\pi}{14}\right)\cdot\sin\left(\frac{13\pi}{14}\right)$

$= \sin\left(\frac{\pi}{14}\right)\sin\left(\frac{3\pi}{14}\right)\sin\left(\frac{5\pi}{14}\right)\sin\left(\frac{\pi}{2}\right)\sin\left(\pi-\frac{5\pi}{14}\right)\sin\left(\pi-\frac{3\pi}{14}\right)\sin\left(\pi-\frac{\pi}{14}\right)$

$= \sin^2\left(\frac{\pi}{14}\right)\sin^2\left(\frac{3\pi}{14}\right)\cdot\sin^2\left(\frac{5\pi}{14}\right)$

$= \left[\sin\left(\frac{\pi}{2}-\frac{3\pi}{7}\right)\sin\left(\frac{\pi}{2}-\frac{2\pi}{7}\right)\cdot\sin\left(\frac{\pi}{2}-\frac{\pi}{7}\right)\right]^2$

$= \left[\cos\frac{3\pi}{7}\cos\frac{2\pi}{7}\cos\frac{\pi}{7}\right]^2$

$= \left[\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\right]^2$

$= \left[\frac{\sin\left(\frac{8\pi}{7}\right)}{8\cdot\sin\left(\frac{\pi}{7}\right)}\right]^2 = \left[\frac{\sin\left(\pi+\frac{\pi}{7}\right)}{8\cdot\sin\left(\frac{\pi}{7}\right)}\right]^2 = \frac{1}{64}$

16. $\sin(B+2C)+\sin(C+2A)+\sin(A+2B) = 4\sin\frac{B-C}{2}\sin\frac{C-A}{2}\sin\frac{A-B}{2}$

Ans.

Sol. $\sin(B+2C)+\sin(C+2A)+\sin(A+2B)$

$= \sin(B+C+C)+\sin(C+A+A)+\sin(A+B+B)$

$= \sin(\pi-A+C)\ \sin(\pi-B+A)+\sin(\pi-C+B)$

$= \sin(A-C)+\sin(B-A)+\sin(C-B)$

$= 2\sin\frac{B-C}{2}\cos\frac{2A-B-C}{2}+2\sin\frac{C-B}{2}\cos\frac{C-B}{2}$

$= 2\sin\frac{B-C}{2}\left\{\cos\frac{2A-B-C}{2}-\cos\frac{B-C}{2}\right\}$

$$= 4\sin\frac{B-C}{2}\sin\frac{A-C}{2}\sin\frac{B-A}{2}$$

$$= 4\sin\frac{B-C}{2}\sin\frac{C-A}{2}\sin\frac{A-B}{2}$$

17. $\cos\frac{A}{2}+\cos\frac{B}{2}-\cos\frac{C}{2}=4\cos\frac{\pi+A}{4}\cos\frac{\pi+B}{4}\cos\frac{\pi-C}{4}$

Ans.

Sol. $4\cos\frac{\pi+A}{4}\cos\frac{\pi+B}{4}\cos\frac{\pi-C}{4}=2\cos\frac{\pi-C}{4}\left\{\cos\frac{2\pi+A+B}{4}+\cos\frac{A-B}{4}\right\}$

$$= 2\cos\frac{A+B}{4}\left\{\cos\left(\frac{\pi}{2}+\frac{A+B}{4}\right)+\cos\left(\frac{A-B}{4}\right)\right\}$$

$$= -2\cos\frac{A+B}{4}\sin\frac{A+B}{4}+2\cos\frac{A+B}{4}\cos\frac{A-B}{4}$$

$$= -\sin\frac{A+B}{2}+\cos\frac{A}{2}+\cos\frac{B}{2}$$

$$= \cos\frac{A}{2}+\cos\frac{B}{2}-\cos\frac{C}{2}$$

RANGE OF TRIGONOMETRIC FUNCTIONS

18. If $y = 10\cos^2x - 6\sin x\cos x + 2\sin^2x$, then find the greatest and least value of y.

यदि $y = 10\cos^2x - 6\sin x\cos x + 2\sin^2x$, तो y का अधिकतम तथा न्यूनतम मान बताओं।

Ans. $y_{max} = 11,\ y_{min} = 1$

Sol. $y = 10\cos^2x - 6\sin x\cos x + 2\sin^2x$

$$= 10\left(\frac{1+\cos 2x}{2}\right)-3\sin 2x+2\left(\frac{1-\cos 2x}{2}\right)$$

$=5 + 5\cos2x - 3\sin2x + 1 - \cos2x$

$= 6 + 4\cos2x - 3\sin2x$

Range of $4\cos2x - 3\sin2x$ is

$$\left[-\sqrt{16+9},\sqrt{16+9}\right]$$

$\Rightarrow y \in [1, 11]$

$\Rightarrow y_{max} = 11 \qquad y_{min.} = 1$

19. If $y = 1 + 2\sin x + 3\cos^2 x$, find the maximum and minimum values of $y \; \forall \; x \in R$.
यदि $y = 1 + 2\sin x + 3\cos^2 x$ तो x के प्रत्येक मान के लिये y का अधिकतम तथा न्यूनतम मान बताओं।

Ans. $y_{max} = \frac{13}{3}, \; y_{min} = -1$

Sol. $y = 1 + 2\sin x + 3\cos^2 x$
$= 1 + 2\sin x + 3(1 - \sin^2 x)$
$= 4 + 2\sin x - 3\sin^2 x$

$$= -3\left[\sin^2 x - \frac{2}{3}\sin x - \frac{4}{3}\right]$$

$$= -3\left[\left(\sin x - \frac{1}{3}\right)^2 - \frac{1}{9} - \frac{4}{3}\right]$$

$$= \frac{1}{3} + 4 - 3\left[\left(\sin x - \frac{1}{3}\right)^2\right] = \frac{13}{3} - 3\left[\left(\sin x - \frac{1}{3}\right)^2\right]$$

$\because -1 \le \sin x \le +1$

$$\Rightarrow -\frac{4}{3} \le \sin x - \frac{1}{3} \le \frac{2}{3}$$

$$\Rightarrow 0 \le \left(\sin x - \frac{1}{3}\right)^2 \le \frac{16}{9}$$

$$\Rightarrow -\frac{16}{3} \le -3\left(\sin x - \frac{1}{3}\right)^2 \le 0$$

$$\Rightarrow \frac{-3}{3} \le \frac{13}{3} - 3\left(\sin x - \frac{1}{3}\right)^2 \le \frac{13}{3}$$

$$\Rightarrow y \in \left[-1, \frac{13}{3}\right]$$

$$\Rightarrow y_{max} = \frac{13}{3} \qquad y_{min.} = -1$$

20. If $y = 9\sec^2 x + 16\,\text{cosec}^2 x$, find the minimum value of $y \; \forall \; x \in R$.
यदि $y = 9\sec^2 x + 16\,\text{cosec}^2 x$, तो x के सभी मान के लिये y का न्यूनतम मान बताओं।

Ans. 49

Sol. $y = 9\sec^2 x + 16\,\text{cosec}^2 x$

$y = 9(1 + \tan^2 x) + 16(1 + \cot^2 x)$
$= 25 + 9\tan^2 x + 16\cot^2 x$
$\because\ a^2\tan^2 x + b^2\cot^2 x \in [2ab, \infty)$
$\Rightarrow 9\tan^2 x + 16\cot^2 x \in [24, \infty)$
$\Rightarrow y \in [49, \infty)$
$\Rightarrow y_{min} = 49$

21. If $a \le 3\cos\left(\theta + \frac{\pi}{3}\right) + 5\cos\theta + 3 \le b$, find a and b.

यदि $a \le 3\cos\left(\theta + \frac{\pi}{3}\right) + 5\cos\theta + 3 \le b$ तो *a* व *b* का मान बताओं।

Ans. $a = -4$ and $b = 10$

Sol. $a \le 3\cos\left(\theta + \frac{\pi}{3}\right) + 5\cos\theta + 3 \le b$

$\Rightarrow a \le 3\left[\cos\theta\cos\frac{\pi}{3} - \sin\theta\sin\frac{\pi}{3}\right] + 5\cos\theta + 3 \le b$

$\Rightarrow a \le 3\left[\frac{\cos\theta}{2} - \frac{\sqrt{3}}{2}\sin\theta\right] + 5\cos\theta + 3 \le b$

$\Rightarrow a \le \frac{3}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta + 5\cos\theta + 3 \le b$

$\Rightarrow a \le \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta + 3 \le b$

$\because \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta \in \left[-\sqrt{\frac{169}{4} + \frac{27}{4}}, \sqrt{\frac{169}{4} + \frac{27}{4}}\right]$

$\Rightarrow 3 + \frac{13}{2}\cos\theta - \frac{3\sqrt{3}}{2}\sin\theta \in [3 - 7, 3 + 7]$

$a = -4,\ b = 10$

22. Show that $2^{\sin x} + 2^{\cos x} \ge 2^{1-\frac{1}{\sqrt{2}}}$.

सिद्ध कीजिए $2^{\sin x} + 2^{\cos x} \ge 2^{1-\frac{1}{\sqrt{2}}}$.

Ans.

Sol. Let $a = 2^{\sin x}, b = 2^{\cos x}$

$$\Rightarrow \frac{a+b}{2} \geq \sqrt{ab}$$

$$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2\sqrt{2^{\sin x + \cos x}}$$

$$\left.\begin{aligned} &-\sqrt{2} \leq \sin x + \cos x \leq \sqrt{2} \\ &\Rightarrow 2^{-\sqrt{2}} \leq 2^{\sin x + \cos x} \leq 2^{\sqrt{2}} \end{aligned}\right\}$$

$$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2^{1-\frac{1}{\sqrt{2}}}$$

23. Show that $\frac{3+\cos x}{\sin x}$ $\forall x \in R$ can not have any value between $-2\sqrt{2}$ and $2\sqrt{2}$. What inference can you draw about the values of $\frac{\sin x}{3+\cos x}$?

सिद्ध करो कि x के प्रत्येक मान के लिये $\frac{3+\cos x}{\sin x}$ का मान $-2\sqrt{2}$ तथा $2\sqrt{2}$ के मध्य कभी नही आयेगा। इस परिणाम की मदद से आप $\frac{\sin x}{3+\cos x}$ के मानों के बारे में क्या कह सकते है?

Ans. $\left[-\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

Sol. $\frac{3+\cos x}{\sin x} = y \Rightarrow y \sin x - 3 = \cos x$

Square both sides

$\Rightarrow y^2 \sin^2 x + 9 - 6y \sin x = 1 - \sin^2 x$

$\Rightarrow (y^2 + 1) \sin^2 x - 6y \sin x + 8 = 0$

For real values of x; $D \geq 0$

$\Rightarrow 36y^2 - 32 (y^2 + 1) \geq 0$

$\Rightarrow y^2 \geq 8$

$$\Rightarrow y \in \left(-\infty, -2\sqrt{2}\right] \cup \left[2\sqrt{2}, +\infty\right)$$

$$\Rightarrow \frac{1}{y} \in \left[-\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$$

24. The maximum value of the expression $\frac{1}{\sin^2\theta+3\sin\theta\cos\theta+5\cos^2\theta}$ is. **[JEE-2010]**

व्यंजक $\frac{1}{\sin^2\theta+3\sin\theta\cos\theta+5\cos^2\theta}$ का अधिकतम मान बताओं। **[JEE-2010]**

Ans. 2

Sol. $f(\theta)=\frac{1}{\sin^2\theta+3\sin\theta\cos\theta+5\cos^2\theta} = \frac{1}{\frac{1-\cos 2\theta}{2}+\frac{3}{2}\sin 2\theta+\frac{5(1+\cos 2\theta)}{2}} = \frac{2}{6+3\sin 2\theta+4\cos 2\theta}$

$\because \quad f(\theta)_{max}=\frac{2}{6-5}=2; \qquad \{\because 3\sin 2\theta+4\cos 2\theta\in[-5,5]\}$

SUMMATION OF SERIES

25. If $f(\theta)=\sum_{n=1}^{6}\text{cosec}\left(\theta+\frac{(n-1)\pi}{4}\right)\text{cosec}\left(\theta+\frac{n\pi}{4}\right)$, where $0<\theta<\frac{\pi}{2}$, then find the minimum value of $f(\theta)$.

यदि $f(\theta)=\sum_{n=1}^{6}\text{cosec}\left(\theta+\frac{(n-1)\pi}{4}\right)\text{cosec}\left(\theta+\frac{n\pi}{4}\right)$, जहाँ $0<\theta<\frac{\pi}{2}$, तो $f(\theta)$ का न्यूनतम मान क्या होगा।

Ans. $2\sqrt{2}$

Sol. $f(\theta)=\sum_{n=1}^{6}\text{cosec}\left(\theta+\frac{(n-1)\pi}{4}\right)\text{cosec}\left(\theta+\frac{n\pi}{4}\right)$

$T_n=\text{cosec}\left(\theta+\frac{(n-1)\pi}{4}\right)\text{cosec}\left(\theta+\frac{n\pi}{4}\right)$

$T_n=\sqrt{2}\frac{\sin\left[\left(\theta+\frac{n\pi}{4}\right)-\left(\theta+\frac{(n-1)\pi}{4}\right)\right]}{\left(\sin\left(\theta+\frac{(n-1)\pi}{4}\right)\sin\left(\theta+\frac{n\pi}{4}\right)\right)}$

$$\Rightarrow f(\theta)=\sqrt{2}\left[\frac{\sin\left(\theta+\frac{n\pi}{4}\right)\cos\left(\theta+\frac{(n-1)\pi}{4}\right)-\cos\left(\theta+\frac{n\pi}{4}\right)\sin\left(\theta+\frac{(n-1)\pi}{4}\right)}{\sin\left(\theta+\frac{(n-1)\pi}{4}\right)\sin\left(\theta+\frac{n\pi}{4}\right)}\right]$$

$$=\sqrt{2}\left[\cot\left(\theta+\frac{(n-1)\pi}{4}\right)-\cot\left(\theta+\frac{n\pi}{4}\right)\right]$$

$$f(\theta)=T_1+T_2+T_3+.....+T_6$$

$$=\sqrt{2}\left[\cot(\theta)-\cot\left(\theta+\frac{\pi}{4}\right)+\cot\left(\theta+\frac{\pi}{4}\right)-\cot\left(\theta+\frac{\pi}{2}\right)+.....+\cot\left(\theta+\frac{5\pi}{4}\right)-\cot\left(\theta+\frac{3\pi}{2}\right)\right]$$

$$=\sqrt{2}\left[\cot\theta+\tan\theta\right]$$

$$=\sqrt{2}\cdot 2\,\mathrm{cosec}2\theta$$

$\because 0<2\theta<\pi$

$\Rightarrow$ Minimum value of cosec2θ is 1

Minimum value of f(θ) is $2\sqrt{2}$

26. Let k = 1°, then prove that $\sum_{n=0}^{88}\frac{1}{\cos nk\cdot\cos(n+1)k}=\frac{\cos k}{\sin^2 k}$.

यदि k = 1° तो सिद्ध करो कि $\sum_{n=0}^{88}\frac{1}{\cos nk\cdot\cos(n+1)k}=\frac{\cos k}{\sin^2 k}$

Ans.

Sol. $\sum_{n=0}^{88}\frac{1}{\cos nk\cdot\cos(n+1)k}$

Let $T_n=\frac{1}{\cos nk\cdot\cos(n+1)k}=\frac{\sin\left[(n+1)k-nk\right]}{\sin k\cos nk\cdot\cos(n+1)k}$

$$\Rightarrow T_n=\frac{1}{\sin k}\left[\frac{\sin(n+1)k\cos(nk)-\cos(n+1)k\sin nk}{\cos(nk)\cdot\cos(n+1)k}\right]$$

$$\Rightarrow T_n=\frac{1}{\sin k}\left[\tan(n+1)k-\tan nk\right]$$

$$\Rightarrow \sum_{n=0}^{88} \frac{1}{\cos(nk)\cdot\cos(n+1)k} = \frac{1}{\sin k}\begin{bmatrix} \tan k - \tan 0 \\ +\tan 2k - \tan k \\ +\tan 3k - \tan 2k \\ + \\ \vdots \\ +\tan 88k - \tan 87k \\ +\tan 89k - \tan 88k \end{bmatrix}$$

$$= \frac{1}{\sin k}[\tan 89k] = \frac{\sin(89k)}{\cos(89k)\sin k}$$

$$= \frac{\sin(89^\circ)}{\cos(89^\circ)\sin 1^\circ} = \frac{\cos(1^\circ)}{\sin(1^\circ)^2} = \frac{\cos k}{\sin^2 k} \qquad \{\because k = 1^\circ\}$$

27. Prove that : $\cos ec\theta + \cos ec2\theta + \cos ec2^2\theta + \ldots\ldots\ldots\ldots + \cos ec2^{n-1}\theta = \cot(\theta/2) - \cot 2^{n-1}\theta$.

सिद्ध कीजिए : $\cos ec\theta + \cos ec2\theta + \cos ec2^2\theta + \ldots\ldots\ldots\ldots + \cos ec2^{n-1}\theta = \cot(\theta/2) - \cot 2^{n-1}\theta$.

Ans.

Sol. $\cos ec\theta + \cos ec2\theta + \cos ec2^2\theta + \ldots\ldots + \cos ec\left(2^{n-1}\theta\right) = \cot\left(\frac{\theta}{2}\right) - \cot\left(2^{n-1}\theta\right)$

Let

$$T_n = \cos ec\left(2^{n-1}\theta\right) = \frac{1}{\sin\left(2^{n-1}\theta\right)}$$

$$= \frac{\sin\left(2^{n-2}\theta\right)}{\sin\left(2^{n-1}\theta\right)} = \frac{\sin\left(2^{n-1}\theta - 2^{n-2}\theta\right)}{\sin\left(2^{n-1}\theta\right)\sin\left(2^{n-2}\theta\right)}$$

$$= \frac{\sin\left(2^{n-1}\theta\right)\cos\left(2^{n-2}\theta\right) - \cos\left(2^{n-1}\theta\right)\sin\left(2^{n-2}\theta\right)}{\sin\left(2^{n-1}\theta\right)\sin\left(2^{n-2}\theta\right)}$$

$$= \cot\left(2^{n-2}\theta\right)\cot\left(2^{n-1}\theta\right)$$

$$\Rightarrow \sum_{r=1}^{n} \operatorname{cosec}\left(2^{r-1}\theta\right) = \begin{bmatrix} \cot\left(\frac{\theta}{2}\right) - \cot\theta \\ +\cot(\theta) - \cot(2\theta) \\ + \\ \vdots \\ + \\ \cot\left(2^{n-3}\theta\right) - \cot\left(2^{n-2}\theta\right) \\ +\cot\left(2^{n-2}\theta\right) - \cot\left(2^{n-1}\theta\right) \end{bmatrix}$$

$$= \cot\left(\frac{\theta}{2}\right) - \cot\left(2^{n-1}\theta\right)$$

28. Let $f_n(\theta) = \sum_{r=1}^{n} \tan\left(\frac{\theta}{2^r}\right)\sec\left(\frac{\theta}{2^{r-1}}\right)$, then compute the value of:

(a) $f_3(2\pi)$, (b) $f_4\left(\frac{4\pi}{3}\right)$, (c) $f_5(4\pi)$ (d) $f_6(48\pi)$.

यदि $f_n(\theta) = \sum_{r=1}^{n} \tan\left(\frac{\theta}{2^r}\right)\sec\left(\frac{\theta}{2^{r-1}}\right)$ तो निम्नलिखित के मान ज्ञात करिये।

(a) $f_3(2\pi)$, (b) $f_4\left(\frac{4\pi}{3}\right)$, (c) $f_5(4\pi)$ (d) $f_6(48\pi)$.

Ans. (a) -1, (b) $2(\sqrt{3}-1)$, (c) $1-\sqrt{2}$, (d) 1

Sol. $f_n(\theta) = \sum_{r=1}^{n} \tan\left(\frac{\theta}{2^r}\right)\sec\left(\frac{\theta}{2^{r-1}}\right)$

$$t_r = \tan\left(\frac{\theta}{2^r}\right)\sec\left(\frac{\theta}{2^{r-1}}\right) = \frac{\sin\left(\frac{\theta}{2^r}\right)}{\cos\left(\frac{\theta}{2^r}\right)\cos\left(\frac{\theta}{2^{r-1}}\right)}$$

$$= \frac{\sin\left[\frac{\theta}{2^{r-1}} - \frac{\theta}{2^{r-2}}\right]}{\cos\left(\frac{\theta}{2^r}\right)\cos\left(\frac{\theta}{2^{r-1}}\right)} = \tan\left(\frac{\theta}{2^{r-1}}\right) - \tan\left(\frac{\theta}{2^r}\right)$$

$$f_n(\theta) = \tan\theta - \tan\frac{\theta}{2} + \tan\frac{\theta}{2} - \tan\frac{\theta}{4} + \ldots\ldots\ldots + \tan\frac{\theta}{2^{n-2}} - \tan\frac{\theta}{2^{n-1}} + \tan\frac{\theta}{2^{n-1}}$$

$$= \tan\theta - \tan\left(\frac{\theta}{2^n}\right)$$

$$(a)\ f_3(2\pi) = \tan 2\pi - \tan\left(\frac{2\pi}{3}\right) = 0 - 1 = -1$$

$$(b)\ f_4\left(\frac{4\pi}{3}\right) = \tan\left(\frac{4\pi}{3}\right) - \tan\left(\frac{4\pi}{3.16}\right) = \tan\frac{\pi}{3} - \tan\left(\frac{\pi}{12}\right)$$

$$= \sqrt{3} - \left(2 - \sqrt{3}\right) = +2\sqrt{3} - 2$$

$$(c)\ f_5(4\pi) = \tan 4\pi - \tan\left(\frac{4\pi}{32}\right) = -\tan\left(\frac{\pi}{8}\right) = -\left(\sqrt{2} - 1\right) = 1 - \sqrt{2}$$

$$(d)\ f_6(48\pi) = \tan(48\pi) - \tan\left(\frac{48\pi}{64}\right) = -\tan\left(\frac{3\pi}{4}\right) = 1$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 29 to 31

29. The minimum value of $16\cos^4\theta - 20\cos^2\theta + 5$ is :

(A) 1 (B) $-\frac{5}{4}$ (C) 0 (D) None of these

$16\cos^4\theta - 20\cos^2\theta + 5$ का न्यूनतम मान क्या हैं :

(A) 1 (B) $-\frac{5}{4}$ (C) 0 (D) इनमें से कोई नहीं

Ans. B

Sol. $y = 16\cos^4\theta - 20\cos^2\theta + 5$

$$\Rightarrow y = 4\left[4\cos^4\theta - 5\cos^2\theta\right] + 5$$

$$\Rightarrow y = 4\left[\left(2\cos^2\theta - \frac{5}{4}\right)^2\right] + 5 - \frac{25}{4}$$

$\because \cos^2\theta \in [0,1]$

$\Rightarrow 0 \le 2\cos^2\theta \le 2$

$\Rightarrow \frac{-5}{4} \le \left(2\cos^2\theta - \frac{5}{4}\right) \le \frac{3}{4}$

$\Rightarrow 0 \le \left(2\cos^2\theta - \frac{5}{4}\right) \le \frac{25}{16}$

$\Rightarrow 0 \le 4\left(2\cos^2\theta - \frac{5}{4}\right)^2 \le \frac{25}{4}$

$\Rightarrow -\frac{5}{4} \le y \le 5$

$\Rightarrow$ Minimum value of y is $-\frac{5}{4}$

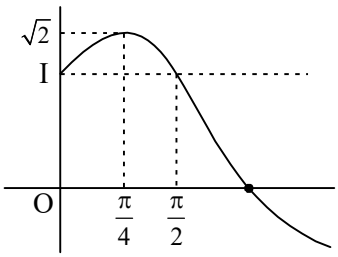
30. The maximum and minimum values of $|\sin x| + \cos x$ when $x \in R$ must be :

|sin x| + cos x जहाँ $x \in R$ का महत्तम व न्यूनतम माान क्या होगा?

(A) [0 , 1] (B) $\left[-1, \sqrt{2}\right]$ (C) $\left[-\sqrt{2}, 1\right]$ (D) $\left[0, \sqrt{2}\right]$

Ans. B

Sol. $y = |\sin x| + \cos x$



$$y = \begin{cases} \sin x + \cos x & ; \quad x \in [0, \pi] \\ \cos x - \sin x & ; \quad x \in [\pi, 2\pi] \end{cases}$$

$\because \sin x + \cos x \in \left[-\sqrt{2}, \sqrt{2}\right] \ \forall x \in R$

In $0 \le x \le \pi; y = \sin x + \cos x$

$\Rightarrow y \in \left[-1, \sqrt{2}\right]$

In $\pi \le x \le 2\pi$

$y = \cos x - \sin x$

$\Rightarrow y \in \left[-1, \sqrt{2}\right]$

$y \in \left[-1, \sqrt{2}\right]$

31. The range of the function |sin x| + |cos x| must be :

(A) $\left[-\sqrt{2}, \sqrt{2}\right]$ (B) $\left[1, \sqrt{2}\right]$ (C) $\left[0, \sqrt{2}\right]$ (D) None of these

$|\sin x| + |\cos x|$ फलन का परिसर होगा :

(A) $\left[-\sqrt{2}, \sqrt{2}\right]$ (B) $\left[1, \sqrt{2}\right]$ (C) $\left[0, \sqrt{2}\right]$ (D) इनमें से कोई नहीं

Ans. B

Sol. $y = |\sin x| + |\cos x|$

$$= \begin{cases} \sin x + \cos x & ; \quad 0 \le x \le \dfrac{\pi}{2} \\ \sin x - \cos x & ; \quad \dfrac{\pi}{2} \le x \le \pi \\ -\sin x - \cos x & ; \quad \pi \le x \le \dfrac{3\pi}{2} \\ -\sin x + \cos x & ; \quad \dfrac{3\pi}{2} \le x \le 2\pi \end{cases}$$

$\because \sin x + \cos x \in \left[-\sqrt{2}, \sqrt{2}\right]$

In $0 \le x \le \dfrac{\pi}{2}; \sin x + \cos x \in \left[1, \sqrt{2}\right]$

In $\dfrac{\pi}{2} \le x \le \pi; \sin x - \cos x \in \left[1, \sqrt{2}\right]$

$\pi \le x \le \dfrac{3\pi}{2}; \sin x - \cos x \in \left[1, \sqrt{2}\right]$

$\dfrac{3\pi}{2} \le x \le 2\pi; \sin x - \cos x \in \left[1, \sqrt{2}\right]$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

32. If $\sec\theta + \tan\theta = 1$, then one root of equation, $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ is :

यदि $\sec\theta + \tan\theta = 1$, तो समीकरण $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ का एक मूल होगा :

(A) $\tan\theta$ (B) $\sec\theta$ (C) $\cos\theta$ (D) $\sin\theta$

Ans. BC

Sol. $(ab - ac)x^2 + (bc - ab)x + (ac - ab) = 0$

$\because$ sum of coefficient is zero $\Rightarrow$ one root will be 1

$\therefore \sec\theta + \tan\theta = 1$ & $\sec\theta - \tan\theta = 1$

$\Rightarrow \sec\theta = 1$ & $\tan\theta = 0$

$\Rightarrow \theta = 0^\circ$

$\Rightarrow \cos\theta = 1, \sec\theta = 1, \sin\theta = 0, \tan\theta = 0$

33. If $\sin\alpha = \frac{3}{5}$, $\frac{\pi}{2} < \alpha < \pi$ and $\cos\beta = \frac{-5}{13}$, $\pi < \beta < \frac{3\pi}{2}$, then the correct statement(s) is(are) :

यदि $\sin\alpha = \frac{3}{5}$, $\frac{\pi}{2} < \alpha < \pi$ तथा $\cos\beta = \frac{-5}{13}$, $\pi < \beta < \frac{3\pi}{2}$ है, तो सत्य कथन होगें :

(A) $\tan(\alpha - \beta) = \frac{63}{16}$ (B) $\tan(\alpha + \beta) = \frac{33}{56}$

(C) $\sin 2\alpha = \frac{-24}{25}$ (D) $\cos 2\beta = \frac{-119}{169}$

Ans. ABCD

Sol. $\sin\alpha = \frac{3}{5}, \frac{\pi}{2} < \alpha < \pi$

$\Rightarrow \cos\alpha = -\sqrt{1-\left(\frac{3}{5}\right)^2} = \sqrt{1-\frac{9}{25}} = \frac{-4}{5}$

Also, $\cos\beta = \frac{-5}{13}$, $\pi < \beta < \frac{3\pi}{2}$

$\Rightarrow \sin\beta = -\sqrt{1-\cos^2\beta} = -\sqrt{1-\frac{25}{169}} = \frac{-12}{13}$

Now,

(A) $\tan(\alpha - \beta) = \frac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta} = \frac{\left(\frac{-3}{4}\right)-\left(\frac{12}{5}\right)}{1+\left(\frac{-3}{4}\right)\left(\frac{12}{5}\right)} = \frac{\frac{-15-48}{20}}{\frac{20-36}{20}} = \frac{-63}{-16} = \frac{63}{16}$

(B) $\tan(\alpha + \beta) = \frac{\tan\alpha + \tan\beta}{1-\tan\alpha\tan\beta} = = \frac{\frac{-15+48}{20}}{\frac{20+36}{20}} = \frac{33}{56}$

(C) $\sin 2\alpha = 2\sin\alpha\cos\alpha = 2\left(\frac{3}{5}\right)\left(\frac{-4}{5}\right) = \frac{-24}{25}$

(D) $\cos 2\beta = 2\cos^2\beta - 1 = 2 \times \frac{25}{169} - 1 = \frac{50-169}{169} = \frac{-119}{169}$

34. If $\sin x = \frac{1}{4}\left(\sqrt{5}-1\right)$ then which of the following is/are correct?

यदि $\sin x = \frac{1}{4}\left(\sqrt{5}-1\right)$ है, तो निम्न में से कौनसा/कौनसे सत्य है?

(A) $\cos 4x = \sin x$ (B) $\sin 3x = \cos 2x$

(C) $\sin 2x + \cos 3x = \frac{1}{2}$ (D) $\sin 5x + \cos 5x = 1$

Ans. ABD

Sol. $\sin x = \frac{\sqrt{5}-1}{4}$

$\Rightarrow x = 18^\circ$

(A) $\cos 4x = \cos 72^\circ = \sin 18^\circ$

(B) $\cos 2x = \cos 36^\circ = \sin (54^\circ) = \sin 3x$

(C) $\sin 36^\circ + \cos 54^\circ = \sin 36^\circ + \sin 36^\circ = 2\sin 36^\circ = 4 \times \sin 18^\circ \cos 18^\circ$

$= 4 \times \frac{\sqrt{5}-1}{4} \times \frac{\sqrt{5}+1}{4} = 1$

(D) $\sin 90^\circ + \cos 90^\circ = 1$

35. If the equation $2\cos^2 x + \cos x - a = 0$ has solutions then a can be :

यदि समीकरण $2\cos^2 x + \cos x - a = 0$ के हल है, तो a हो सकता है :

(A) $\frac{-1}{4}$ (B) $\frac{-1}{8}$ (C) 2 (D) 5

Ans. BC

Sol. $2\cos^2 x + \cos x - a = 0$

$\Rightarrow \cos x = \frac{-1 \pm \sqrt{1+8a}}{4}$

$\Rightarrow$ For real root $1 + 8a \geq 0$

$\Rightarrow a \geq \frac{-1}{8}$

(C) If $a = 2$

then $\cos x = \frac{-1 \pm \sqrt{17}}{4}$

(D) If $a = 5$

then $\cos x = \frac{-1 \pm \sqrt{41}}{4} \in (-\infty, -1) \cup (1, +\infty)$

36. $\frac{\sin^2 50° - \sin^2 40°}{\cos^2 50° \cdot \cos^2 40°}$ is equal to :

$\frac{\sin^2 50° - \sin^2 40°}{\cos^2 50° \cdot \cos^2 40°}$ बराबर होगा :

(A) 4 tan 10° sec 10°
(B) 2 tan 10° sec 10°
(C) 4 tan 10°(tan 10° + cot 40°)
(D) 4 tan 10°(tan 10° + tan 40°)

Ans. AD

Sol. $\frac{\sin^2 50° - \sin^2 40°}{\cos^2 50° \cdot \cos^2 40°} = \frac{4(\sin^2 50° - \sin^2 40°)}{(2\cos 50° \cdot \cos 40°)^2} = \frac{4\sin 90° \cdot \sin 10°}{(\cos 90° + \cos 10°)^2} = \frac{4\sin 10°}{\cos 10° \cdot \cos 10°}$

$= 4 \tan 10° \sec 10°$

now (D) $\rightarrow 4 \tan 10° \left[\frac{\sin 10° \cos 40° + \sin 40° \cos 10°}{\cos 10° \cdot \cos 40°}\right] = 4 \tan 10° \left[\frac{\sin 50°}{\cos 10° \cdot \cos 40°}\right]$

$= 4 \tan 10° \frac{1}{\cos 10°} = 4\tan 10° \sec 10°$

37. Which of following functions have the maximum value unity?

निम्न में से कौनसे फलन का अधिकतम मान इकाई है?

(A) $\sin^2 x - \cos^2 x$
(B) $\sqrt{\frac{6}{5}}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{3}}\cos x\right)$
(C) $\cos^6 + \sin^6 x$
(D) $\cos^2 x + \sin^4 x$

Ans. ABCD

Sol. (A) $\sin^2 x - \cos^2 x = 1 - 2\cos^2 x = -\cos 2x$
$\Rightarrow \sin^2 x - \cos^2 x \in [-1, 1]$

(B) Let $y = \sqrt{\frac{6}{5}}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{3}}\cos x\right)$

$\because \frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{3}}\cos x \in \left[-\sqrt{\frac{1}{2}+\frac{1}{3}}, +\sqrt{\frac{1}{2}+\frac{1}{3}}\right] \quad \Rightarrow y \in [-1, 1]$

(C) $\cos^6 x + \sin^6 x = (\cos^2 x + \sin^2 x)^3 - 3\cos^2 x \sin^2 x (\cos^2 x + \sin^2 x)$
$= 1 - 3\sin^2 x \cos^2 x$

$= 1 - \frac{3}{4}\sin^2 2x$

$\Rightarrow \frac{1}{4} \le \cos^6 x + \sin^6 x \le 1 \qquad \left\{\because 0 \le \sin^2 2x \le 1\right\}$

(D) $\cos^2x + \sin^4x$

$= \cos^2x + (\sin^2x)^2$

$= \cos^2x + (1 - \cos^2x)^2$

$= \cos^4x + \cos^2x + 1 - 2\cos^2x$

$= \cos^4x - \cos^2x + 1$

$= \cos^2x (\cos^2x - 1) + 1$

$= -\sin^2x \cos^2x + 1$

$= 1 - \frac{1}{4}\sin^2 2x$

$\because 0 \le \sin^2 2x \le 1$

$\Rightarrow -\frac{1}{4} \le -\frac{1}{4}\sin^2 2x \le 0$

$\Rightarrow \frac{3}{4} \le \cos^2 x + \sin^4 x \le 1$

38. If $\cos\beta$ is geometric mean between $\sin\alpha$ and $\cos\alpha$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\cos 2\beta =$

यदि $\sin\alpha$ तथा $\cos\alpha$ का गुणोतर माध्य $\cos\beta$ है, जहाँ $0 < \alpha, \beta < \frac{\pi}{2}$, तो $\cos 2\beta =$

(A) $-2\sin^2\left(\frac{\pi}{4} - \alpha\right)$ (B) $-2\cos^2\left(\frac{\pi}{4} + \alpha\right)$ (C) $2\sin^2\left(\frac{\pi}{4} + \alpha\right)$ (D) $2\cos^2\left(\frac{\pi}{4} - \alpha\right)$

Ans. AB

Sol. $\cos\beta$ is geometric mean between $\sin\alpha$ and $\cos\alpha$, where $0 < \alpha, \beta < \frac{\pi}{2}$

$\Rightarrow \cos\beta = \sqrt{\sin\alpha\cos\alpha} \quad \Rightarrow \cos^2\beta = \sin\alpha\cos\alpha$

$\Rightarrow \cos 2\beta = 2\cos^2\beta - 1$

$= 2\sin\alpha\cos\alpha - 1$

$= \sin 2\alpha - 1 = -\left[\sin\alpha - \cos\alpha\right]^2$

Now

(A) $-2\sin^2\left(\frac{\pi}{4} - \alpha\right) = -2\left[\sin\left(\frac{\pi}{4} - \alpha\right)\right]^2 = -\left(\cos\alpha - \sin\alpha\right)^2$

(B) $-2\cos^2\left(\frac{\pi}{4} + \alpha\right) = -2\left[\cos\left(\frac{\pi}{4} + \alpha\right)\right]^2$

$$= -2\left[\frac{1}{\sqrt{2}}\cos\alpha - \frac{\sin\alpha}{\sqrt{2}}\right]^2 = -\left[\cos^2\alpha + \sin^2\alpha - 2\sin\alpha\cos\alpha\right]$$

$$= 2\sin\alpha\cos\alpha - 1$$

$$= \sin 2\alpha - 1 = \cos 2\beta$$

(C) $2\sin^2\left(\frac{\pi}{4}+\alpha\right) = 2\left[\frac{\cos\alpha}{\sqrt{2}} + \frac{\sin\alpha}{\sqrt{2}}\right]^2 = 1 + \sin 2\alpha$

(D) $2\cos^2\left(\frac{\pi}{4}-\alpha\right) = 2\left[\frac{\cos\alpha}{\sqrt{2}} + \frac{\sin\alpha}{\sqrt{2}}\right]^2 = 1 + \sin 2\alpha$

39. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then : **[JEE-2009]**

यदि $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$ हो, तो : **[JEE-2009]**

(A) $\tan^2 x = \frac{2}{3}$

(B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$

(D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

Ans. AB

Sol. $\frac{\tan^4 x}{2} + \frac{1}{3} = \frac{\sec^4 x}{5}$

Let $\tan^2 x = t$

$$\Rightarrow \frac{t^2}{2} + \frac{1}{3} = \frac{(1+t)^2}{5}$$

$$\Rightarrow \frac{3t^2+2}{6} = \frac{t^2+2t+1}{5}$$

$$\Rightarrow t = \frac{2}{3}$$

$$\Rightarrow \sin^2 x = \frac{2}{5} \ \& \ \cos^2 x = \frac{3}{5}$$

$$\Rightarrow \frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$$

40. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right)\operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is (are) : **[JEE-2009]**

यदि $0 < \theta < \frac{\pi}{2}$ तो समीकरण $\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right)\operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ के हल होगें : **[JEE-2009]**

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{12}$ (D) $\frac{5\pi}{12}$

Ans. CD

Sol. $$\sqrt{2}\sum_{m=1}^{6} \frac{\sin\left[\left(\theta + \frac{m\pi}{4}\right) - \left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right)\right]}{\sin\left(\theta + \frac{m\pi}{4} - \frac{\pi}{4}\right)\sin\left(\theta + \frac{m\pi}{4}\right)} = 4\sqrt{2}$$

$$\Rightarrow \sqrt{2}\sum_{m=1}^{6}\left[\cot\left(\theta + \frac{(m-1)\pi}{4}\right) - \cot\left(\theta + \frac{m\pi}{4}\right)\right] = 4\sqrt{2}$$

$\Rightarrow \cot\theta + \tan\theta = 4.$

$\Rightarrow \tan\theta = 2 \pm \sqrt{3}$

$\Rightarrow \theta = \frac{\pi}{12}$ or $\frac{5\pi}{12}$

MATCH THE COLUMN TYPE QUESTION(S)

41. If α and β satisfy the equation, $a\cos\theta + b\sin\theta = c$, then :

Column-I	**Column-II**
(A) the value of $\tan\frac{\alpha}{2} \cdot \tan\frac{\beta}{2}$ is	(P) $\frac{2b}{c+a}$
(B) the value of $\tan\frac{\alpha}{2} + \tan\frac{\beta}{2}$ is	(Q) $\frac{c-a}{c+a}$
(C) the value of $\tan^2\frac{\alpha}{2} + \tan^2\frac{\beta}{2}$ is	(R) $\frac{4\left(b^2+a^2-c^2\right)}{(c+a)^2}$
(D) the value of $\left(\tan\frac{\alpha}{2} - \tan\frac{\beta}{2}\right)^2$ is	(S) $\frac{2\left(2b^2+a^2-c^2\right)}{(c+a)^2}$

यदि α एवं β समीकरण $a\cos\theta + b\sin\theta = c$ को संतुष्ट करते हो तब :

Column-I	**Column-II**
(A) $\tan\frac{\alpha}{2}\,.\,\tan\frac{\beta}{2}$ का मान होगा	(P) $\frac{2b}{c+a}$
(B) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}$ का मान होगा	(Q) $\frac{c-a}{c+a}$
(C) $\tan^2\frac{\alpha}{2}+\tan^2\frac{\beta}{2}$ का मान होगा	(R) $\frac{4\left(b^2+a^2-c^2\right)}{(c+a)^2}$
(D) $\left(\tan\frac{\alpha}{2}-\tan\frac{\beta}{2}\right)^2$ का मान होगा	(S) $\frac{2\left(2b^2+a^2-c^2\right)}{(c+a)^2}$

Ans. (A) Q (B) P (C) S (D) R

Sol. $a\cos\theta+b\sin\theta=c \left\langle \begin{matrix}\alpha\\ \beta\end{matrix}\right.$

$$\Rightarrow a\left(\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right)+\frac{2b\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}=c$$

let $\tan\frac{\theta}{2}=t$

$a(1-t^2)+2bt=c(1+t^2)$

$\Rightarrow c+ct^2-a+at^2-2bt=0$

$$\Rightarrow (a+c)t^2-2bt+(c-a)=0 \left\langle \begin{matrix}\tan\frac{\alpha}{2}\\ \tan\frac{\beta}{2}\end{matrix}\right.$$

(A) $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}=\frac{c-a}{a+c}$ (Q)

(B) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}=\frac{2b}{a+c}$ (P)

(C) $\tan^2\frac{\alpha}{2}+\tan^2\frac{\beta}{2}=\left(\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}\right)^2-2\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$

$=\frac{4b^2}{(a+c)^2}-\frac{2(c-a)}{a+c}$

$$= \frac{4b^2 - 2\left(c^2 - a^2\right)}{(a+c)^2} = \frac{2\left(2b^2 + a^2 - c^2\right)}{(c+a)^2} \qquad \text{(S)}$$

(D) $\left(\tan\frac{\alpha}{2} - \tan\frac{\beta}{2}\right)^2 = \tan^2\frac{\alpha}{2} + \tan^2\frac{\beta}{2} - 2\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$

$$= \frac{4b^2 + 2a^2 - 2c^2}{(a+c)^2} - \frac{2(c-a)}{a+c}$$

$$= \frac{4b^2 + 2a^2 - 2c^2 - 2c^2 + 2a^2}{(a+c)^2}$$

$$= \frac{4\left(b^2 + a^2 - c^2\right)}{(a+c)^2} \qquad \text{(R)}$$

42. Match the following column :

Column-I	**Column-II**
(A) The value of $\operatorname{cosec}\frac{\pi}{18} - \sqrt{3}\sec\frac{\pi}{18}$ is	(P) 1
(B) $\frac{\cos 20° + 8\sin 70°\sin 50°\sin 10°}{\sin^2 80°}$ is equal to	(Q) 2
(C) Exact value of $\cos 20^\circ + 2\sin^2 55^\circ - \sqrt{2}\sin 65^\circ$ is	(R) 3
(D) Let $f(\theta) = \frac{1}{2} + \frac{2}{3}\operatorname{cosec}^2\theta + \frac{3}{8}\sec^2\theta$. The least value of f(θ) for all permissible values of θ, is greater than	(S) 4

निम्न कॉलम को सुमेलित कीजिए :

Column-I	**Column-II**
(A) $\operatorname{cosec}\frac{\pi}{18} - \sqrt{3}\sec\frac{\pi}{18}$ का मान होगा	(P) 1
(B) $\frac{\cos 20° + 8\sin 70°\sin 50°\sin 10°}{\sin^2 80°}$ का मान होगा	(Q) 2
(C) $\cos 20^\circ + 2\sin^2 55^\circ - \sqrt{2}\sin 65^\circ$ का मान होगा	(R) 3
(D) यदि $f(\theta) = \frac{1}{2} + \frac{2}{3}\operatorname{cosec}^2\theta + \frac{3}{8}\sec^2\theta$ तो θ के सभी संभावित मान f(θ) हमेशा किस–किस से बड़ा होगा।	(S) 4

Ans. (A) S (B) Q (C) P (D) PQ

Sol. (A) $\dfrac{1}{\sin\pi/18}-\dfrac{\sqrt{3}}{\cos\pi/18}=\dfrac{2\left[\dfrac{1}{2}\cos\dfrac{\pi}{18}-\dfrac{\sqrt{3}}{2}\sin\dfrac{\pi}{18}\right]}{\dfrac{\sin(\pi/9)}{2}}$

$$=\frac{4\left[\sin\frac{\pi}{6}\cos\frac{\pi}{18}-\cos\frac{\pi}{6}\sin\frac{\pi}{18}\right]}{\sin\frac{\pi}{9}}=4$$

(B) Nr. $=\cos20^0+4(\cos20^0-\cos120^0)\sin10^0$

$=\cos20^0+2(\sin30^0-\sin10^0)-4(-1/2)\sin10^0$

$=1+\cos20^0=2\cos^210^0\Rightarrow 2\cos^210^0/\sin^280^0=2$

(C) $\cos20^\circ+1-\cos110^\circ-\sqrt{2}\sin65^\circ$

$2\sin65^0\sin45^0-\sqrt{2}\sin65^\circ+1=1$ Ans]

(D) $\dfrac{1}{2}+\dfrac{2}{3}+\dfrac{3}{8}+\dfrac{2}{3}\cot^2\theta+\dfrac{3}{8}\tan^2\theta\geq\dfrac{37}{24}+1=\dfrac{61}{24}$ by AM-GM in equality

$$\left\{\because\frac{2}{3}\cot^2\theta+\frac{3}{8}\tan^2\theta\geq2\sqrt{\frac{2}{3}\cdot\cot^2\theta\cdot\frac{3}{8}\tan^2\theta}\geq1\right\}$$

Exercise J-M

1. If $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$, then the value of $\left(a + \frac{1}{a}\right)$ is : **[JEE(Main)jan.2023]**

यदि $\tan 15^\circ + \frac{1}{\tan 75^\circ} + \frac{1}{\tan 105^\circ} + \tan 195^\circ = 2a$ है, तो $\left(a + \frac{1}{a}\right)$ का मान है :

(A) $4 - 2\sqrt{3}$ (B) 4 (C) 2 (D) $5 - \frac{3}{2}\sqrt{3}$

Ans. B

Sol. $\Rightarrow \tan 15^\circ + \tan 15^\circ - \tan 15 + \tan 15 = 2a$

$\Rightarrow a = \tan 15^\circ$

Now $\left(a + \frac{1}{a}\right) = \tan 15^\circ + \frac{1}{\tan 15^\circ}$

$= \frac{1 + \tan^2 15^\circ}{\tan 15^\circ} = \frac{2}{\sin 30^\circ} = 4$

2. The value of $36\left(4\cos^2 9^\circ - 1\right)\left(4\cos^2 27^\circ - 1\right)\left(4\cos^2 81^\circ - 1\right)\left(4\cos^2 243^\circ - 1\right)$ is :**[JEE(Main)April,2023]**

$36\left(4\cos^2 9^\circ - 1\right)\left(4\cos^2 27^\circ - 1\right)\left(4\cos^2 81^\circ - 1\right)\left(4\cos^2 243^\circ - 1\right)$ का मान है :**[JEE(Main)April,2023]**

(A) 54 (B) 27 (C) 18 (D) 36

Ans. D

Sol. $4\cos^2\theta - 1 = 4\left(1 - \sin^2\theta\right) - 1 = 3 - 4\sin^2\theta = \frac{\sin 3\theta}{\sin\theta}$

so given expression can be written as

$36 \times \frac{\sin 27^\circ}{\sin 9^\circ} \times \frac{\sin 81^\circ}{\sin 27^\circ} \times \frac{\sin 243^\circ}{\sin 81^\circ} \times \frac{\sin 729^\circ}{\sin 243^\circ}$

$36 \times \frac{\sin 729^\circ}{\sin 9^\circ} = 36$

3. $96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ is equal to : **[JEE(Main)April,2023]**

$96\cos\frac{\pi}{33}\cos\frac{2\pi}{33}\cos\frac{4\pi}{33}\cos\frac{8\pi}{33}\cos\frac{16\pi}{33}$ बराबर है : **[JEE(Main)April,2023]**

(A) 4 (B) 1 (C) 2 (D) 3

Ans. D

Sol. Let $\frac{\pi}{33} = A$

$96\cos A\cos 2A\cos 2^2A\cos 2^3A\cos 2^4A$

$$= 96\frac{\sin 2^5 A}{2^5 \sin A} = \frac{96.\sin\frac{32\pi}{33}}{32\sin\frac{\pi}{33}} = \frac{96.\sin\left(\pi-\frac{\pi}{33}\right)}{32\sin\frac{\pi}{33}} = \frac{96.\sin\frac{\pi}{33}}{32\sin\frac{\pi}{33}} = 3 \text{ Ans.}$$

4. The value of $\tan9° - \tan27° - \tan63° + \tan81°$ is _____. **[JEE(Main)April,2023]**

$\tan9° - \tan27° - \tan63° + \tan81°$ का मान है _____ **[JEE(Main)April,2023]**

Ans. 4

Sol. $\left(\tan 9^\circ + \cot 9^\circ\right) - \left(\tan 27^\circ + \cot 27^\circ\right)$

$$\frac{1}{\sin 9^\circ \cos 9^\circ} - \frac{1}{\sin 27^\circ \cos 27^\circ}$$

$$\frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ}$$

$$\frac{2(4)}{\sqrt{5}-1} - \frac{2(4)}{(\sqrt{5}+1)}$$

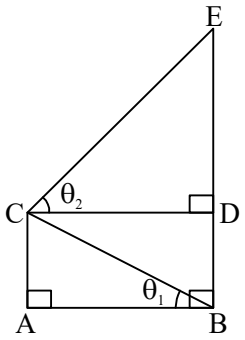
$$\frac{8(\sqrt{5}+1)}{4} - \frac{8(\sqrt{5}-1)}{4}$$

$$2[(\sqrt{5}+1)-(\sqrt{5}-1)]$$

$$= 4$$

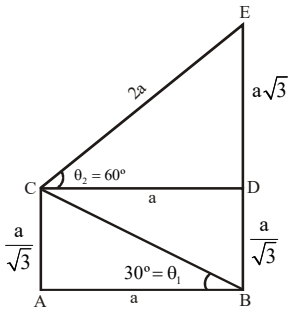
5. In the figure, $\theta_1 + \theta_2 = \frac{\pi}{2}$ and $\sqrt{3}(BE) = 4(AB)$. If the area of ΔCAB is $2\sqrt{3}-3$ unit^2, when $\frac{\theta_2}{\theta_1}$ is the largest, then the perimeter (in unit) of ΔCED is equal to _____. **[JEE(Main)April,2023]**

दी गई आकृति में $\theta_1 + \theta_2 = \frac{\pi}{2}$ तथा $\sqrt{3}(BE) = 4(AB)$ है। यदि ΔCAB का क्षेत्रफल $2\sqrt{3}-3$ वर्ग इकाई है, जब $\frac{\theta_2}{\theta_1}$ अधिकतम है, तो ΔCED का परिमाप (इकाई में) बराबर है : **[JEE(Main)April,2023]**



Ans. 6

Sol.



$$\tan\theta_2 = \frac{DE}{CD} = \frac{BE - AC}{AB}$$

$$\tan\theta_2 = \frac{4}{\sqrt{3}} - \tan\theta_1 \Rightarrow \tan\theta_1 + \tan\theta_2 = \frac{4}{\sqrt{3}} \ldots\ldots$$

$$\text{Now } \theta_1 + \theta_2 = \frac{\pi}{2} \Rightarrow \tan\theta_1 \tan\theta_2 = 1 \ldots (2)$$

Let $\tan\theta_1 = x$ and $\tan\theta_2 = y$ and $\frac{\theta_2}{\theta_1}$ is the largest

$$\Rightarrow x + y = \frac{4}{\sqrt{3}} \qquad x + \frac{1}{x} = \frac{4}{\sqrt{3}} \qquad \sqrt{3}x^2 - 4x + \sqrt{3} = 0$$

$$xy = 1 \qquad \sqrt{3}x\left(x - \sqrt{3}\right) - 1\left(x - \sqrt{3}\right) = 0$$

$$x = \sqrt{3}, x = \frac{1}{\sqrt{3}}$$

$$y = \frac{1}{\sqrt{3}}, y = \sqrt{3}$$

$x = 60^\circ$, $y = 30^\circ$

$$\frac{x}{y} = 2$$

6. For $\alpha, \beta \in (0, \pi/2)$, let $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$ and a real number k be such that $\tan\alpha = k\tan\beta$. Then the value of k is equal to: **[JEE(Main)Jan.2024]**

$\alpha, \beta \in (0, \pi/2)$ के लिए मान $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$ है। माना एक वास्तविक संख्या k के लिए $\tan\alpha = k\tan\beta$ है। तो k का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) $-\frac{2}{3}$ (B) 5 (C) –5 (D) $\frac{2}{3}$

Ans. C

Sol. $\frac{\sin(\alpha+\beta)}{\sin\alpha-\beta}=\frac{2}{3}$

$\frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{\sin(\alpha+\beta)-\sin(\alpha-\beta)}=\frac{2+3}{2-3}$

$\frac{2\sin\alpha\cos\beta}{2\cos\alpha\sin\beta}=\frac{5}{-1}$

$\tan\alpha = -5\tan\beta$

7. If $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}$, $\tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan C=\left(x^{-3}+x^{-2}+x^{-1}\right)^{\frac{1}{2}}$, $0<A,B,C<\frac{\pi}{2}$, then A + B is equal to : **[JEE(Main)Jan.2024]**

यदि $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}$, $\tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ तथा $\tan C=\left(x^{-3}+x^{-2}+x^{-1}\right)^{\frac{1}{2}}$, $0<A,B,C<\frac{\pi}{2}$ हैं, तो A + B बराबर है : **[JEE(Main)Jan.2024]**

(A) $\pi - C$ (B) $\frac{\pi}{2}-C$ (C) $2\pi - C$ (D) C

Ans. D

Sol. $\tan A=\frac{1}{\sqrt{x\left(x^2+x+1\right)}}, \tan B=\frac{\sqrt{x}}{\sqrt{x^2+x+1}}$

$\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\cdot\tan B}$

$\tan(A+B)=\frac{\frac{1}{\sqrt{x\left(x^2+x+1\right)}}+\frac{\sqrt{x}}{\sqrt{x^2+x+1}}}{1-\frac{1}{\sqrt{x\left(x^2+x+1\right)}}\cdot\frac{\sqrt{x}}{\sqrt{x^2+x+1}}}$

$\tan(A+B)=\frac{1+x}{\sqrt{x}\cdot\sqrt{x^2+x+1}}\times\frac{\left(x^2+x+1\right)}{x^2+x}$

$\tan(A+B)=\frac{\sqrt{x^2+x+1}}{x\sqrt{x}}$ ...(1)

Given $\tan C = \left(\frac{1}{x^3}+\frac{1}{x^2}+\frac{1}{x}\right)^{\frac{1}{2}}$

$\tan C = \frac{\sqrt{1+x+x^2}}{x\sqrt{x}}$ ...(2)

from equation (1) & (2)
$\tan(A+B) = \tan C$
or $A + B = C$

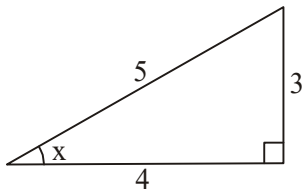
8. If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to : **[JEE(Main)April,2024]**

यदि $\sin x = -\frac{3}{5}$, जहाँ $\pi < x < \frac{3\pi}{2}$, तब $80(\tan^2 x - \cos x)$ बराबर होगा : **[JEE(Main)April,2024]**

(A) 108 (B) 18 (C) 109 (D) 19

Ans. C

Sol.



$\tan x = \frac{3}{4}$, $\cos x = -\frac{4}{5} \Rightarrow 80(\tan^2 x - \cos x) = 80\left(\frac{9}{16}+\frac{4}{5}\right) = 5\times 9 + 16 \times 4 = 109$

9. If the value of $\frac{3\cos 36^\circ + 5\sin 18^\circ}{5\cos 36^\circ - 3\sin 18^\circ}$ is $\frac{a\sqrt{5}-b}{c}$, where a, b, c are natural numbers and gcd(a, c) = 1, then a + b + c is equal to : **[JEE(Main)April,2024]**

यदि $\frac{3\cos 36^\circ + 5\sin 18^\circ}{5\cos 36^\circ - 3\sin 18^\circ} = \frac{a\sqrt{5}-b}{c}$ है, जहाँ a, b, c हैं, तथा gcd(a, c) = 1 है, तो a + b + c बराबर है : **[JEE(Main)April,2024]**

(A) 52 (B) 50 (C) 40 (D) 54

Ans. A

Sol. $$\frac{\frac{3\left(\sqrt{5}+1\right)}{4}+5\left(\frac{\sqrt{5}-1}{4}\right)}{5\left(\frac{\sqrt{5}+1}{4}\right)-3\left(\frac{\sqrt{5}-1}{4}\right)}=\frac{8\sqrt{5}-2}{2\sqrt{5}+8}$$

$$=\frac{4\sqrt{5}-1}{\sqrt{5}+4}\times\frac{\sqrt{5}-4}{\sqrt{5}-4}$$

$$=\frac{20-16\sqrt{5}-\sqrt{5}+4}{-11}$$

$$=\frac{17\sqrt{5}-24}{11} \Rightarrow a = 17, b = 24, c = 11$$

$a + b + c = 52$

10. Let $\left|\cos\theta\cdot\cos\left(60-\theta\right)\cdot\cos\left(60+\theta\right)\right|\le\frac{1}{8},\theta\in\left[0,2\pi\right]$. Then, then sum of all $\theta\in\left[0,2\pi\right]$, where $\cos3\theta$ attains its maximum value of is : **[JEE(Main)April,2024]**

माना $\left|\cos\theta\cdot\cos\left(60-\theta\right)\cdot\cos\left(60+\theta\right)\right|\le\frac{1}{8},\theta\in\left[0,2\pi\right]$ है। तो $\theta\in\left[0,2\pi\right]$ के सभी मानों, जिनके लिए $\cos3\theta$ का मान उच्चतम है, का योग है : **[JEE(Main)April,2024]**

(A) 18π (B) 9π (C) 15π (D) 6π

Ans. D

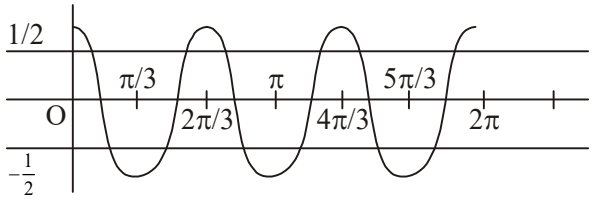
Sol. $$\cos\theta.\cos\left(60-\theta\right)\cos\left(60+\theta\right)=\frac{1}{4}\cos3\theta$$

$$\therefore|\cos3\theta|\le\frac{1}{2}$$

$0 \le 3\theta \le 6\pi$

$$-\frac{1}{2}\le\cos3\theta\le\frac{1}{2}$$

$$\frac{\pi}{6},\frac{\pi}{2},\frac{5\pi}{6},\frac{7\pi}{6},\frac{9\pi}{6},\frac{11\pi}{6}$$



θ satisfying condition are $\frac{\pi}{9},\frac{\pi}{2}+\frac{\pi}{18}+\frac{5\pi}{6}-\frac{\pi}{18},\frac{7\pi}{6}+\frac{\pi}{16},\frac{3\pi}{2}-\frac{\pi}{18},\frac{11\pi}{6}+\frac{\pi}{18}$

$$\sum\theta=\frac{\pi}{9},\frac{\pi}{2}+\frac{\pi}{6}+\frac{7\pi}{6}+\frac{3\pi}{2}+\frac{11\pi}{6}+\frac{\pi}{18}$$

$$=\frac{(2+9+15+21+27+33+1)\pi}{18}=6\pi$$

11. If the range of $f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta},\theta\in R$ is $[\alpha,\beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____. **[JEE(Main)April,2024]**

यदि $f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta},\theta\in R$ का परिसर $[\alpha,\beta]$ है, तब अनंत गुणोत्तर श्रेणी का योग क्या होगा, जिसका प्रथम पद 64 तथा सार्वनुपात $\frac{\alpha}{\beta}$ है, _____ **[JEE(Main)April,2024]**

Ans. 96

Sol. $$f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta}$$

$$f(\theta)=1+\frac{2\cos^2\theta}{\sin^4\theta+\cos^2\theta}$$

$$f(\theta)=\frac{2\cos^2\theta}{\cos^4\theta-\cos^2\theta+1}+1$$

$$f(\theta)=\frac{2}{\cos^2\theta+\sec^2\theta-1}+1$$

$f(\theta)|_{min}=1$

$f(\theta)_{max}=3$

$$S=\frac{64}{1-1/3}=96$$

Exercise J - A

1. Let $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$. Suppose α_1 and β_1 are the roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ and α_2 and β_2 are the roots of the equation $x^2 + 2x\tan\theta - 1 = 0$. If $\alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$, then $\alpha_1 + \beta_2$ equals : **[JEEAdv-2016]**

माना कि $-\frac{\pi}{6} < \theta < -\frac{\pi}{12}$ है। मान लिजिये कि α_1 और β_1 समीकरण $x^2 - 2x\sec\theta + 1 = 0$ के मूल (roots) हैं और α_2 और β_2 समीकरण $x^2 + 2x\tan\theta - 1 = 0$ के मूल हैं। यदि $\alpha_1 > \beta_1$ और $\alpha_2 > \beta_2$ हैं, जब $\alpha_1 + \beta_2$ का मान है : **[JEEAdv-2016]**

(A) $2(\sec\theta - \tan\theta)$ (B) $2\sec\theta$ (C) $-2\tan\theta$ (D) 0

Ans. C

Sol. Roots of the equation $x^2 - 2x\sec\theta + 1 = 0$ are $\sec\theta - \tan\theta$, $\sec\theta + \tan\theta$

$\alpha_1 = \sec\theta - \tan\theta$, $\beta_1 = \sec\theta + \tan\theta$ $(\because \alpha_1 > \beta_1)$

Roots of the equation $x^2 + 2x\tan\theta - 1 = 0$ are $-\tan\theta + \sec\theta$, $-\tan\theta - \sec\theta$

$\alpha_2 = \sec\theta - \tan\theta$, $\beta_2 = -\sec\theta - \tan\theta$ $(\because \alpha_2 > \beta_2)$

Hence $\alpha_1 + \beta_2 = -2\tan\theta$

2. The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to : **[JEEAdv-2016]**

$\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right)\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ का मान है : **[JEEAdv-2016]**

(A) $3 - \sqrt{3}$ (B) $2\left(3 - \sqrt{3}\right)$ (C) $2\left(\sqrt{3} - 1\right)$ (D) $2\left(2 + \sqrt{3}\right)$

Ans. C

Sol. $$\sum_{k=1}^{13} \frac{\sin\left[\left(\frac{\pi}{4} + \frac{k\pi}{6}\right) - \left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right)\right]}{\sin\frac{\pi}{6}\left(\sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)\sin\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right)\right)} = 2\sum_{k=1}^{13}\left(\cot\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right) - \cot\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)\right)$$

$$= 2\left(\cot\frac{\pi}{4} - \cot\left(\frac{\pi}{4} + \frac{13\pi}{6}\right)\right) = 2\left(1 - \cot\left(\frac{29\pi}{12}\right)\right) = 2\left(1 - \cot\left(\frac{5\pi}{12}\right)\right) = 2(1 - (2 - \sqrt{3})) = 2(-1 + \sqrt{3})$$

$$= 2\left(\sqrt{3} - 1\right)$$

3. Let α and β be non-zero real numbers such that $2(\cos\beta - \cos\alpha) + \cos\alpha\cos\beta = 1$. Then which of the following is/are true : **[JEEAdv-2017]**

माना कि α तथा β इस प्रकार की अशून्य वास्तविक संख्यायें हैं, कि $2(\cos\beta - \cos\alpha) + \cos\alpha\cos\beta = 1$. तब निम्न में से कौनसा/कौनसे कथन सत्य है/हैं : **[JEEAdv-2017]**

(A) $\tan\left(\frac{\alpha}{2}\right) + \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$

(B) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right) = 0$

(C) $\tan\left(\frac{\alpha}{2}\right) - \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$

(D) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 0$

Ans. AC

Sol. $\cos\alpha = \left(\frac{1-a}{1+a}\right); \qquad a = \tan^2\frac{\alpha}{2}$

$\cos\beta = \left(\frac{1-b}{1+b}\right); \qquad b = \tan^2\frac{\beta}{2}$

$$2\left(\left(\frac{1-b}{1+b}\right) - \left(\frac{1-a}{1+a}\right)\right) + \left(\left(\frac{1-a}{1+a}\right)\left(\frac{1-b}{1+b}\right)\right) = 1$$

$\Rightarrow$ $2((1 - b)(1 + a) - (1 - a)(1 + b)) + (1 - a)(1 - b) = (1 + a)(1 + b)$
$\Rightarrow 2(1 + a - b - ab - (1 + b - a - ab)) + 1 - a - b + ab = 1 + a + b + ab$
$\Rightarrow 4(a - b) = 2(a + b)$
$\Rightarrow 2a - 2b = a + b$
$\Rightarrow a = 3b$

$$\tan^2\frac{\alpha}{2} = 3\tan^2\frac{\beta}{2}$$

$$\tan\frac{\alpha}{2} = \pm\sqrt{3}\tan\left(\frac{\beta}{2}\right)$$

4. Let $f:[0,2] \to R$ be the function defined by $f(x) = (3 - \sin(2\pi x))\sin(\pi x - \frac{\pi}{4}) - \sin(3\pi x + \frac{\pi}{4})$. If $\alpha, \beta \in [0, 2]$ are such that $\{x \in [0, 2]: f(x) \geq 0\} = [\alpha, \beta]$, then the value of $\beta - \alpha$ is. **[JEEAdv-2020]**

माना कि फलन $f:[0,2] \to R$ को $f(x) = (3 - \sin(2\pi x)) \sin(\pi x - \frac{\pi}{4}) - \sin(3\pi x + \frac{\pi}{4})$ द्वारा परिभाषित किया जाता है। यदि $\alpha, \beta \in [0,2]$ इस प्रकार से हैं कि $\{x \in [0, 2]: f(x) \geq 0\} = [\alpha, \beta]$, तब $\beta - \alpha$ का मान है। **[JEEAdv-2020]**

Ans. 1

Sol. $f(x) = (3 - \sin 2\pi x) \sin\left(\pi x - \frac{\pi}{4}\right) - \sin\left(3\pi x + \frac{\pi}{4}\right)$

Let $\pi x - \frac{\pi}{4} = \pi y \Rightarrow x = y + \frac{1}{4}$

$f(x) = \left(3 - \sin 2\pi\left(y + \frac{1}{4}\right)\right) \sin y\pi - \sin\left(3\pi\left(y + \frac{1}{4}\right) + \frac{\pi}{4}\right)$

$= \left(3 - \sin\left(2\pi y + \frac{\pi}{2}\right)\right) \sin y\pi - \sin\left(3\pi y + \frac{3\pi}{4} + \frac{\pi}{4}\right)$

$= (3 - \cos 2\pi y) \sin\pi y + \sin(3\pi y)$

$= (3 - (1 - 2\sin^2 \pi y)) \sin \pi y + 3 \sin \pi y - 4 \sin^3 \pi y$

$= 5 \sin \pi y + 2 \sin^3 \pi y = \sin \pi y (5 + 2\sin^2 \pi y)$

Since $f(x) \geq 0 \sin \pi y \geq 0$

$\Rightarrow \sin \pi \left(x - \frac{1}{4}\right) \geq 0 \Rightarrow \pi \left(x - \frac{\pi}{4}\right) \in [2n\pi, (2n + 1)\pi]$

$x \in [0, 2] \Rightarrow \pi x - \frac{\pi}{4} \in \left[-\frac{\pi}{4}, \frac{7\pi}{4}\right]$

From both of these,

$\pi x - \frac{\pi}{4} \in [0, \pi]$

$\Rightarrow x \in \left[\frac{1}{4}, \frac{5}{4}\right]$

So, $\alpha = \frac{1}{4}, \beta = \frac{5}{4}$

$\beta - \alpha = \frac{5}{4} - \frac{1}{4} = 1$

5. Let α and β be real numbers such that $-\frac{\pi}{4} < \beta < 0 < \alpha < \frac{\pi}{4}$. If $\sin(\alpha + \beta) = \frac{1}{3}$ and $\cos(\alpha - \beta) = \frac{2}{3}$, then the greatest integer less than or equal to $\left(\frac{\sin\alpha}{\cos\beta} + \frac{\cos\beta}{\sin\alpha} + \frac{\cos\alpha}{\sin\beta} + \frac{\sin\beta}{\cos\alpha}\right)^2$ is _____.

[JEEAdv-2022]

माना कि α एवं β ऐसी वास्तविक संख्यायें हैं कि $-\frac{\pi}{4}<\beta<0<\alpha<\frac{\pi}{4}$ है। यदि $\sin(\alpha+\beta)=\frac{1}{3}$ एवं $\cos(\alpha-\beta)=\frac{2}{3}$ है, तब

$\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\cos\alpha}{\sin\beta}+\frac{\sin\beta}{\cos\alpha}\right)^2$ से कम या बराबर महत्तम पूर्णांक _____ है। **[JEEAdv-2022]**

Ans. 1

Sol. $\left(\frac{\sin\alpha}{\cos\beta}+\frac{\cos\alpha}{\sin\beta}+\frac{\cos\beta}{\sin\alpha}+\frac{\sin\beta}{\cos\alpha}\right)^2$

$=\left(\frac{\cos(\alpha-\beta)}{\sin\beta\cos\beta}+\frac{\cos(\alpha-\beta)}{\sin\alpha\cdot\cos\alpha}\right)^2$

$=\left(\frac{4}{3}\left\{\frac{1}{\sin2\beta}+\frac{1}{\sin2\alpha}\right\}\right)^2$

$=\frac{16}{9}\left(\frac{2\sin(\alpha+\beta)\cdot\cos(\alpha-\beta)}{\sin2\alpha\cdot\sin2\beta}\right)^2$

$=\frac{16}{9}\left(\frac{4\cdot\frac{1}{3}\cdot\frac{2}{3}}{\cos(2\alpha-2\beta)-\cos(2\alpha+2\beta)}\right)^2$

$=\frac{16}{9}\left(\frac{\frac{8}{9}}{2\cos^2(\alpha-\beta)-1-1+2\sin^2(\alpha+\beta)}\right)^2$

$=\frac{16}{9}\left(\frac{\frac{8}{9}}{\frac{8}{9}-2+\frac{2}{9}}\right)^2 \qquad =\frac{16}{9}$

6. Let $\frac{\pi}{2}<x<\pi$ be such that $\cot x=\frac{-5}{\sqrt{11}}$. Then

$\left(\sin\frac{11x}{2}\right)(\sin6x-\cos6x)+\left(\cos\frac{11x}{2}\right)(\sin6x+\cos6x)$ is equal to : **[JEE(Adv.)2024]**

माना कि $\frac{\pi}{2} < x < \pi$ इस प्रकार है कि $\cot x = \frac{-5}{\sqrt{11}}$ है। तब

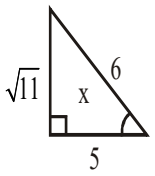
$\left(\sin\frac{11x}{2}\right)(\sin 6x - \cos 6x) + \left(\cos\frac{11x}{2}\right)(\sin 6x + \cos 6x)$ बराबर है : : **[JEE(Adv.)2024]**

(A) $\frac{\sqrt{11}-1}{2\sqrt{3}}$ (B) $\frac{\sqrt{11}+1}{2\sqrt{3}}$ (C) $\frac{\sqrt{11}+1}{3\sqrt{2}}$ (D) $\frac{\sqrt{11}-1}{3\sqrt{2}}$

Ans. B

Sol. $\frac{\pi}{2} < x < \pi$

$\cot x = -\frac{5}{\sqrt{11}}$



$\left(\sin\frac{11x}{2}\right)(\sin 6x - \cos 6x) + \left(\cos\frac{11x}{2}\right)(\sin 6x + \cos 6x)$

$\left(\cos 6x\cos 11\frac{x}{2} + \sin 6x\sin 11\frac{x}{2}\right) + \left(\sin 6x\cos 11\frac{x}{2} - \cos 6x\sin 11\frac{x}{2}\right)$

$\Rightarrow \cos\left(6x - 11\frac{x}{2}\right) + \sin\left(6x - 11\frac{x}{2}\right) = \cos\frac{x}{2} + \sin\frac{x}{2}$

$\Rightarrow \sqrt{(1+\sin x)^2}$

$= \sqrt{\left(1+\frac{\sqrt{11}}{6}\right)^2}$

$\Rightarrow \sqrt{\left(1+\frac{\sqrt{11}}{6}\right)^2} = \sqrt{\frac{12+2\sqrt{11}}{12}}$

$= \frac{\sqrt{11}+1}{2\sqrt{3}}$

7. Let $\alpha = \frac{1}{\sin 60^\circ \sin 61^\circ} + \frac{1}{\sin 62^\circ \sin 63^\circ} + \cdots + \frac{1}{\sin 118^\circ \sin 119^\circ}$. Then the value of $\left(\frac{\text{cosec}\,1^\circ}{\alpha}\right)^2$ is _____.

[JEE(Adv.)2025]

मान लीजिए कि $\alpha = \frac{1}{\sin 60^\circ \sin 61^\circ} + \frac{1}{\sin 62^\circ \sin 63^\circ} + \cdots + \frac{1}{\sin 118^\circ \sin 119^\circ}$ है। तब $\left(\frac{\text{cosec}\,1^\circ}{\alpha}\right)^2$ का मान _____ है।

[JEE(Adv.)2025]

Ans. 3

Sol. $(\alpha)\ \sin 1^\circ = \frac{\sin(61^\circ - 60^\circ)}{\sin 60^\circ \sin 61^\circ} + \frac{\sin(63^\circ - 62^\circ)}{\sin 63^\circ \sin 62^\circ} + \ldots\ldots + \frac{\sin(119^\circ - 118^\circ)}{\sin 118^\circ \sin 119^\circ}$

$= (\cot 60° - \cot 61°) + (\cot 62° - \cot 63°) + \ldots\ldots\ldots + (\cot 118° - \cot 119°)$

$119° = 180° - 61° \quad \Rightarrow \cot 119° = -\cot 61°$

$\Rightarrow (\cot 60° + \cot 119°) + (-\cot 118° + \cot 117°) + \ldots\ldots\ldots\ldots + (\cot 118° - \cot 119°)$

$\Rightarrow \alpha = \frac{\cot 60^\circ}{\sin 10^\circ}$

$\text{RSH} \left(\frac{\text{cosec}\,1°}{\cot 60°} \sin 1°\right)^2 = 3$

8. Let $\alpha = \left(1 - 2\cos\left(\frac{\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{3\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{9\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{27\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{81\pi}{11}\right)\right)$. Then the value of $5 - \alpha^2$ is _____.

[JEE(Adv.)2026]

मान लीजिए कि $\alpha = \left(1 - 2\cos\left(\frac{\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{3\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{9\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{27\pi}{11}\right)\right)\left(1 - 2\cos\left(\frac{81\pi}{11}\right)\right)$ है। तब $5 - \alpha^2$ का मान _____ है।

[JEE(Adv.)2026]

Ans. 4

Sol. $\alpha = \left(1 - 2\cos\frac{\pi}{11}\right)\left(1 - 2\cos\frac{3\pi}{11}\right)\left(1 - 2\cos\frac{9\pi}{11}\right)\left(1 - 2\cos\frac{27\pi}{11}\right)\left(1 - 2\cos\frac{81\pi}{11}\right)$

Let $\theta = \frac{\pi}{11}$

$\alpha = (1 - 2\cos\theta)(1 - 2\cos 3\theta)(1 - 2\cos 9\theta)(1 - 2\cos 27\theta)(1 - 2\cos 81\theta)$

Now $1-2\cos\theta$

$$1-2\left\{2\cos^2\frac{\theta}{2}-1\right\}$$

$$3-4\cos^2\theta/2$$

$$\frac{\left(3-4\cos^2\theta/2\right)\cdot\cos\theta/2}{\cos\theta/2}$$

$$\Rightarrow -\frac{\cos\frac{3\theta}{2}}{\cos\frac{\theta}{2}}$$

$$\therefore\ \alpha=\left(-\frac{\cos\frac{3\theta}{2}}{\cos\frac{\theta}{2}}\right)\left(-\frac{\cos\frac{9\theta}{2}}{\cos\frac{3\theta}{2}}\right).....\left(-\frac{\cos\frac{243\theta}{2}}{\cos\frac{81\theta}{2}}\right)\quad=-\frac{\cos\frac{243\theta}{2}}{\cos\frac{\theta}{2}}$$

$$\text{Now }\theta=\frac{\pi}{11}\Rightarrow\ \frac{\theta}{2}\ =\frac{\pi}{22}\Rightarrow\quad 243\frac{\theta}{2}\ =\frac{\pi}{22}\times 243\qquad =11\pi+\frac{\pi}{22}$$

$$\cos 243\frac{\theta}{2}\ =-\cos\frac{\pi}{22}\ =-\cos\frac{\theta}{2}$$

$$\therefore\ \alpha=1\qquad\Rightarrow\qquad 5-\alpha^2=4$$

Exercise C - I

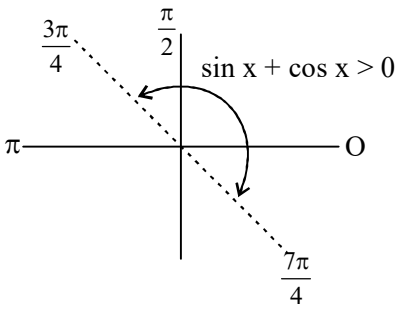
1. Let $\frac{3\pi}{4} < \alpha < \pi$, then $\sqrt{2\cot\alpha + \frac{1}{\sin^2\alpha}}$ is equal to :

यदि $\frac{3\pi}{4} < \alpha < \pi$ तब $\sqrt{2\cot\alpha + \frac{1}{\sin^2\alpha}}$ किस के बराबर है :

(A) $\frac{\sin\alpha - \cos\alpha}{\sin\alpha}$ (B) $\frac{\sin\alpha + \cos\alpha}{\sin\alpha}$ (C) $\frac{-\sin\alpha + \cos\alpha}{\sin\alpha}$ (D) $\frac{-\sin\alpha - \cos\alpha}{\sin\alpha}$

Ans. D

Sol. $\frac{3\pi}{4} < \alpha < \pi$



$$\sqrt{2\cot\alpha + \frac{1}{\sin^2\alpha}} = \frac{\sqrt{2\cot\alpha\cdot\sin^2\alpha + 1}}{|\sin\alpha|}$$

$$= \frac{\sqrt{1+\sin 2\alpha}}{\sin\alpha}$$

$$= \frac{|\sin\alpha + \cos\alpha|}{\sin\alpha}$$

$$= \frac{-\sin\alpha - \cos\alpha}{\sin\alpha} \qquad \left\{\because \frac{3\pi}{4} < \alpha < \pi\right\}$$

2. If $\cos^2 A + \cos^2 B + \cos^2 C = 1$, then triangle ABC is :

(A) right angled (B) equilateral (C) isosceles (D) None of these

यदि $\cos^2 A + \cos^2 B + \cos^2 C = 1$ तब त्रिभुज ABC एक :

(A) समकोण त्रिभुज होगा (B) समबाहु त्रिभुज होगा (C) समद्विबाहु त्रिभुज होगा (D) इनमें से कोई नहीं

Ans. A

Sol. $\cos^2A + \cos^2B + \cos^2C = 1$

$\Rightarrow 1 + \cos2A + 1 + \cos 2B + 1 + \cos2C = 2$

$\Rightarrow \cos2A + \cos2B + \cos2C = -1$

$\Rightarrow 2\cos(A + B)\cos(A - B) + \cos 2C = -1$

$\Rightarrow -2\cos(C)\cos(A - B) + 2\cos^2C = 0$

$\Rightarrow -2\cos C[\cos(A - B) - \cos C] = 0$

$\Rightarrow 2\cos C[\cos(A - B) + \cos(A + B)] = 0$

$\Rightarrow 4\cos C \cos A \cos B = 0$

$\Rightarrow A = \frac{\pi}{2}$ or $B = \frac{\pi}{2}$ or $C = \frac{\pi}{2}$

3. If $\frac{\tan 3A}{\tan A} = k$, then $\frac{\sin 3A}{\sin A}$ is equal to :

(A) $\frac{2k}{k-1}$ (B) $\frac{2k}{k+1}$ (C) $\frac{3k}{k+1}$ (D) None of these

यदि $\frac{\tan 3A}{\tan A} = k$, then $\frac{\sin 3A}{\sin A}$ किस के बराबर है :

(A) $\frac{2k}{k-1}$ (B) $\frac{2k}{k+1}$ (C) $\frac{3k}{k+1}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\frac{\tan 3A}{\tan A} = k$

Apply C & D rule

$\Rightarrow \frac{\tan 3A + \tan A}{\tan 3A - \tan A} = \frac{K+1}{K-1}$

$\Rightarrow \frac{\sin 3A \cos A + \cos 3A \sin A}{\sin 3A \cos A - \cos 3A \sin A} = \frac{K+1}{K-1}$

$\Rightarrow \frac{\sin 4A}{\sin 2A} = \frac{K+1}{K-1}$

$\Rightarrow 2\cos 2A = \frac{K+1}{K-1}$

$\Rightarrow \frac{\sin 3A}{\sin A} = \frac{3\sin A - 4\sin^3 A}{\sin A}$

$= 3 - 4\sin^2 A$

$= 3 - 4\left[\frac{1-\cos 2A}{2}\right]$

$= 3 - 2 + 2\cos 2A$

$= 1 + \frac{K+1}{K-1} = \frac{2K}{K-1}$

4. $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x) =$

(A) 11 (B) 12 (C) 13 (D) 14

Ans. C

Sol. $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$

$= 3[\sin^2 x + \cos^2 x - \sin 2x]^2 + 6[1 + \sin 2x] + 4[1 - 3\sin^2 x \cos^2 x]$

$\Rightarrow 3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x)$

$= 3[1 + 4\sin^2 x \cos^2 x - 4\sin x \cos x] + 6 + 12\sin x \cos x + 4 - 12\sin^2 x \cos^2 x$

$= 3 + 12\sin^2 x \cos^2 x - 12\sin x \cos x + 6 + 12\sin x \cos x + 4 - 12\sin^2 x \cos^2 x$

$= 13$

5. $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$ is true if and only if :

$\sec^2\theta = \dfrac{4xy}{(x+y)^2}$ सत्य होगा यदि और केवल यदि :

(A) $x + y \neq 0$ (B) $x = y,\ x \neq 0$ (C) $x = y$ (D) $x \neq 0, y \neq 0$

Ans. B

Sol. $\sec^2\theta = \dfrac{4xy}{(x+y)^2}$

$\because \sec^2\theta \in [1, \infty)$

$\Rightarrow \dfrac{4xy}{(x+y)^2} \geq 1$

$\Rightarrow \dfrac{4xy - x^2 - y^2 - 2xy}{(x+y)^2} \geq 0$

$\Rightarrow \dfrac{-(x^2 + y^2 - 2xy)}{(x+y)^2} \geq 0$

$\Rightarrow \dfrac{(x-y)^2}{(x+y)^2} \leq 0$

$\Rightarrow x - y = 0 \ \& \ x + y \neq 0$

$\Rightarrow x = y;\ x \neq 0$

6. If $\alpha + \beta = \dfrac{\pi}{2}$ and $\beta + \gamma = \alpha$, then $\tan\alpha$ equals :

यदि $\alpha + \beta = \frac{\pi}{2}$ एवं $\beta + \gamma = \alpha$ तब $\tan\alpha$ किस के बराबर होगा :

(A) $2(\tan\beta + \tan\gamma)$ (B) $\tan\beta + \tan\gamma$ (C) $\tan\beta + 2\tan\gamma$ (D) $2\tan\beta + \tan\gamma$

Ans. C

Sol. $\alpha + \beta = \frac{\pi}{2}$ & $\beta + \gamma = \alpha$

$\tan(\beta + \gamma) = \tan\alpha$

$\Rightarrow \frac{\tan\beta + \tan\gamma}{1 - \tan\beta\tan\gamma} = \tan\alpha$

$\Rightarrow \tan\beta + \tan\gamma = \tan\alpha - \tan\alpha\tan\beta\tan\gamma$

$\Rightarrow \tan\beta + \tan\gamma = \tan\alpha - \tan\left(\frac{\pi}{2} - \beta\right)\tan\beta\tan\gamma$

$\Rightarrow \tan\alpha = \tan\beta + 2\tan\gamma$

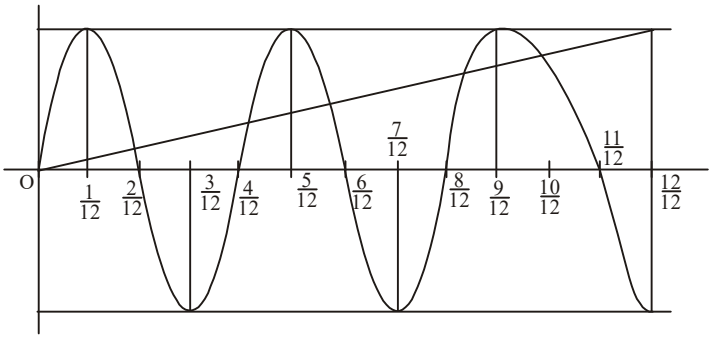
7. Number of real solutions to the equation $\sin(6\pi x) = x$, is :

समीकरण $\sin(6\pi x) = x$ के वास्तविक मानों की संख्या होगी :

(A) 13 (B) 11 (C) 9 (D) 7

Ans. B

Sol. Draw graphs of $y = x$ and $y = \sin 6\pi x$ and then interpret.



8. The value of x satisfying the equation, $x = \sqrt{2 + \sqrt{2 - \sqrt{2 + x}}}$ is :

समीकरण $x = \sqrt{2 + \sqrt{2 - \sqrt{2 + x}}}$ को सन्तुष्ट करने वाले x का मान होगा :

(A) $2\cos 10°$ (B) $2\cos 20°$ (C) $2\cos 40°$ (D) $2\cos 80°$

Ans. C

Sol. Note that $x \in [-2, 2]$ for x to be real

Let $x = 2\cos\theta$ where $\theta \in [0, \pi]$

$$x = \sqrt{2+\sqrt{2-\sqrt{2+2\cos\theta}}}$$

$$2\cos\theta = \sqrt{2+\sqrt{2-2\cos\frac{\theta}{2}}} = \sqrt{2+\sqrt{2\left(1-\cos\frac{\theta}{2}\right)}} = \sqrt{2+2\sin\frac{\theta}{4}} = \sqrt{2+2\cos\left(\frac{\pi}{2}-\frac{\theta}{4}\right)}$$

$$= \sqrt{2\left(1+\cos\left(\frac{\pi}{2}-\frac{\theta}{4}\right)\right)} = \sqrt{4\cos^2\left(\frac{\pi}{4}-\frac{\theta}{8}\right)}$$

$2\cos\theta = 2\cos \quad \Rightarrow \quad \theta = \frac{\pi}{4}-\frac{\theta}{8} \quad \Rightarrow \quad \frac{9\theta}{8} = \frac{\pi}{4} \quad \Rightarrow \quad \theta = \frac{2\pi}{9}$

Hence $x = 2\cos\frac{2\pi}{9} = 2\cos 40°$

9. The value of, $(\tan 18°)(\sin 36°)(\cos 54°)(\tan 72°)(\tan 108°)(\cos 126°)(\sin 144°)(\tan 162°)(\cos 180°)$ is $k\sin^2 18°$ then k has the value equal to :

यदि $(\tan 18°)(\sin 36°)(\cos 54°)(\tan 72°)(\tan 108°)(\cos 126°)(\sin 144°)(\tan 162°)(\cos 180°) = k\sin^2 18°$ हो, तो k का मान होगा :

(A) $\frac{5}{16}$ (B) $\frac{5}{4}$ (C) $\frac{1}{4}$ (D) $\frac{1}{16}$

Ans. B

Sol. $(\tan 18°)(\tan 72°)(\tan 108°)(\tan 162°) = 1$

& $\sin 36° \cos 54° \cos(126°)\sin(144°) = \sin^4 36°$

$$\Rightarrow k\sin^2 18° = \left(\frac{1-\cos 72°}{2}\right)^2 = \frac{(1-\sin 18°)^2}{4}$$

$$= \frac{1}{4}\left(1-\frac{\sqrt{5}-1}{4}\right)^2 = \frac{1}{4}\left(\frac{5-\sqrt{5}}{4}\right)^2 = \frac{5}{4}\left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{5}{4}\sin^2 18° \Rightarrow \quad k = \frac{5}{4}$$

10. If $x\sec\alpha + y\tan\alpha = x\sec\beta + y\tan\beta = a$, then $\sec\alpha \cdot \sec\beta =$

यदि $x\sec\alpha + y\tan\alpha = x\sec\beta + y\tan\beta = a$, तो $\sec\alpha \cdot \sec\beta =$

(A) $\frac{a^2+y^2}{x^2+y^2}$ (B) $\frac{a^2+y^2}{x^2-y^2}$ (C) $\frac{x^2+y^2}{a^2+y^2}$ (D) $\frac{x^2-y^2}{a^2-y^2}$

Ans. B

Sol. α and β satisfy the equation

$$x \sec\theta + y \tan\theta = a$$

$$(x\sec\theta - a)^2 = y^2 \tan^2\theta = y^2(\sec^2\theta - 1)$$

$$\sec^2\theta\,(x^2 - y^2) - 2ax \sec\theta + a^2 + y^2 = 0$$

This is a quadratic in $\sec\theta$, whose roots are $\sec\alpha$ and $\sec\beta$

$$\Rightarrow \quad \sec\alpha \cdot \sec\beta = \frac{a^2 + y^2}{x^2 - y^2}$$

11. The graphs of $y = \sin x$, $y = \cos x$, $y = \tan x$ and $y = \text{cosec}\, x$ are drawn on the same axes from 0 to $\pi/2$. A vertical line is drawn through the point where the graphs of $y = \cos x$ and $y = \tan x$ cross, intersecting the other two graphs at points A and B. The length of the line segment AB is :

यदि $[0, \pi/2]$ के अन्तराल में $y = \sin x$, $y = \cos x$, $y = \tan x$ तथा $y = \text{cosec}\, x$ का आरेख खींचा जाता है। $y = \cos x$ एवं $y = \tan x$ के प्रतिच्छेद बिन्दु से y अक्ष के समान्तर एक रेखा खींची जाती है जो बाकी दो आरेख A एवं B पर काटती हो तो AB की लम्बाई क्या होगी :

(A) 1 (B) $\frac{\sqrt{5}-1}{2}$ (C) $\sqrt{2}$ (D) $\frac{\sqrt{5}+1}{2}$

Ans. A

Sol. $\tan x = \cos x$

$\Rightarrow \cos^2 x = \sin x$

$\Rightarrow 1 - \sin^2 x = \sin x$

$\Rightarrow \sin^2 x + \sin x - 1 = 0$

$$\Rightarrow \sin x = \frac{-1 \pm \sqrt{5}}{2}$$

$$\Rightarrow \sin x = \frac{\sqrt{5}-1}{2} \qquad \left\{\because 0 < x < \frac{\pi}{2}\right\}$$

$$\text{cosec}\, x = \frac{2}{\sqrt{5}-1}$$

$$\Rightarrow \left|\frac{2}{\sqrt{5}-1} - \frac{\sqrt{5}-1}{2}\right|$$

$$= \left|\frac{4 - (5 + 1 - 2\sqrt{5})}{2(\sqrt{5}-1)}\right|$$

$= 1$

12. The value of $4\cos\frac{\pi}{10} - 3\sec\frac{\pi}{10} - 2\tan\frac{\pi}{10}$ is equal to :

(A) 1 (B) $\sqrt{5}-1$ (C) $\sqrt{5}+1$ (D) zero

$4\cos\frac{\pi}{10} - 3\sec\frac{\pi}{10} - 2\tan\frac{\pi}{10}$ का मान होगा :

(A) 1 (B) $\sqrt{5}-1$ (C) $\sqrt{5}+1$ (D) शून्य

Ans. D

Sol. $4\cos 18^0 - \frac{3}{\cos 18^0} - 2\tan 18^0$

$$\frac{4\cos^2 18^0 - 3 - 2\sin 18^0}{\cos 18^0} = \frac{2(1+\cos 36^0) - 2\sin 18^0 - 3}{\cos 18^0}$$

$$\Rightarrow \quad \frac{2(1+\cos 36^0 - \sin 18^0) - 3}{\cos 18^0} = \frac{2\left(1+\frac{1}{2}\right) - 3}{\cos 18^0} = 0$$

13. The value of $\cot x + \cot(60^\circ + x) + \cot(120^\circ + x)$ is equal to :

व्यंजक $\cot x + \cot(60^\circ + x) + \cot(120^\circ + x)$ का मान होगा :

(A) $\cot 3x$ (B) $\tan 3x$ (C) $3\tan 3x$ (D) $\frac{3 - 9\tan^2 x}{3\tan x - \tan^3 x}$

Ans. D

Sol. $\cot x + \frac{\cos(60+x)}{\sin(60+x)} + \frac{\cos(x-60)}{\sin(x-60)}$

$$= \frac{\cos x}{\sin x} + \frac{\sin(2x)}{\sin(x+60)\sin(x-60)}$$

$$= \frac{\cos x}{\sin x} + \frac{8\sin x\cos x}{4\sin^2 x - 3} = \frac{4\sin^2 x\cos x - 3\cos x + 8\sin^2 x\cos x}{4\sin^3 x - 3\sin x}$$

$$= \frac{3[3\cos x - 4\cos^3 x]}{\sin 3x} = 3\cot 3x = \frac{3[1-3\tan^2 x]}{3\tan x - \tan^3 x}$$

14. The minimum value of the function $f(x) = \frac{\sin x}{\sqrt{1-\cos^2 x}} + \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{\tan x}{\sqrt{\sec^2 x - 1}} + \frac{\cot x}{\sqrt{\operatorname{cosec}^2 x - 1}}$ as x varies over all numbers in the largest possible domain of $f(x)$ is :

x के सभी सम्भावित प्रान्त के लिये फलन $f(x) = \frac{\sin x}{\sqrt{1-\cos^2 x}} + \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{\tan x}{\sqrt{\sec^2 x - 1}} + \frac{\cot x}{\sqrt{\operatorname{cosec}^2 x - 1}}$ का न्यूनतम मान होगा :

(A) 4 (B) -2 (C) 0 (D) 2

Ans. B

Sol. Let $y = f(x) = \frac{\sin x}{|\sin x|} + \frac{\cos x}{|\cos x|} + \frac{\tan x}{|\tan x|} + \frac{\cot x}{|\cot x|}$

$$\left.\begin{array}{l} \text{if } x \in 1^{st} \text{ quadrant then } y = 1+1+1+1 = 4 \\ \text{if } x \in 2^{nd} \text{ quadrant then } y = 1-1-1-1 = -2 \\ \text{if } x \in 3^{rd} \text{ quadrant then } y = -1-1+1+1 = 0 \\ \text{if } x \in 4^{th} \text{ quadrant then } y = -1+1-1-1 = -2 \end{array}\right\} \Rightarrow y_{min} = -2$$

15. If $a > b > 0$, then the minimum value of $a\sec\theta - b\tan\theta \; \forall \theta \in \left(0, \frac{\pi}{2}\right)$:

यदि $a > b > 0$ तो $\forall \theta \in \left(0, \frac{\pi}{2}\right)$ के लिये $a\sec\theta - b\tan\theta$ का न्यूनतम मान होगा :

(A) $\sqrt{a^2 - b^2}$ (B) $a - b$ (C) $\sqrt{a-b}$ (D) $\left(a^2 - b^2\right)^{3/2}$

Ans. A

Sol. $y + b\tan\theta = a\sec\theta$

$y^2 + 2by\tan\theta + b^2\tan^2\theta = a^2 + a^2\tan^2\theta$

$(b^2 - a^2)\tan^2\theta + 2by\tan\theta + (y^2 - a^2) = 0$

$D \geq 0$

$4b^2y^2 - 4(b^2 - a^2)(y^2 - a^2) \geq 0$

$b^2y^2 - [b^2y^2 - b^2a^2 - a^2y^2 + a4] \geq 0$

$b^2 + y^2 - a^2 \geq 0$

$y^2 \geq a^2 - b^2$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

16. If $\sin(x-y) = \cos(x+y) = \frac{1}{2}$, then the values of 'x' and 'y' lying between 0° and 180° are given by :

यदि $\sin(x-y) = \cos(x+y) = \frac{1}{2}$ तब 'x' एवं 'y' के वे मान जो 0° एवं 180° के बीच में हो होंगे :

(A) $x = 45°, y = 15°$ (B) $x = 45°, y = 135°$ (C) $x = 165°, y = 15°$ (D) $x = 165°, y = 135°$

Ans. AD

Sol. $\sin(x-y) = \cos(x+y) = \frac{1}{2}$

$\sin x \cos y - \cos x \sin y = \cos x \cos y - \sin x \sin y$

$\Rightarrow (\sin x - \cos x)\cos y = \sin y(\cos x - \sin x)$

$\Rightarrow (\cos y + \sin y)(\sin x - \cos x) = 0$

$\Rightarrow \sin x = \cos x$ or $\cos y = -\sin y$

$\Rightarrow x = \frac{\pi}{4}$ or $y = \frac{3\pi}{4}$

If $x = \frac{\pi}{4}$ then $\sin\left(\frac{\pi}{4} - y\right) = \frac{1}{2} \Rightarrow y = 15^\circ$

If $y = \frac{3\pi}{4}$ then $\sin\left(x - \frac{3\pi}{4}\right) = \frac{1}{2} \Rightarrow x = 165^\circ$

17. For a positive integer 'n', let : $f_n(\theta) = \left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta)\ldots\ldots(1+\sec 2^n\theta)$. Then :

किसी धनात्मक पूर्णांक 'n' के लिए माना $f_n(\theta) = \left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta)\ldots\ldots(1+\sec 2^n\theta)$ तब:

(A) $f_2\left(\frac{\pi}{16}\right) = 1$ (B) $f_3\left(\frac{\pi}{32}\right) = 1$ (C) $f_4\left(\frac{\pi}{64}\right) = 1$ (D) $f_5\left(\frac{\pi}{128}\right) = 1$

Ans. ABCD

Sol. $f_n(\theta) = \left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)\ldots\ldots\ldots\ldots\left(1+\sec 2^n\theta\right)$

$\Rightarrow f_n(\theta) = \left(\tan\frac{\theta}{2}\right)\left(\frac{1+\cos\theta}{\cos\theta}\right)\left(\frac{1+\cos 2\theta}{\cos 2\theta}\right)\ldots\ldots\ldots\ldots\left(\frac{1+\cos 2^n\theta}{\cos 2^n\theta}\right)$

$$\Rightarrow f_n(\theta)=\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\cdot\frac{2\cos^2\frac{\theta}{2}}{\cos\theta}\cdot\frac{2\cos^2\theta}{\cos 2\theta}\cdot\frac{2\cos^2 2\theta}{\cos 4\theta}\cdot\frac{.....}{\cos 2^{n-1}\theta}\cdot\frac{2\cos^2 2^{n-1}\theta}{\cos 2^n\theta}$$

$= \tan 2^n\theta$

$$f_2\left(\frac{\pi}{16}\right)=\tan\left(4\cdot\frac{\pi}{16}\right)=1$$

$$f_3\left(\frac{\pi}{32}\right)=\tan\left(8\cdot\frac{\pi}{32}\right)=1$$

$$f_4\left(\frac{\pi}{64}\right)=\tan\left(16\cdot\frac{\pi}{64}\right)=1$$

$$f_5\left(\frac{\pi}{128}\right)=\tan\left(\frac{32\pi}{128}\right)=1$$

18. If S is the solution set of all values of x for which $2\cos t=\frac{3x+1}{x-1}$, then :

(A) number of integral values in S is 5 (B) number of integral values in S is 4.

(C) sum of all integral values in S is –6 (D) sum of all integral values in S is –5

यदि $2\cos t=\frac{3x+1}{x-1}$ को संतुष्ट करने वाले x के हलों का समुच्चय S है, तो :

(A) S में पूर्णांक मानों की संख्या 5 है (B) S में पूर्णांक मानों की संख्या 4 है

(C) S में पूर्णांक मानों का योग –6 है (D) S में पूर्णांक मानों का योग –5 है

Ans. BC

Sol. $-2 \le 2\cos t \le 2$

$$-2 \le \frac{3x+1}{x-1} \le 2$$

$$\Rightarrow \frac{3x+1}{x-1}+2 \ge 0 \ \& \ \Rightarrow \frac{3x+1}{x-1}-2 \le 0$$

$$\Rightarrow \frac{5x+1}{x-1} \ge 0 \ \& \ \frac{x+3}{x-1} \le 0$$

$$\Rightarrow x \in \left[-3,-\frac{1}{5}\right] \Rightarrow x=-3,-2,-1,0$$

19. Let $f_n(\theta) = 2(\sin^2\theta + \sin 2\theta \sin 4\theta + \sin 3\theta \sin 9\theta + ... + \sin n\theta \sin n^2\theta)$ then which of the following hold(s) good :

माना कि $f_n(\theta) = 2(\sin^2\theta + \sin 2\theta \sin 4\theta + \sin 3\theta \sin 9\theta + ... + \sin n\theta \sin n^2\theta)$, तो निम्न में से कौनसे कथन सत्य

है :

(A) $f_3\left(\frac{\pi}{6}\right)=0$ (B) $f_4\left(\frac{\pi}{3}\right)=2$ (C) $f_8\left(\frac{\pi}{2}\right)=0$ (D) $f_{11}\left(\frac{\pi}{4}\right)=2$

Ans. ACD

Sol. $f_n(\theta)=\sum_{r=1}^{n}2\sin r\theta\,\sin r^2\theta$

$=\sum_{r=1}^{n}\left[\cos(r^2-r)\theta-\cos(r^2+r)\theta\right]$

$=\cos 0^o-\cos(n^2+n)\theta=1-\cos(n^2+n)\theta$

$\therefore\ f_3\left(\frac{\pi}{6}\right)=1-\cos(9+3)\frac{\pi}{6}=1-\cos 2\pi=0$

$f_8\left(\frac{\pi}{2}\right)=1-\cos(64+8)\frac{\pi}{2}=1-\cos 36\pi=0$

$f_{11}\left(\frac{\pi}{4}\right)=1-\cos(121+11)\frac{\pi}{4}=1-\cos 33\pi=2$

$f_4\left(\frac{\pi}{3}\right)=1-\cos\left(\frac{19\pi}{3}\right)=1-\cos\frac{\pi}{3}=\frac{1}{2}$

Exercise C - II

1. If $\cos(x-y)$, $\cos x$ and $\cos(x+y)$ are in H.P., then $\cos x \sec\left(\frac{y}{2}\right) =$ _____.

यदि $\cos(x-y)$, $\cos x$ एवं $\cos(x+y)$ हरात्मक श्रेढी में हो तो $\cos x \sec\left(\frac{y}{2}\right) =$ _____

Ans. $\pm\sqrt{2}$

Sol. $\cos(x-y), \cos x, \cos(x+y) \rightarrow$ H.P.

$$\Rightarrow \cos x = \frac{2\cos(x-y)\cos(x+y)}{\cos(x-y)+\cos(x+y)}$$

$$\Rightarrow \cos x = \frac{\cos^2 x - \sin^2 y}{\cos x \cdot \cos y}$$

$\Rightarrow \cos^2 x \cos y = \cos^2 x - \sin^2 y$

$\Rightarrow \sin^2 y = \cos^2 x (1 - \cos y)$

$$\Rightarrow 4\sin^2\frac{y}{2}\cos^2\frac{y}{2} = \cos^2 x \cdot 2 \cdot \sin^2\frac{y}{2}$$

$$\Rightarrow \left[\cos x \sec\left(\frac{y}{2}\right)\right]^2 = 2$$

$$\Rightarrow \cos x \sec\left(\frac{y}{2}\right) = \pm\sqrt{2}$$

2. If in ΔABC, $\sin A \sin B \sin C + \cos A \cos B = 1$, then the value of $\sin C =$ _____.

किसी Δ ABC में $\sin A \ \sin B \ \sin C + \cos A \ \cos B = 1$ तब $\sin C =$ _____

Ans. 1

Sol. $\cos A \ \cos B + \sin A \sin B \sin C = 1$

$\Rightarrow 2 \cos A \cos B + 2 \sin A \sin B \sin C = 2$

$\Rightarrow 2 \cos A \cos B + 2 \sin A \sin B \sin C = \sin^2 A + \cos^2 A + \sin^2 B + \cos^2 B$

$\Rightarrow \cos^2 A + \cos^2 B - 2 \cos A \cos B + \sin^2 A + \sin^2 B - 2 \sin A \sin B + 2 \sin A \sin B - 2 \sin B \sin A \sin C = 0$

$\Rightarrow (\cos A - \cos B)^2 + (\sin A - \sin B)^2 + 2\sin A \sin B (1 - \sin C) = 0$

$\because$ In ΔABC; it is true if

$\cos A = \cos B$ & $\sin A = \sin B$ & $\sin C = 1$

$\Rightarrow A = B$ & $\sin C = 1$

3. Given that for a, b, c, d ∈ R, if a sec(200°) – c tan(200°) = d and b sec(200°) + d tan(200°) = c, then find the value of $\left(\frac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 20°$.

यदि a sec(200°) – c tan(200°) = d तथा b sec(200°) +d tan(200°) = c जहां a, b, c तथा d ∈ R वास्तविक संख्याऐं है, तो $\left(\frac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 20°$ का मान बताओं।

Ans. 2

Sol. Given

a sec (200º) – c tan (200º) = d

& b sec (200º) + d tan (200º) = c

$\Rightarrow$ a sec (180º + 20º) – c tan (180º + 20º) = d

& b sec (180º + 20º) + d tan (180º + 20º) = c

$\Rightarrow$ –a sec 20º – c tan 20º = d ...(i)

& –b sec 20º + d tan 20º = c ...(ii)

By (i) · d + (ii) · c; we get

$$\sec 20^\circ = \frac{d^2+c^2}{-ad-bc} \quad ...(iii)$$

By (i) · b – (ii) · a; we get

$$\tan 20^\circ = \frac{bd-ac}{-bc-ad} \quad ...(iv)$$

By (iii) & (iv)

$$\sec 20^\circ = \frac{d^2+c^2}{(bd-ac)}\cdot\tan 20^\circ$$

$$\Rightarrow \left(\frac{d^2+c^2}{bd-ac}\right)\sin 20^\circ = 1 \quad ...(v)$$

Now, by (i) · a + (ii) · b; we get

– ($a^2 + b^2$) sec 20º + tan 20º (bd – ac) = ad + bc

$\Rightarrow$ ($a^2 + b^2$) sec20º = tan 20º (bd – ac) – (ad + bc)

$$\Rightarrow \left(\frac{a^2+b^2}{bd-ac}\right)\sec 20^\circ = \tan 20^\circ - \left(\frac{ad+bc}{bd-ac}\right)$$

$$\because \tan 20^\circ = \frac{bd-ac}{-(bc+ad)}$$

$$\Rightarrow \left(\frac{a^2+b^2}{bd-ac}\right)\sec 20^\circ = \tan 20^\circ + \frac{1}{\tan 20^\circ} = \frac{\sec^2 20^\circ}{\tan 20^\circ}$$

$$\Rightarrow \left(\frac{a^2+b^2}{bd-ac}\right)\sin 20^\circ = 1 \qquad \text{...(vi)}$$

By (v) + (vi); we get

$$\left(\frac{a^2+b^2+c^2+d^2}{bd-ac}\right)\sin 20^\circ = 2$$

4. In a right angled triangle, acute angles A and B satisfy $\tan A + \tan B + \tan^2 A + \tan^2 B + \tan^3 A + \tan^3 B = 70$. Find the angle A and B in radians.

एक समकोणीय त्रिभुज ABC में कोण A तथा कोण B न्यून कोण है तथा $\tan A + \tan B + \tan^2 A + \tan^2 B + \tan^3 A + \tan^3 B = 70$ को संतुष्ट करते है तो A व B के रेडियन में मान बताओं।

Ans. $\frac{\pi}{12}$ and $\frac{5\pi}{12}$

Sol. Given, in right angled triangle ΔABC; acute angles A and B satisfy
$\tan A + \tan B + \tan^2 A + \tan^2 B + \tan^3 A + \tan^3 B = 70$
$\Rightarrow \tan A + \tan^2 A + \tan^3 A + \tan B + \tan^2 B + \tan^3 B = 70$

$\because A + B = \frac{\pi}{2}$

$\Rightarrow \tan B = \cot A$
$\Rightarrow (\tan A + \cot A) + (\tan^2 A + \cot^2 A) + (\tan^3 A + \cot^3 A) = 70$
$\Rightarrow (\tan A + \cot A) + (\tan A + \cot A)^2 - 2 + (\tan A + \cot A)^3 - 3(\tan A + \cot A) = 70$
$\Rightarrow (\tan A + \cot A)^3 + (\tan A + \cot A)^2 - 2(\tan A + \cot A) = 72$
Let $\tan A + \cot A = x$
$\Rightarrow x^3 + x^2 - 2x - 72 = 0$
$\Rightarrow (x-4)(x^2 + 5x + 18) = 0$
$\Rightarrow$ Only possible root is 4.

$$\Rightarrow \tan A + \frac{1}{\tan A} = 4$$
$$\Rightarrow \tan^2 A - 4\tan A + 1 = 0$$
$$\Rightarrow \tan A = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm 2\sqrt{3}}{2}$$
$$\Rightarrow \tan A = 2 + \sqrt{3} \text{ or } \Rightarrow \tan A = 2 - \sqrt{3}$$
$\Rightarrow A = 15^\circ$ or 75°
$$\Rightarrow A = \frac{\pi}{12} \text{ or } \frac{5\pi}{12}$$

$\Rightarrow B = \frac{5\pi}{12}$ or $\frac{\pi}{12}$

5. If the product $(\sin 1°)(\sin 3°)(\sin 5°)(\sin 7°) \ldots\ldots\ldots(\sin 89°) = \frac{1}{2^n}$, then find the value of n.

यदि $(\sin 1°)(\sin 3°)(\sin 5°)(\sin 7°) \ldots\ldots\ldots(\sin 89°) = \frac{1}{2^n}$ तो n का मान बताओ।

Ans. $\frac{89}{2}$

Sol. $(\sin 1^o)(\sin 3^o)(\sin 5^o) \ldots\ldots \sin(89^o) = \frac{1}{2^n}$

$$\Rightarrow \sum_{r=1}^{45} \sin\left(\frac{(2r-1)\pi}{180}\right) = \frac{1}{2^n}$$

$$\Rightarrow \frac{(\sin 1^o)(\sin 89^o)(\sin 2^o)(\sin 88^o)\ldots\ldots\ldots\sin 44^o \sin 46^o \cdot \sin 45^o}{\sin 2^o \ldots\ldots\ldots \sin 88^o} = \frac{1}{2^n}$$

$$\Rightarrow \frac{(\sin 1^o \cos 1^o)(\sin 2^o \cos 2^o)\ldots\ldots\ldots(\sin 44^o \cos 44^o)\cdot \sin(45^o)}{(\sin 2^o \sin 4^o \ldots\ldots\ldots \sin 88^o)} = \frac{1}{2^n}$$

$$\Rightarrow \frac{1}{2^{44}} \cdot \frac{1}{\sqrt{2}}\left(\frac{\sin 2^o \sin 4^o \ldots\ldots\ldots \sin 88^o}{\sin 2^o \sin 4^o \ldots\ldots\ldots \sin 88^o}\right) = \frac{1}{2^n}$$

$$\Rightarrow n = 44 + \frac{1}{2} = \frac{89}{2}$$

6. If $\tan\alpha = p/q$ where $\alpha = 6\beta$, α being an acute angle, prove that : $\frac{1}{2}(p\,\text{cosec}\,2\beta - q\sec 2\beta) = \sqrt{p^2+q^2}$.

यदि $\tan\alpha = p/q$ जहां $\alpha = 6\beta$ तथा α एक न्यून कोण हो, तो सिद्ध करो : $\frac{1}{2}(p\,\text{cosec}\,2\beta - q\sec 2\beta) = \sqrt{p^2+q^2}$.

Ans.

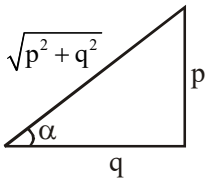
Sol. $\frac{1}{2}\left(\frac{p}{\sin 2\beta} - \frac{q}{\cos 2\beta}\right) = \frac{p\cos 2\beta - q\sin 2\beta}{2\sin 2\beta \cos 2\beta}$

Put $p = \sqrt{p^2+q^2}\ \sin\alpha$ and $q = \sqrt{p^2+q^2}\ \cos\alpha$ using $\tan\alpha = p/q$

$$= \frac{\sqrt{p^2+q^2}\sin\alpha\cos 2\beta - \sqrt{p^2+q^2}\cos\alpha\sin 2\beta}{2\sin 2\beta\cos 2\beta}$$

$$= \sqrt{p^2+q^2}\ \frac{(\sin\alpha\cos 2\beta - \cos\alpha\sin 2\beta)}{\sin 4\beta}$$

$$= \sqrt{p^2+q^2}\ \frac{\sin(\alpha-2\beta)}{\sin 4\beta}$$



$$= \sqrt{p^2+q^2}$$

7. Determine the smallest positive value of x (in degrees) for which $\tan(x+100°) = \tan(x+50°)\tan x\ \tan(x-50°)$.

x का न्यूनतम धनात्मक मान (डिग्री में) ज्ञात करें, जो $\tan(x+100°) = \tan(x+50°)\tan x\ \tan(x-50°)$ को संतुष्ट करता है।

Ans. $x = 30°$

Sol. $\tan(x+100^o) = \tan(x+50^o)\tan x\tan(x-50^o)$

$$\frac{\tan(x+100^o)}{\tan(x-50^o)} = \tan x\tan(x+50^o)$$

$$\Rightarrow \frac{\sin(x+100^o)\cos(x-50^o)}{\cos(x+100^o)\sin(x-50^o)} = \frac{\sin x\sin(x+50^o)}{\cos x\cos(x+50^o)}$$

Apply componendo–dividendo

$$\Rightarrow \frac{\sin(x+100^o)\cos(x-50^o)+\cos(x+100^o)\sin(x-50^o)}{\sin(x+100^o)\cos(x-50^o)-\cos(x+100^o)\sin(x-50^o)} = \frac{\sin x\sin(x+50^o)+\cos x\cos(x+50^o)}{\sin x\sin(x+50^o)-\cos x\cos(x+50^o)}$$

$$\Rightarrow \frac{\sin(x+100^o+x-50^o)}{\sin(x+100^o-x+50^o)} = \frac{\cos(x+50^o-x)}{-\cos(x+50^o+x)}$$

$$\Rightarrow \frac{\sin(2x+50^o)}{\sin(150^o)} = \frac{-\cos(50^o)}{\cos(2x+50^o)}$$

$$\Rightarrow \sin(2x+50^o)\cos(2x+50^o) = -\frac{1}{2}\cos 50^o$$

$$\Rightarrow \sin(4x+100^o) = -\cos(50^o)$$

$$\Rightarrow \sin(4x+100^o) = \sin\left(\frac{3\pi}{2}-50^o\right) = \sin(-40^o)$$

$\Rightarrow 4x + 100^o = n\pi - (-1)^n\, 40^o$

for smallest positive number

substitute n = 1

$4x + 100^o = \pi + 40^o = 220^o$

$\Rightarrow 4x = 120^o$

$\Rightarrow x = 30^o$

8. $\forall\, x \in R$, find the range of the function, $f(x) = \cos x\,(\sin x + \sqrt{\sin^2 x + \sin^2 \alpha}\,)$; $\alpha \in [0, \pi]$

यदि $\forall\, x \in R$ तथा $\alpha \in [0, \pi]$ तो फलन $f(x) = \cos x\,(\sin x + \sqrt{\sin^2 x + \sin^2 \alpha}\,)$ का परिसर बताओं।

Ans. $-\sqrt{1+\sin^2\alpha} \le y \le \sqrt{1+\sin^2\alpha}$

Sol. $y = f(x) = \cos x\left(\sin x + \sqrt{\sin^2 x + \sin^2 \alpha}\right)\ \forall x \in R; \alpha \in [0, \pi]$

$y\sec x - \sin x = \sqrt{\sin^2 x + \sin^2 \alpha}$

$\Rightarrow y^2 \sec^2 x + \sin^2 x - 2\tan xy = \sin^2 x + \sin^2 \alpha$

$\Rightarrow y^2\left(1 + \tan^2 x\right) - 2\tan xy = \sin^2 \alpha$

$\Rightarrow \underbrace{\tan^2 x\left(y^2\right) + \tan x(-2y) + y^2 - \sin^2 \alpha = 0}_{\downarrow\ \tan x \in (-\infty, \infty)}$

$\Rightarrow D \ge 0$ for real values of $\tan x$

$\Rightarrow 4y^2\left(1 - \left(y^2 - \sin^2 \alpha\right)\right) \ge 0$

$\Rightarrow \left(1 + \sin^2 \alpha - y^2\right) \ge 0 \Rightarrow y \in \left[-\sqrt{1+\sin^2\alpha}, \sqrt{1+\sin^2\alpha}\right]$

9. Prove that : $4\sin 27° = \left(5 + \sqrt{5}\right)^{1/2} - \left(3 - \sqrt{5}\right)^{1/2}$.

सिद्ध कीजिए : $4\sin 27° = \left(5 + \sqrt{5}\right)^{1/2} - \left(3 - \sqrt{5}\right)^{1/2}$.

Ans.

Sol. $4\sin 27^\circ = 4\sin(45^\circ - 18^\circ) = 4[\sin(45^\circ)\cos(18^\circ) - \cos(45^\circ)\sin(18^\circ)]$

$$= \frac{4}{\sqrt{2}}\left[\cos(18^\circ) - \sin(18^\circ)\right]$$

$$= \frac{4}{\sqrt{2}}\left[\sqrt{\frac{10+2\sqrt{5}}{16}} - \left(\frac{\sqrt{5}-1}{4}\right)\right]$$

$$= \frac{1}{\sqrt{2}}\left[\sqrt{10+2\sqrt{5}} - \left(\sqrt{5}-1\right)\right]$$

$$= \left[\sqrt{5+\sqrt{5}} - \left(\frac{\sqrt{\left(\sqrt{5}-1\right)^2}}{\sqrt{2}}\right)\right]$$

$$= \left(5+\sqrt{5}\right)^{\frac{1}{2}}\left(3-\sqrt{5}\right)^{\frac{1}{2}}$$

10. Let $x_1 = \prod_{r=1}^{5}\cos\frac{r\pi}{11}$ and $x_2 = \sum_{r=1}^{5}\cos\frac{r\pi}{11}$, then show that $x_1 \cdot x_2 = \frac{1}{64}\left(\operatorname{cosec}\frac{\pi}{22} - 1\right)$, where Π denotes the continued product.

यदि $x_1 = \prod_{r=1}^{5}\cos\frac{r\pi}{11}$ तथा $x_2 = \sum_{r=1}^{5}\cos\frac{r\pi}{11}$, तो सिद्ध करो कि $x_1 \cdot x_2 = \frac{1}{64}\left(\operatorname{cosec}\frac{\pi}{22} - 1\right)$, जहां Π सतत गुणन (continued product) है।

Ans.

Sol. $x_1 = \prod_{r=1}^{5}\cos\left(\frac{r\pi}{11}\right) = \frac{\cos\pi}{11}\cdot\frac{\cos 2\pi}{11}\cdot\frac{\cos 3\pi}{11}\cdot\frac{\cos 4\pi}{11}\cdot\frac{\cos 5\pi}{11}$

$$\Rightarrow x_1 = \cos\frac{\pi}{11}\cdot\cos\frac{2\pi}{11}\cdot\cos\frac{4\pi}{11}\cdot\cos\frac{8\pi}{11}\cdot\cos\frac{5\pi}{11}$$

$$\Rightarrow x_1 = \frac{1}{16}\sin\left(\frac{16\pi}{11}\right)\cdot\frac{1}{\sin\left(\frac{\pi}{11}\right)}\cdot\cos\left(\frac{5\pi}{11}\right)$$

$$\Rightarrow x_1 = \frac{1}{16}\frac{\sin\left(\frac{5\pi}{11}\right)\cdot\cos\left(\frac{5\pi}{11}\right)}{\sin\left(\frac{\pi}{11}\right)} = \frac{1}{32}\frac{\sin\left(\frac{10\pi}{11}\right)}{\sin\left(\frac{\pi}{11}\right)} = \frac{1}{32}$$

$$x_2 = \sum_{r=1}^{5} \cos\left(\frac{r\pi}{11}\right) = \cos\left(\frac{\pi}{11}\right) + \cos\left(\frac{2\pi}{11}\right) + ... + \cos\left(\frac{5\pi}{11}\right)$$

$$= \cos\left(\frac{\frac{\pi}{11} + \frac{5\pi}{11}}{2}\right) \frac{\sin\left(\frac{5\pi}{22}\right)}{\sin\left(\frac{\pi}{22}\right)}$$

$$\Rightarrow x_2 = \frac{\cos\left(\frac{6\pi}{22}\right) \cdot \sin\left(\frac{5\pi}{22}\right)}{\sin\left(\frac{\pi}{22}\right)}$$

$$\Rightarrow x_2 = \frac{\sin\left(\frac{5\pi}{22} + \frac{6\pi}{22}\right) + \sin\left(\frac{5\pi}{22} - \frac{6\pi}{22}\right)}{2\sin\left(\frac{\pi}{22}\right)}$$

$$\Rightarrow x_2 = \frac{1 - \sin\left(\frac{\pi}{22}\right)}{2\sin\left(\frac{\pi}{22}\right)} = \frac{1}{2}\left[\operatorname{cosec}\left(\frac{\pi}{22}\right) - 1\right]$$

$$\Rightarrow x_1 \cdot x_2 = \frac{1}{64}\left(\operatorname{cosec}\left(\frac{\pi}{22}\right) - 1\right)$$

11. If $(1 + \sin t)(1 + \cos t) = \frac{5}{4}$. Find the value of $(1 - \sin t)(1 - \cos t)$.

यदि $(1 + \sin t)(1 + \cos t) = \frac{5}{4}$ तो $(1 - \sin t)(1 - \cos t)$ का मान बताओं।

Ans. $\frac{13}{4} - \sqrt{10}$

Sol. Given

$$(1 + \sin t)(1 + \cos t) = \frac{5}{4}$$

$$\Rightarrow 1 + \sin t + \cos t + \sin t \cos t = \frac{5}{4}$$

$$\Rightarrow 2+2\sin t\cos t+2\left(\sin t+\cos t\right)=\frac{5}{2}$$

$$\Rightarrow 1+2\sin t\cos t+2\left(\sin t+\cos t\right)=\frac{3}{2}$$

$$\Rightarrow \sin^2 t+\cos^2 t+2\sin t\cos t+2\left(\sin t+\cos t\right)=\frac{3}{2}$$

$$\Rightarrow \left(\sin t+\cos t\right)^2+2\left(\sin t+\cos t\right)-\frac{3}{2}=0$$

$$\Rightarrow \sin t+\cos t=\frac{-2\pm\sqrt{4+6}}{2}$$

$$\because \sin t+\cos t\in\left[-\sqrt{2},\sqrt{2}\right]$$

$$\Rightarrow \sin t+\cos t=\frac{\sqrt{10}}{2}-1$$

$\therefore (1-\sin t)(1-\cos t)=1-(\sin t+\cos t)+\sin t\cos t$

$$=\frac{5}{4}-2\left(\sin t+\cos t\right)=\frac{5}{4}-\sqrt{10}+2$$

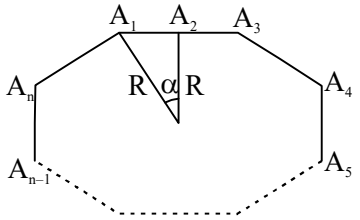
$$=\frac{13}{4}-\sqrt{10}$$

12. Let $A_1, A_2, A_3 \ldots\ldots\ldots A_n$ are the vertices of a regular n sided polygon inscribed in a circle of radius R. If $(A_1A_2)^2+(A_1A_3)^2+\ldots\ldots\ldots+(A_1A_n)^2=14R^2$, find the number of sides in the polygon.

यदि $A_1, A_2, A_3 \ldots\ldots\ldots A_n$, n भुजाओं वाले समबाहुभुज (Regular polygon) के शीर्ष है, जिसके परिवृत की त्रिज्या R है। यदि $(A_1A_2)^2+(A_1A_3)^2+\ldots\ldots\ldots+(A_1A_n)^2=14R^2$ तो बहुभुज में भुजाओं की संख्या बताओं।

Ans. n = 7

Sol. $\alpha=\frac{\pi}{n}$



$A_1A_2=2R\sin\alpha$

$A_1A_3=2R\sin 2\alpha$

$A_1A_4=2R\sin 3\alpha$

$\vdots$

$A_1A_n=2R\sin(n-1)\alpha$

$$\Rightarrow \left(A_1A_2\right)^2+\left(A_1A_3\right)^2+\ldots\ldots\ldots+\left(A_1A_n\right)^2$$

$$=4R^2\left[\sin^2\alpha+\sin^2 2\alpha+\ldots\ldots\ldots+\sin^2\left(n-1\right)\alpha\right]$$

$$= 2R^2\left[1-\cos 2\alpha+1-\cos 4\alpha+1-\cos 6\alpha+.....+1-\cos 2(n-1)\alpha\right]$$

$$= 2R^2\left[(n-1)-\left[\cos(2\alpha)+\cos(4\alpha)+\cos 6\alpha+.....+\cos 2(n-1)\alpha\right]\right]$$

$$= 2R^2\left[(n-1)-\frac{\sin\left(\frac{(n-1)2\alpha}{2}\right)}{\sin\left(\frac{2\alpha}{2}\right)}\cdot\cos\left(\frac{2n\alpha}{2}\right)\right]$$

$$= 2R^2\left[(n-1)-\frac{\sin((n-1)\alpha)}{\sin\alpha}\cdot\cos(n\alpha)\right]$$

$$= 2R^2\,[(n-1)+1] = 14R^2 \quad \left\{\because \alpha = \frac{\pi}{n}\right\} = 2nR^2$$

$$\Rightarrow n = 7$$

13. Let $\theta \in R$ and $\sum_{k=2}^{\infty} \sin\left(2^k\theta\right) = a$. Find the value $\sum_{k=0}^{\infty} \left[\cot^3\left(2^k\theta\right)-\cot\left(2^k\theta\right)\right]\sin^4\left(2^k\theta\right)$ in terms of 'a'.

यदि $\sum_{k=2}^{\infty} \sin\left(2^k\theta\right) = a$ जहां $\theta \in R$ तो व्यंजक $\sum_{k=0}^{\infty} \left[\cot^3\left(2^k\theta\right)-\cot\left(2^k\theta\right)\right]\sin^4\left(2^k\theta\right)$ का मान 'a' के पदों में बताओं।

Ans. $\frac{a}{4}$

Sol. Given

$$\theta \in R \ \& \ \sum_{k=2} \sin\left(2^k\theta\right) = a$$

$$\sum_{k=0}^{\infty}\left[\cot^3\left(2^k\theta\right)-\cot\left(2^k\theta\right)\right]\sin^4\left(2^k\theta\right)$$

$$= \sum_{k=0}^{\infty}\cot\left(2^k\theta\right)\left(\cot^2\left(2^k\theta\right)-1\right)\cdot\sin^4\left(2^k\theta\right)$$

$$= \sum_{k=0}^{\infty}\cot\left(2^k\theta\right)\cdot\frac{\left(\cos^2\left(2^k\theta\right)-\sin^2\left(2^k\theta\right)\right)}{\sin^2\left(2^k\theta\right)}\cdot\sin^4\left(2^k\theta\right)$$

$$= \sum_{k=0}^{\infty} \cos\left(2^k\theta\right)\cdot\sin\left(2^k\theta\right)\cdot\cos\left(2^{k+1}\theta\right)$$

$$= \frac{1}{4}\sum_{k=0}^{\infty}\sin\left(2^{k+2}\theta\right) = \frac{1}{4}\sum_{k=2}^{\infty}\sin\left(2^k\theta\right)$$

$$= \frac{a}{4}$$

14. If x and y are real number such that $x^2 + 2xy - y^2 = 6$, find the minimum value of $(x^2 + y^2)^2$.

यदि वास्तविक संख्याऐं x तथा y सम्बन्ध $x^2 + 2xy - y^2 = 6$ को संतुष्ट करती है तो $(x^2 + y^2)^2$ का न्यूनतम मान बताओ।

Ans. 18

Sol. $x^2 + 2xy - y^2 = 6$

$(x^2 - y^2)^2 = (6 - 2xy)^2$

$\Rightarrow x^4 + y^4 - 2x^2y^2 + 4x^2y^2 = 36 + 4x^2y^2 + 4x^2y^2 - 24xy$

$\Rightarrow (x^2 + y^2)^2 = 8x^2y^2 - 24xy + 36$

$\Rightarrow (x^2 + y^2)^2 = 4\,(2x^2y^2 - 6xy + 9)$

$$\underbrace{2(xy)^2 - 6(xy) + 9 = 0}_{\text{quad in xy}}$$

$$\text{Minimum value is } \left(\frac{-D}{4a}\right) = -\left(\frac{36-72}{8}\right) = \frac{36}{8}$$

$$\Rightarrow \text{Minimum value of } (x^2 + y^2)^2 \text{ is } 4\times\frac{36}{8} = 18$$

15. Find the exact value of $\tan^2\frac{\pi}{16} + \tan^2\frac{3\pi}{16} + \tan^2\frac{5\pi}{16} + \tan^2\frac{7\pi}{16}$

$\tan^2\frac{\pi}{16} + \tan^2\frac{3\pi}{16} + \tan^2\frac{5\pi}{16} + \tan^2\frac{7\pi}{16}$ का आंकिक मान बताओं।

Ans. 28

Sol. $\tan^2\frac{\pi}{16} + \tan^2\frac{3\pi}{16} + \tan^2\frac{5\pi}{16} + \tan^2\frac{7\pi}{16}$

$$= \tan^2\frac{\pi}{16} + \tan^2\frac{7\pi}{16} + \tan^2\frac{3\pi}{16} + \tan^2\frac{5\pi}{16}$$

$$= \tan^2\frac{\pi}{16} + \tan^2\left(\frac{\pi}{2} - \frac{\pi}{16}\right) + \tan^2\left(\frac{3\pi}{16}\right) + \tan^2\left(\frac{\pi}{2} - \frac{3\pi}{16}\right)$$

$$= \tan^2\frac{\pi}{16} + \cot^2\frac{\pi}{16} + \tan^2\frac{3\pi}{16} + \cot^2\frac{3\pi}{16}$$

$$= \left(\tan\frac{\pi}{16} + \cot\frac{\pi}{16}\right)^2 + \left(\tan\frac{3\pi}{16} + \cot\frac{3\pi}{16}\right)^2 - 4$$

$$= \left(2\,\text{cosec}\left(\frac{2\pi}{16}\right)\right)^2 + \left(2\,\text{cosec}\left(\frac{3\pi}{18}\right)\right)^2 - 4$$

$$= 4\left[\text{cosec}^2\frac{\pi}{8} + \text{cosec}^2\frac{3\pi}{8}\right] - 4$$

$$= 4\left[\text{cosec}^2\frac{\pi}{8} + \text{cosec}^2\left(\frac{\pi}{2} - \frac{3\pi}{8}\right)\right] - 4$$

$$= 4\left[\text{cosec}^2\frac{\pi}{8} + \text{sec}^2\frac{\pi}{8}\right] - 4$$

$$= 4\left[\frac{1}{\sin^2\frac{\pi}{8}\cdot\cos^2\frac{\pi}{8}}\right] - 4$$

$$= 16\left[\frac{1}{\sin^2\frac{\pi}{4}}\right] - 4$$

$= 32 - 4 = 28$

16. If 'θ' is eliminated from the equations $\cos\theta - \sin\theta = b$ and $\cos 3\theta + \sin 3\theta = a$, find the eliminant.

यदि समीकरण $\cos\theta - \sin\theta = b$ तथा $\cos 3\theta + \sin 3\theta = a$ में से 'θ' विलुप्त किया जाये, तो a तथा b में प्रतिबन्ध ज्ञात किजिए।

Ans. $a = 3b - 2b^3$

Sol. $\cos\theta - \sin\theta = b$

$\cos 3\theta + \sin 3\theta = a$

$\Rightarrow a = \cos 3\theta + \sin 3\theta$

$= 4\cos^3\theta - 3\cos\theta + 3\sin\theta - 4\sin^3\theta$

$= 4\,[\cos^3\theta - \sin^3\theta] + 3\,(\sin\theta - \cos\theta)$

$= 4\,[(\cos\theta - \sin\theta)^3 + 3\sin\theta\cos\theta\,(\cos\theta - \sin\theta)] + (\sin\theta - \cos\theta)$

$= 4\,[b^3 + 3\sin\theta\cos\theta\,(b)] + - 3b$

$\Rightarrow a = 4b^3 - 3b + 12b \sin\theta \cos\theta$

$\because \cos\theta - \sin\theta = b$

$\Rightarrow \cos^2\theta + \sin^2\theta - \sin 2\theta = b^2$

$$\Rightarrow a = 4b^3 - 3b + 12b\left(\frac{1-b^2}{2}\right)$$

$$\Rightarrow a = 4b^3 - 3b + 6b - 6b^3$$

$$\Rightarrow a = 3b - 2b^3$$

17. Show that elliminating x and y from the equations, $\sin x + \sin y = a$; $\cos x + \cos y = b$ and $\tan x + \tan y = c$ gives

$$\frac{8ab}{\left(a^2+b^2\right)^2 - 4a^2} = c.$$

सिद्ध करिये कि समीकरण $\sin x + \sin y = a$; $\cos x + \cos y = b$ तथा $\tan x + \tan y = c$ से x व y विलुप्त करने पर $\frac{8ab}{\left(a^2+b^2\right)^2 - 4a^2}$ = c प्राप्त होगा।

Ans.

Sol. Given

$\sin x + \sin y = a$

$\cos x + \cos y = b$

$\tan x + \tan y = c$

$\Rightarrow \sin^2 x + \sin^2 y + 2 \sin x \sin y = a^2$ ...(i)

$\cos^2 x + \cos^2 y + 2 \cos x \cos y = b^2$ ...(ii)

By (i) + (ii); we get

$1 + 1 + 2 (\sin x \sin y + \cos x \cos y) = a^2 + b^2$

$$\Rightarrow a^2 + b^2 = 2 + 2\cos(x - y) = 4\cos^2\left(\frac{x-y}{2}\right)$$

Now $\tan x + \tan y = c$

$$\Rightarrow \frac{\sin x \cos y + \cos x \sin y}{\cos x \cos y} = c$$

$$\because a = \sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$

$$\Rightarrow \left(a^2 + b^2\right)^2 - 4a^2 = 16\cos^4\left(\frac{x-y}{2}\right) - 16\sin^2\left(\frac{x+y}{2}\right)\cos^2\left(\frac{x-y}{2}\right)$$

$$= 16\cos^2\left(\frac{x-y}{2}\right)\left(\cos^2\left(\frac{x-y}{2}\right) - \sin^2\left(\frac{x+y}{2}\right)\right)$$

$$\left(a^2+b^2\right)^2-4a^2=16\cos^2\left(\frac{x-y}{2}\right)\cos x\cos y \qquad ...(i)$$

Now

$2ab = 2 (\sin x + \sin y) (\cos x + \cos y)$

$= \sin 2x + \sin 2y + 2 \sin (x + y)$

$= 2 \sin (x + y) \cos (x - y) + 2 \sin (x + y)$

$$= 4\sin\left(x+y\right)\cos^2\left(\frac{x-y}{2}\right)$$

$$\Rightarrow 8ab=16\sin\left(x+y\right)\cos^2\left(\frac{x-y}{2}\right) \qquad ...(ii)$$

By (i) & (ii)

$$\frac{8ab}{\left(a^2+b^2\right)^2-4a^2}=\frac{16\sin\left(x+y\right)\cos^2\left(\frac{x-y}{2}\right)}{16\cos^2\left(\frac{x-y}{2}\right)\cos x\cos y}$$

$= \tan x + \tan y = c$

18. Given that $3 \sin x + 4 \cos x = 5$ where $x \in (0, \pi/2)$. Find the value of $2 \sin x + \cos x + 4 \tan x$.

दिया है कि $3 \sin x + 4 \cos x = 5$ जहां $x \in (0, \pi/2)$ तो $2 \sin x + \cos x + 4 \tan x$ का मान बताओं।

Ans. 5

Sol. Given

$$3 \sin x + 4 \cos x = 5;\ x \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow 3\sin x = 5 - 4\cos x$$

Square both sides

$\Rightarrow 9 \sin^2 x = (5 - 4 \cos x)^2 = 25 + 16 \cos^2 x - 40 \cos x$

$\Rightarrow 9 (1 - \cos^2 x) = 25 + 16 \cos^2 x - 40 \cos x$

$\Rightarrow 25 \cos^2 x - 40 \cos x + 16 = 0$

$$\Rightarrow \cos x = \frac{40 \pm \sqrt{1600 - 1600}}{50}$$

$$\Rightarrow \cos x = \frac{4}{5}$$

$$\Rightarrow \sin x = \frac{3}{5}$$

$$\Rightarrow \tan x = \frac{3}{4}$$

$$\Rightarrow 2\sin x + \cos x + 4\tan x = \frac{6}{5} + \frac{4}{5} + 3 = 5$$

19. Prove that the triangle ABC is equilateral iff, $\cot A + \cot B + \cot C = \sqrt{3}$.

यदि त्रिभुज ABC एक समबाहु त्रिभुज हो तो यदि और केवल यदि कि $\cot A + \cot B + \cot C = \sqrt{3}$.

Ans.

Sol. If ΔABC is equilateral then $A = B = C = \frac{\pi}{3}$

$$\therefore \cot\frac{\pi}{3} + \cot\frac{\pi}{3} + \cot\frac{\pi}{3} = \sqrt{3}$$

Now if $\cot A + \cot B + \cot C = \sqrt{3}$
sq. both sides we get
$\Sigma \cot^2 A + 2\Sigma \cot A \cot B = 3(1)$
we know that $\Sigma \cot A \cot B = 1$
$\therefore \Sigma \cot^2 A + 2\Sigma \cot A \cot B = 3\ (\Sigma \cot A \cot B)$
$\Rightarrow \Sigma \cot^2 A - \Sigma \cot A \cot B = 0$
$\Rightarrow (\cot A - \cot B)^2 + (\cot B - \cot C)^2 + (\cot C - \cot A)^2 = 0$
$\Rightarrow \cot A = \cot B = \cot C \Rightarrow A = B = C$
$\Rightarrow \Delta ABC$ is equilateral.

20. If $A+B+C = \pi$; prove that $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1$.

यदि $A+B+C = \pi$; तो सिद्ध करो कि $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2} \geq 1$.

Ans.

Sol. Given $A + B + C = \pi$

$$\Rightarrow \frac{A}{2} + \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2}$$

$$\Rightarrow \tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = I$$

$$\because \left(\tan\frac{A}{2} - -\tan\frac{B}{2}\right)^2 \geq 0 \qquad ...(i)$$

$$\left(\tan\frac{B}{2} - \tan\frac{C}{2}\right)^2 \geq 0 \qquad ...(ii)$$

$$\left(\tan\frac{C}{2}-\tan\frac{A}{2}\right)^2 \geq 0 \qquad \text{...(iii)}$$

By (i) + (ii) + (iii); we get

$$2\left(\tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2}\right) \geq 2\sum\tan\frac{A}{2}\cdot\tan\frac{B}{2}$$

$$\Rightarrow \tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2} \geq 1$$

21. If $\tan\left(\frac{\pi}{4}+\frac{y}{2}\right)=\tan^3\left(\frac{\pi}{4}+\frac{x}{2}\right)$, prove that $\frac{\sin y}{\sin x}=\frac{3+\sin^2 x}{1+3\sin^2 x}$.

यदि $\tan\left(\frac{\pi}{4}+\frac{y}{2}\right)=\tan^3\left(\frac{\pi}{4}+\frac{x}{2}\right)$, तो सिद्ध करो कि $\frac{\sin y}{\sin x}=\frac{3+\sin^2 x}{1+3\sin^2 x}$.

Ans.

Sol. $$\tan\left(\frac{\pi}{4}+\frac{y}{2}\right)=\sqrt{\frac{1-\cos\left(\frac{\pi}{2}+y\right)}{1+\cos\left(\frac{\pi}{2}+y\right)}}==\sqrt{\frac{1+\sin y}{1-\sin y}}$$

$$\Rightarrow \frac{1+\sin y}{1-\sin y}=\left(\frac{1+\sin x}{1-\sin x}\right)^3 \Rightarrow \frac{\sin y+1}{\sin y-1}=\left(\frac{\sin x+1}{\sin x-1}\right)^3$$

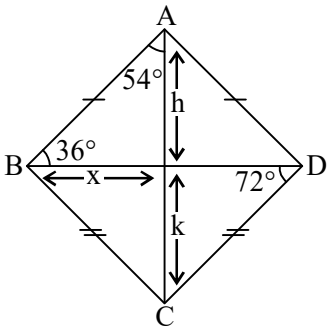
Now apply C and D

$$\Rightarrow \frac{\sin y}{1}=\frac{2\sin^3 x+6\sin x}{2+6\sin^2 x}=\frac{\sin x\left(3+\sin^2 x\right)}{1+3\sin^2 x}$$

22. In a kite ABCD, AB = AD and CB = CD. If $\angle A = 108°$ and $\angle C = 36°$ then the ratio of the area of ΔABD to the area of ΔCBD can be written in the form $\frac{a-b\tan^2 36°}{c}$ where a, b and c are relatively prime positive integers.determine the ordered triple (a, b, c). Also find the numerical value of the ratio. एक पतंग (kite) ABCD में AB = AD तथा CB = CD है। यदि $\angle A = 108°$ तथा $\angle C = 36°$ तो ΔABD व ΔCBD के क्षेत्रफलों का अनुपात $\frac{a-b\tan^2 36°}{c}$ के रूप में लिखा जा सकता है जहाँ a, b तथा c सापेक्षिक अभाज्य (relatively prime) संख्याऐं हो, तो a, b, c का मान बताओं तथा उपरोक्त अनुपात का आंकिक मान भी बताओं।

Ans. $(1,1,2);\ \sqrt{5}-2$

Sol. Since ΔABD and ΔCBD have a common base.
$\Rightarrow$ ratio of their areas equals the ratio of their heights.



$\because \tan 36^\circ = \dfrac{h}{x}$ & $\tan 72^\circ = \dfrac{k}{x}$

$$\Rightarrow \frac{h}{k} = \frac{x\tan 36^\circ}{x\tan 72^\circ} = \frac{\tan 36^\circ}{2\tan 36^\circ}\left(1-\tan^2 36^\circ\right)$$

$$\Rightarrow \frac{h}{k} = \frac{1-\tan^2 36^\circ}{2}$$

$\Rightarrow a = 1, b = 1, c = 2$

$$\text{Now } \frac{1-\tan^2 36^\circ}{2} = \frac{\cos^2 36^\circ - \sin^2 36^\circ}{2\cos^2 36^\circ}$$

$$= \frac{\cos 72^\circ}{2\cos^2 36^\circ}$$

$$= \frac{\cos 72^\circ}{1+\cos 72^\circ} = \frac{\sin 18^\circ}{1+\sin 18^\circ} = \frac{\sqrt{5}-1}{4+\sqrt{5}-1}$$

$$= \frac{\sqrt{5}-1}{\sqrt{5}+3} = \frac{\left(\sqrt{5}-1\right)\left(\sqrt{5}-3\right)}{-4}$$

$$= \sqrt{5}-2$$

23. If $4\sin x \cdot \cos y + 2\sin x + 2\cos y + 1 = 0$ where $x, y \in [0, 2\pi]$ find the largest possible value of the sum $(x + y)$.

यदि $4\sin x \cdot \cos y + 2\sin x + 2\cos y + 1 = 0$ जहाँ $x, y \in [0, 2\pi]$ तो $(x + y)$ का अधिकतम सम्भव मान ज्ञात कीजिए।

Ans. $\dfrac{19\pi}{6}$

Sol. $4\sin x\cos y + 2\sin x + 2\cos y + 1 = 0;\ x, y \in [0, 2\pi]$
$2[\sin x + \cos y + 2\sin x\cos y] + 1 = 0$
$\Rightarrow (2\sin x + 1)(2\cos y + 1) = 0$

$$\Rightarrow \sin x = \frac{-1}{2} \text{ or } \cos y = \frac{-1}{2}$$

$$\Rightarrow x = \frac{7\pi}{6}, \frac{11\pi}{6} \text{ or } y = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$\therefore$ Largest possible value of $x + y$ is $\frac{11\pi}{6} + \frac{4\pi}{3} = \frac{19\pi}{6}$

24. Let $x > 1, y > 1$ and $(\ell n\, x)^2 + (\ell n\, y)^2 = \ell n\, x^2 + \ell n\, y^2$, then find the maximum value of $x^{\ell n\, y}$.

माना $x > 1, y > 1$ तथा $(\ell n\, x)^2 + (\ell n\, y)^2 = \ell n\, x^2 + \ell n\, y^2$ तो $x^{\ell n\, y}$ का अधिकतम मान ज्ञात कीजिए।

Ans. e^4

Sol. $(\ell nx)^2 + (\ell ny)^2 = \ell nx^2 + \ell ny^2$

$\Rightarrow (\ell nx)^2 + (\ell n(y))^2 = 2(\ell nx + \ell ny) \qquad \{\because x > 1, y > 1\}$

$\Rightarrow (\ell nx - 1)^2 + (\ell ny - 1)^2 = 2$

$\Rightarrow \left(\frac{\ell nx - 1}{\sqrt{2}}\right)^2 + \left(\frac{\ell ny - 1}{\sqrt{2}}\right)^2 = 1$

Let $\frac{\ell nx - 1}{\sqrt{2}} = \sin\theta$ & $\frac{\ell ny - 1}{\sqrt{2}} = \cos\theta$

$\Rightarrow \ell nx = 1 + \sqrt{2}\sin\theta$ & $\Rightarrow \ell ny = 1 + \sqrt{2}\cos\theta$

$\Rightarrow x^{\ell ny} = e^{(1+\sqrt{2}\sin\theta)(1+\sqrt{2}\cos\theta)}$

$= e^{1+2\sin\theta\cos\theta+\sqrt{2}\cos\theta+\sqrt{2}\sin\theta}$

$\because \cos\theta + \sin\theta \in \left[-\sqrt{2}, \sqrt{2}\right]$

$\Rightarrow$ Max value of $x^{\ell ny} = e^{1+1+2} = e^4$

25. Prove that $1 + \cot\theta \le \cot\frac{\theta}{2}$ for $0 < \theta < \pi$. Find θ when equality sign holds.

सिद्ध कीजिए $1 + \cot\theta \le \cot\frac{\theta}{2}$ for $0 < \theta < \pi$. θ का मान ज्ञात करों जब बराबर का चिन्ह है।

Ans. $\frac{\pi}{2}$

Sol. $1 + \cot\theta - \cot\frac{\theta}{2}$

$$= 1 + \frac{\cot^2\frac{\theta}{2} - 1}{2\cot\frac{\theta}{2}} - \cot\frac{\theta}{2}$$

$$= \frac{2\cot\frac{\theta}{2} - 2\cot^2\frac{\theta}{2} + \cot^2\frac{\theta}{2} - 1}{2\cot\frac{\theta}{2}}$$

$$= -\frac{\left(\cot\frac{\theta}{2} - 1\right)^2}{2\cot\frac{\theta}{2}} \le 0. \text{ for } 0 < \theta < \pi.$$

$$\Rightarrow 1 + \cot\theta \le \cot\frac{\theta}{2}$$

Equality holds if

$$\cot\frac{\theta}{2} = 1$$

$$\Rightarrow \frac{\theta}{2} = \frac{\pi}{4}$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

26. For all θ in $\left[0, \frac{\pi}{2}\right]$, show that $\cos(\sin\theta) > \sin(\cos\theta)$.

सिद्ध कीजिए $\cos(\sin\theta) > \sin(\cos\theta)$, $\forall\ \theta \in \left[0, \frac{\pi}{2}\right]$

Ans.

Sol. $\because \cos\theta + \sin\theta = \sqrt{2}\left[\frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta\right]$

$$= \sqrt{2}\left[\sin\left(\frac{\pi}{4} + \theta\right)\right]$$

$$\Rightarrow \cos\theta + \sin\theta \le \sqrt{2} < \frac{\pi}{2}$$

$\Rightarrow \cos\theta < \frac{\pi}{2} - \sin\theta$

As $\theta \in \left[0, \frac{\pi}{2}\right]$ in which sinθ increases.

$\Rightarrow \sin(\cos\theta) < \sin\left(\frac{\pi}{2} - \sin\theta\right)$

$\Rightarrow \cos(\sin\theta) > \sin(\cos\theta) \forall \theta \in \left[0, \frac{\pi}{2}\right]$

27. If $x + y + z = xyz$

Prove that $\frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{4xyz}{(1-x^2)(1-y^2)(1-z^2)}$.

यदि $x + y + z = xyz$

सिद्ध कीजिए $\frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{4xyz}{(1-x^2)(1-y^2)(1-z^2)}$.

Ans.

Sol. $x + y + z = xyz$

$$\frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{x(1-y^2)(1-z^2) + y(1-x^2)(1-z^2) + z(1-x^2)(1-y^2)}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{x\left[1-y^2-z^2+y^2z^2\right] + y\left[1-x^2-z^2+x^2z^2\right] + z\left[1-x^2-y^2+x^2y^2\right]}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{x+y+z-xy^2-xz^2+xy^2z^2-x^2y-z^2y+x^2yz^2+x^2y^2z-x^2z-y^2z}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{x+y+z-xy^2+xz^2y^2+x^2y^2z-y^2z-x^2y+x^2yz^2-x^2z-xz^2-z^2y}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{x+y+z-y^2(x+z)+xyz(yz+xy)-x^2(y+z)+z^2(x^2y-x-y)}{(1-x^2)(1-y^2)(1-z^2)}$$

$$= \frac{xyz - y^2(xyz - y) + xyz\left(xy^2z - y\right) - x^2(xyz - x) + z^2\left(x^2y + z - xyz\right)}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{xyz\left[1 - y^2 + xy^2z - x^2 - z^2\right] + \left[y^3 - xy^3z + x^3 + x^2yz^2 + z^3\right]}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{xyz\left[1 + xy^2z - \left(x^2 + y^2 + z^2\right)\right] + \left[y^3 + x^3 + z^3 - xy^3z + x^2yz^2\right]}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{xyz\left[1 + xy^2z - \left[x^2y^2z^2 - 2xy - 2xz - 2yz\right]\right] + \left[x^3y^3z^3 - 3x^2y^2z - 3x^2yz^2 - 3xy^2z^2 + 3xyz\right] - xy^3z + x^2yz^2}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{xyz + x^2y^3z^2 - x^3y^3z^3 + 2x^2y^2z + 2x^2yz^2 + 2y^2z^2x + x^3y^3z^3 - 3x^2y^2z - 3x^2yz^2 - 3xy^2z^2 + 3xyz - xy^3z + x^2yz^2}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz + x^2y^3z^2 - x^2y^2z - x^2yz^2 - xy^2z^2 - xy^3z + x^2yz^2}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz + x^2y^3z^2 - x^2y^2z - xy^2z^2 - xy^3z}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz + xyz\left[xy^2z - xy - yz - y^2\right]}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz + xyz\left[y(x + y + z) - xy - yz - y^2\right]}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

$$= \frac{4xyz}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

Method–IInd

$x + y + z = xyz$

Let $x = \tan A$, $y = \tan B$, $z = \tan C$

$\Rightarrow A + B + C = \pi$

$\Rightarrow (2A + 2B + 2C) = 2\pi$

$\Rightarrow \tan(2A + 2B) = \tan(2\pi - 2C)$

$$\frac{\tan 2A + \tan 2B}{1 - \tan 2A \tan 2B} = -\tan 2C$$

$$\Rightarrow \tan 2A + \tan 2B = -\tan 2C + \tan 2A \tan 2B \tan 2C$$

$$\Rightarrow \frac{2\tan A}{1-\tan^2 A} + \frac{2\tan B}{1-\tan^2 B} + \frac{2\tan C}{1-\tan^2 C} = \frac{8\tan A \tan B \tan C}{\left(1-\tan^2 A\right)\left(1-\tan^2 B\right)\left(1-\tan^2 C\right)}$$

$$\Rightarrow \frac{x}{1-x^2} + \frac{y}{1-y^2} + \frac{z}{1-z^2} = \frac{4xyz}{\left(1-x^2\right)\left(1-y^2\right)\left(1-z^2\right)}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 28 to 30

If $\cos\alpha + \cos\beta = a$ and $\sin\alpha + \sin\beta = b$ and θ is arithmatic mean between α and β, then

$$\sin 2\theta + \cos 2\theta = 1 + \frac{nb(a-b)}{a^2+b^2}$$

where n is some integer then answer the following questions :

यदि $\cos\alpha + \cos\beta = a$ तथा $\sin\alpha + \sin\beta = b$ एवं θ, α तथा β के मध्य समान्तर माध्य है, तो

$$\sin 2\theta + \cos 2\theta = 1 + \frac{nb(a-b)}{a^2+b^2}$$

जहाँ n कोई पूर्णांक है, तो निम्न प्रश्नों के उत्तर दीजिए।

28. The value of n is :

n का मान है :

(A) 0 (B) 1 (C) 2 (D) –2

Ans. C

Sol. Given $\cos\alpha + \cos\beta = a$, $\sin\alpha + \sin\beta = b$ & $\frac{\alpha+\beta}{2} = \theta$

$$\Rightarrow 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = a \quad ...(i)$$

$$2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = b \quad ...(ii)$$

By (i) ÷ (ii); we get

$$\frac{a}{b} = \cot\theta \Rightarrow \tan\theta = \frac{b}{a}$$

$$\Rightarrow \sin\theta = \frac{b}{\sqrt{a^2+b^2}}, \cos\theta = \frac{a}{\sqrt{a^2+b^2}}$$

$\sin 2\theta + \cos 2\theta = 2\sin\theta\cos\theta + 2\cos^2\theta - 1$

$$= \frac{2ab}{a^2+b^2} + \frac{2a^2}{a^2+b^2} - 1$$

$$= \frac{a^2-b^2+2ab}{a^2+b^2} = 1 + \frac{2ab-2b^2}{a^2+b^2}$$

$$= 1 + \frac{2b(a-b)}{a^2+b^2}$$

$\Rightarrow n = 2$

29. If for n obtained in above question, $\sin^n A = x$, then $\sin A \sin 2A \sin 3A \sin 4A$ is a polynomial in x, of degree :
यदि ऊपर दिये गये प्रश्न में निकाले गये n के मान के लिए $\sin^n A = x$ हो, तो $\sin A \sin 2A \sin 3A \sin 4A$, x में एक बहुपद है जिसकी घात है :
(A) 5 (B) 6 (C) 7 (D) 8

Ans. A

Sol. Now $\sin^2 A = x$
$\sin A \sin 2A \sin 3A \sin 4A$
$= \sin A \cdot (3\sin A - 4\sin^3 A) \cdot \sin^2 2A \cdot \cos 2A$
$= (3\sin^2 A - 4\sin^4 A) \cdot 2 \cdot 4 \cdot \sin^2 A \cdot \cos^2 A \cdot (-2\sin^2 A + 1)$
$= (3x - 4x^2) \cdot 8 \cdot x \cdot (1-x)(1-2x)$
$= 8(3x - 4x^2) \cdot (x - x^2)(1-2x)$
$= 8(3x - 4x^2) \cdot (x - 3x^2 + 2x^3)$
degree is 5.

30. If degree of polynomial obtained in previous questins is p and $(p-5) + \sin x, \cos x, \tan x$ are in G.P., then $\cos^9 x + \cos^6 x + 3\cos^5 x - 1 =$
(A) –1 (B) 0 (C) 1 (D) none of these
यदि पहले वाले प्रश्न में निकाली गई बहुपद की घात p है तथा $(p-5) + \sin\theta, \cos\theta, \tan\theta$ गुणोत्तर श्रेढ़ी में है, तो $\cos^9 x + \cos^6 x + 3\cos^5 x - 1 =$
(A) –1 (B) 0 (C) 1 (D) इनमें से कोई नहीं

Ans. B

Sol. P = 5
$0 + \sin x, \cos x, \tan x \rightarrow$ G.P.

$$\Rightarrow \cos^2 x = \frac{\sin^2 x}{\cos x}$$

$\Rightarrow \cos^3 x = \sin^2 x$
$\Rightarrow \cos^9 x + \cos^6 x + 3\cos^2 x \cdot \cos^3 x - 1$
$= (\sin^2 x)^3 + \cos^6 x + 3\sin^2 x \cos^2 x - 1$
$= \sin^6 x + \cos^6 x + 3\sin^2 x \cos^2 x - 1$
$= (\sin^2 x + \cos^2 x)^3 - 3\sin^2 x \cos^2 x (\sin^2 x + \cos^2 x) + 3\sin^2 x \cos^2 x - 1$
$= 0$

Paragraph for Question No. 31 to 33

Let p be the product of the sines of the angles of a triangle ABC and q is the product of the cosines of the angles.

मानाकि एक त्रिभुज ABC के कोणों की ज्याओं (sines) का गुणनफल p है और कोणों की कोज्याओं (cosines) का गुणनफल q है।

31. In this triangle $\tan A + \tan B + \tan C$ is equal to :

(A) $p + q$ (B) $p - q$ (C) $\frac{p}{q}$ (D) none of these

इस त्रिभुज में $\tan A + \tan B + \tan C$ =

(A) $p + q$ (B) $p - q$ (C) $\frac{p}{q}$ (D) इनमें से कोई नहीं

Ans. C

Sol. Given $\sin A \sin B \sin C = P$ $\cos A \cos B \cos C = q$

in ΔABC

$A + B + C = \pi$

$\Rightarrow \tan A + \tan B + \tan C = \tan A \cdot \tan B \cdot \tan C$

$$= \frac{\sin A \cdot \sin B \cdot \sin C}{\cos A \cos B \cos C}$$

$$= \frac{p}{q}$$

32. $\tan A \tan B + \tan B \tan C + \tan C \tan A =$

$\tan A \tan B + \tan B \tan C + \tan C \tan A =$

(A) $1 + q$ (B) $\frac{1+q}{q}$ (C) $1 + p$ (D) $\frac{1+p}{p}$

Ans. B

Sol. $\tan A \tan B + \tan B \tan C + \tan C \tan A$

$$= \frac{\sin A \sin B}{\cos A \cos B} + \frac{\sin B \sin C}{\cos B \cos C} + \frac{\sin A \sin C}{\cos A \cos C}$$

$$= \frac{\cos C \sin A \sin B + \cos A \sin B \sin C + \sin A \cos B \sin C}{\cos A \cos B \cos C}$$

$$= \frac{\sin B \cdot (\sin A \cos C + \cos A \sin C) + \sin A \cos B \sin C}{q}$$

$$= \frac{1}{q}\left[\sin B \cdot \sin(A+C) + \sin A \cdot \cos B \cdot \sin C\right]$$

$$= \frac{1}{q}\left[\sin^2 B + \sin A \cos B \cdot \sin C\right]$$

$$= \frac{1}{q}\left[1 - \cos^2 B + \sin A \cos B \sin C\right]$$

$$= \frac{1}{q}\left[1 - \cos B\left(\cos B - \sin A \cdot \sin C\right)\right]$$

$$= \frac{1}{q}\left[1 - \cos B\left\{-\cos(A+C) - \sin A \sin C\right\}\right]$$

$$= \frac{1}{q}\left[1 + \cos B \cdot \cos A \cdot \cos C\right] = \frac{1+q}{q}$$

33. The value of $\tan^3 A + \tan^3 B + \tan^3 C$ is :

$\tan^3 A + \tan^3 B + \tan^3 C$ का मान है :

(A) $\frac{p^3 - 3pq^2}{q^3}$ (B) $\frac{q^3}{p^3}$ (C) $\frac{p^3}{q^3}$ (D) $\frac{p^3 - 3pq}{q^3}$

Ans. D

Sol. $(\tan A + \tan B + \tan C)^3$

$= \tan^3 A + \tan^3 B + \tan^3 C + 3(\tan A + \tan B + \tan C) \cdot (\tan A \cdot \tan B + \tan B \tan C + \tan C \tan A) - 3\tan A \tan B \tan C$

$$\Rightarrow \frac{p^3}{q^3} = \tan^3 A + \tan^3 B + \tan^3 C + 3\left(\frac{1+q}{q}\right)\cdot\frac{p}{q} - 3\frac{p}{q}$$

$$\Rightarrow \frac{p^3}{q^3} - \frac{3p}{q^2} = \tan^3 A + \tan^3 B + \tan^3 C$$

$$\Rightarrow \tan^3 A + \tan^3 B + \tan^3 C = \frac{p^3 - 3pq}{q^3}$$

MATCH THE COLUMN TYPE QUESTION(S)

34. Match the following column :

Column I	**Column II**
(A) The value of $\cos\frac{\pi}{19}+\cos\frac{3\pi}{19}+\cos\frac{5\pi}{19}+.....+\cos\frac{17\pi}{19}$ is	(P) – 3
(B) Exact value of $\cos^4\frac{\pi}{16}+\cos^4\frac{3\pi}{16}+\cos^4\frac{5\pi}{16}+\cos^4\frac{7\pi}{16}$ is	(Q) $\frac{3}{16}$
(C) The value of $6(\sin^6\theta+\cos^6\theta)-9(\sin^4\theta+\cos^4\theta)$ is	(R) $\frac{1}{2}$
(D) The value of the expression $\sin 20°\sin 40°\sin 60°\sin 100°$ is	(S) $\frac{3}{2}$

निम्न कॉलम को सुमेलित कीजिए :

Column I	**Column II**
(A) $\cos\frac{\pi}{19}+\cos\frac{3\pi}{19}+\cos\frac{5\pi}{19}+.....+\cos\frac{17\pi}{19}$ का मान होगा	(P) – 3
(B) $\cos^4\frac{\pi}{16}+\cos^4\frac{3\pi}{16}+\cos^4\frac{5\pi}{16}+\cos^4\frac{7\pi}{16}$ का मान होगा	(Q) $\frac{3}{16}$
(C) $6(\sin^6\theta+\cos^6\theta)-9(\sin^4\theta+\cos^4\theta)$ का मान होगा	(R) $\frac{1}{2}$
(D) $\sin 20°\sin 40°\sin 60°\sin 100°$ का मान होगा	(S) $\frac{3}{2}$

Ans (A) R; (B) S; (C) P; (D) Q

Sol. (A) $\cos\left(\frac{\pi}{19}\right)+\cos\left(\frac{3\pi}{19}\right)+.....+\cos\left(\frac{17\pi}{19}\right)$

$=\cos\left(\frac{\pi}{19}\right)+\cos\left(\frac{\pi}{19}+\frac{2\pi}{19}\right)+\cos\left(\frac{\pi}{19}+\frac{4\pi}{19}\right)+.....+\cos\left(\frac{\pi}{19}+\frac{16\pi}{19}\right)$

$=\frac{\sin\left(\frac{9}{2}\cdot\frac{2\pi}{19}\right)}{\sin\left(\frac{\pi}{19}\right)}\cos\frac{\left(\frac{2\pi}{19}+\frac{16\pi}{19}\right)}{2}$

$=\frac{\sin\left(\frac{9\pi}{19}\right)\cos\left(\frac{9\pi}{19}\right)}{\sin\left(\frac{\pi}{19}\right)}=\frac{\sin\left(\frac{18\pi}{19}\right)}{2\sin\left(\frac{\pi}{19}\right)}=\frac{1}{2}$

(B) $\cos^4\frac{\pi}{16}+\cos^4\frac{3\pi}{16}+\cos^4\frac{5\pi}{16}+\cos^4\frac{7\pi}{16}$

$=\cos^4\frac{\pi}{16}+\cos^4\frac{5\pi}{16}+\cos^4\left(\frac{\pi}{2}-\frac{5\pi}{16}\right)+\cos^4\left(\frac{\pi}{2}-\frac{7\pi}{16}\right)$

$=\cos^4\frac{\pi}{16}+\sin^4\frac{\pi}{16}+\cos^4\frac{5\pi}{16}+\sin^4\frac{5\pi}{16}$

$=\left(\sin^2\frac{\pi}{16}+\cos^2\frac{\pi}{16}\right)^2-2\sin^2\frac{\pi}{16}\cos^2\frac{\pi}{16}+\left(\sin^2\frac{5\pi}{16}+\cos^2\frac{5\pi}{16}\right)^2-2\sin^2\frac{5\pi}{16}\cos^2\frac{5\pi}{16}$

$=1+1-\frac{1}{2}\left[\sin^2\frac{\pi}{8}+\sin^2\frac{5\pi}{8}\right]$

$=2-\frac{1}{2}\left[\sin^2\frac{\pi}{8}+\sin^2\left(\frac{\pi}{2}+\frac{\pi}{8}\right)\right]$

$=2-\frac{1}{2}=\frac{3}{2}$

(C) $6\,[\sin^6\theta+\cos^6\theta]-9\,(\sin^4\theta+\cos^4\theta)$

$=6\,[(\sin^2\theta+\cos^2\theta)^3-3\sin^2\theta\cos^2\theta]-9\,((\sin^2\theta+\cos^2\theta)^2-2\sin^2\theta\cos^2\theta)$

$=6-18\sin^2\theta\cos^2\theta-9+18\sin^2\theta\cos^2\theta$

$=-3$

(D) $\sin 20^\circ\sin 40^\circ\sin 60^\circ\sin 100^\circ$

$=\frac{\sqrt{3}}{2}\cdot\sin 40^\circ\sin\left(60^\circ+40^\circ\right)\sin\left(60^\circ-40^\circ\right)$

$=\frac{\sqrt{3}}{2}\cdot\frac{1}{4}\cdot\sin\left(120^\circ\right)=\frac{1}{4}\cdot\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2}$

$=\frac{3}{16}$